Kenya ENSO Explorer: Citable Scientific Methods Monograph & Companion Documentation

Theoretical Foundations, Empirical Baselines, and Remote Sensing Analytics for Subnational Climate Risk Assessment

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1 Overview & Citation Guide

2 Executive Summary

This scientific monograph provides formal methodological documentation, empirical baselines, statistical derivations, and data provenance for the **Kenya ENSO Explorer** (https://africaadaptationatlas.cgiar.org/).

The platform provides subnational decision-support across Kenya's 47 counties to evaluate how equatorial ocean dynamics (El Niño–Southern Oscillation, Indian Ocean Dipole, and Western Pacific anomalies) govern precipitation regimes, rangeland pasture biomass, physical flood inundation, and agricultural capital at risk.

2.1 Suggested Citation

> Steward, P., African Agriculture Adaptation Atlas Science Team, & Kenya Meteorological Department (2026). *Kenya ENSO Explorer: Citable Scientific Methods Monograph & Companion Documentation* (Version 3.11.0). CGIAR Climate Action / Alliance of Bioversity International and CIAT. https://doi.org/10.5281/zenodo.10849201

2.1.1 BibTeX Entry

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2.2 Structure of this Monograph

- **Chapter 1: County Context, Demography & Social Inclusion**: Demographic baselines, population densities, subcounty area geometry, and gender equity indicators.

- **Chapter 2: Equatorial Ocean Teleconnections & Seasonal Forecasting**: Sea surface temperature indices (RONI, ONI, DMI, Western-V), forecast plume verification, and ranked probability skill scores (RPSS).
- **Chapter 3: Precipitation Climatology, Empirical Quantiles & Regimes**: Mathematical rationale for non-parametric quantiles, positive rainfall skewness, and independent seasonal sequence classification.
- **Chapter 4: Satellite Earth Observation, Hazards & Exposure**: Sentinel-1 SAR flood extent (Copernicus GFM), JRC return period hydrodynamics, and MODIS NDVI pasture biomass anomalies.
- **Chapter 5: Agricultural Production & Economic Valuation**: KNBS agricultural survey detrending, subcounty capital weighting, and shoat-to-maize terms of trade.
- **Chapter 6: Master Dataset Catalogue & Provenance**: Full metadata contracts, producers, and licensing for all 23 platform datasets.
- **Chapter 7: Controlled Vocabulary & Glossary**: Standardized definitions, equations, and institutional references for 65 domain terms.

3 County Context, Demography & Social Inclusion

4 County Context, Demography & Social Inclusion

This chapter details the demographic, administrative, and gender vulnerability baselines across Kenya's 47 counties and sub-counties. It documents spatial geometric intersections over UN OCHA COD-AB boundaries, population density models from the KNBS 2019 National Census, and the multidimensional Gender Equality and Social Inclusion (GESI) indicators.

4.1 Figure 1.3: Socio-Economic & Gender Vulnerability Profile

Section in Platform: Section 1.3 | **Target Identifier:** `fig_1_3` | **Output Type:** Chart

> **Policy / Scientific Question:** *How does the county compare against the national benchmark across key social and economic vulnerability indicators?*

4.1.1 Scientific Purpose & Context

Quantify and communicate Socio-Economic & Gender Vulnerability Profile across Kenyan counties to inform anticipatory action and climate adaptation planning.

4.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Gender Equality and Social Inclusion (GESI) indicators extracted from KNBS County Gender Data Sheets (Oct 2025 edition). Evaluates female headship, agricultural land ownership titles, agricultural labor share, and clean cooking fuel access relative to national medians.

4.1.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$\Delta_{\text{GESI}} = I_c - \text{Median}_{\text{Kenya}}$$

4.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-county-gender-datasheets`	Kenya County Gender Data Sheets, 2025	kenya	2025 edition (indicators from KDHS 2022, KPHC 2019 and the 2022 general election)

Served Parquet Files in Platform: `gesi.parquet`, `gesi_v2.parquet`

4.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

4.1.5 Controlled Vocabulary Keywords

`GESI` • `KNBS`

4.2 Table 1.1: Sub-county Baseline Geography & Rainfall Climatology

Section in Platform: Section 1.0 | **Target Identifier:** `table_1_0` | **Output Type:** Table

> **Policy / Scientific Question:** *What are the baseline geographic dimensions, population, and rainfall patterns across sub-counties?*

4.2.1 Scientific Purpose & Context

Quantify and communicate Sub-county Baseline Geography & Rainfall Climatology across Kenyan counties to inform anticipatory action and climate adaptation planning.

4.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Spatial polygon intersection over UN OCHA COD-AB boundaries; population density = headcount / area (km²); 1981–2025 CHIRPS v3 monthly accumulation aggregated to subcounty administrative boundaries.

4.2.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$\rho_{sc} = \frac{\text{Pop}_{sc}}{\text{Area}_{sc}}$$

$$\bar{R}_{s,sc} = \frac{1}{N} \sum_{y=1}^{N} \sum_{m \in s} R_{y,m,sc}$$

4.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
kenya-population-knbs-census	KNBS 2019 Census Population Counts by Co	kenya — counties (47 admin-1 units); a sub-county table is included in the KNBS universe	2019
africa-precipitation-monthly-seasonal	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)
kenya-admin-codab	Kenya Administrative Boundaries, Countie	kenya — Admin 1 (47 counties) and admin 2 (290 sub-counties)	2019-10 to 2025-01

Served Parquet Files in Platform: `population_knbs_census_adm1.parquet`, `subcounty_rainfall_climatology.parquet`

4.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

4.2.5 Controlled Vocabulary Keywords

`KNBS` • `COD-AB` • `MAM` • `OND` • `WMO Normal`

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5 Equatorial Ocean Teleconnections & Seasonal Forecasting

6 Equatorial Ocean Teleconnections & Seasonal Forecasting

This chapter presents the physical oceanographic indices, dynamical model plumes, and statistical verification frameworks governing Kenyan seasonal climate predictability. It establishes the detrended Relative Oceanic Niño Index (RONI), Indian Ocean Dipole Mode Index (DMI), multi-model forecast plume ensembles (SINTEX-F, IRI), and Ranked Probability Skill Score (RPSS) cross-validation protocols.

6.1 Advisory Card 2.0: Kenya Meteorological Department (KMD) ClimWeb CAP Weather Advisories

Section in Platform: Section 2 | **Target Identifier:** `box_2_0` | **Output Type:** Table

> **Policy / Scientific Question:** *What active severe weather alerts, heavy rainfall warnings, and precautionary advisories are currently in effect for this county from the Kenya Meteorological Department?*

6.1.1 Scientific Purpose & Context

Quantify and communicate Kenya Meteorological Department (KMD) ClimWeb CAP Weather Advisories across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Official Kenya Meteorological Department (KMD) seasonal climate advisory synthesis and sectoral contingency protocols.

6.1.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

6.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kmd-official-advisories`	kmd-official-advisories	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `kmd_cap_county_active.parquet`

6.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.1.5 Controlled Vocabulary Keywords

`KMD • AA • Common Alerting Protocol`

6.2 Figure 2.1: Empirical Tercile Probabilities & Analogue Matching (Short Rains or Long Rains Outlook)

Section in Platform: Section 2 | **Target Identifier:** `fig_4_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *Given the current and forecast state of the equatorial Pacific and Indian Ocean, how were the coming season's rainfall terciles distributed in the eight most similar historical years for this county, and how well has that analogue method performed when tested against 45 years of history?*

6.2.1 Scientific Purpose & Context

Figure 2.1 is the Explorer's seasonal outlook card. For the selected county and target season (the Short Rains, OND, or the Long Rains, MAM) it finds the eight historical years whose ocean state most closely resembles the present and reports how often each of those years produced a wetter, near-normal or drier season in that county. The result is shown as a three-segment probability bar with dashed markers at the one-third climatological shares, the tercile bounds in millimetres from the county's 1991–2020 record, dots for the individual analogue years, and a verification strip summarising the method's leave-one-out hindcast skill for that county, season and matching criterion.

The approach is deliberately empirical and transparent. A planner can see which years drove the outlook, check what happened in those years against local memory and records, and judge the outlook against the track record shown beside it. Three matching criteria are offered: Full Trajectory, which weights the pre-season ocean state and the forecast in-season state equally; Lead-in Observed, which uses only the observed pre-season state; and Projected Season, which uses only the forecast in-season state.

The card does not replace the official seasonal forecast of the Kenya Meteorological Department (KMD), which should govern operational decisions. Its purpose is to make the historical base rate behind ENSO-conditioned outlooks visible at county scale and to state plainly where that base rate carries skill. In OND the analogue method beats climatology in a minority of counties, mostly in the east and northeast. In MAM it does not beat climatology anywhere in a meaningful way, and every MAM outlook carries a standing notice to that effect.

6.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: Tercile bounds and distance standardization from the 1991–2020 WMO standard normal; analogue candidates 1981 to 2025

Target season and candidates. The user selects OND or MAM. Candidate years are every year from 1981 to 2025 in the outlook table with the required ocean coordinates. Each candidate carries its observed county rainfall tercile (Drier, Near normal, Wetter), assigned offline from the county's 1991–2020 seasonal totals with empirical tercile bounds (linear interpolation between order statistics), a total at or below the lower bound classed Drier and a total strictly above the upper bound classed Wetter.

Current ocean state. The lead-in coordinates are the most recently published CPC Relative Oceanic Niño Index (RONI) three-month value and the most recent single month of the CPC ERSSTv6 Dipole Mode Index (DMI). The in-season (peak) coordinates are the median across members of the IRI Niño 3.4 multi-model plume for the target season, shifted into RONI space by subtracting a fixed 0.38 °C, and the median of the JAMSTEC SINTEX-F IOD plume members for the same season. Plume seasons are located by their label (the target season first, then OND).

Distance and selection. Each coordinate difference between the current state and a candidate year is divided by that coordinate's 1991–2020 sample standard deviation for the target season (for OND: RONI lead-in 0.74, DMI lead-in 0.37, RONI in-season 1.12, DMI

in-season 0.37 °C; for MAM: 1.11, 0.19, 0.64, 0.22 °C). Under Full Trajectory the distance is the square root of half the summed squared lead-in differences plus half the summed squared in-season differences; Lead-in Observed and Projected Season use only one pair. Candidates are ranked by distance and the eight nearest are kept. Years missing a required coordinate are excluded rather than scored as zero difference.

Probabilities. The forecast probability of each tercile is the share of the eight analogue years whose observed tercile fell in it, displayed as a whole-number percentage. There is no distance weighting and no shrinkage toward one third. The modal category is the tercile with the largest count; ties are reported as no single modal category. The millimetre bounds printed with the bar are the empirical 33.3rd and 66.7th percentiles of the county's 1991–2020 seasonal totals, shown only when at least ten baseline years exist.

Verification strip. The card reads the leave-one-out hindcast record for the same county, season and matching criterion with eight analogues (the method and the full skill results are documented in the Table 2.3 card). It shows the Ranked Probability Skill Score (RPSS), the modal hit rate and its gain over the 33.3 percent baseline, the Heidke Skill Score, the Brier Skill Score for the wet tercile and the severe wet false-alarm rate. The skill class (High, Moderate, Marginal or No skill) sets the badge colour and the guidance text. Probabilities are displayed in every class; MAM outlooks always carry a notice that they are a historical frequency guide rather than predictive guidance.

6.2.2.1 Methodological Rationale

Why analogues. Dynamical seasonal forecasts for East Africa are issued at regional scale and in terms of tercile shifts; county planners need the same information for their own county and season. An analogue ensemble provides it directly from each county's observed rainfall record, with no statistical model between the ocean state and the county outcome. It also keeps the evidence inspectable: the eight years are listed, and anyone can check what happened in them.

Why standardized distance over RONI and DMI. The Short Rains respond to both the Pacific (ENSO) and the Indian Ocean Dipole, which often co-vary but not always. Using both indices in the distance lets analogues reflect the joint state. Dividing each difference by its 1991–2020 standard deviation stops the index with the larger raw variance (RONI) from dominating. RONI is used rather than raw Niño 3.4 because subtracting the tropical-mean sea surface temperature anomaly removes the secular warming that would otherwise make recent years look like El Niño states; the fixed 0.38 °C shift applied to the IRI plume approximates that tropical-mean correction for a plume issued in raw Niño 3.4 units.

Why eight analogues and no smoothing. Eight years is a compromise between sharpness and sampling noise in a 45-year record: fewer analogues give jumpy probabilities, more drift toward climatology and dilute the ocean signal. The probabilities are left as raw frequencies so that they remain interpretable as counts of real seasons. The cost is coarseness (steps of 12.5 percent) and overconfidence when the method has no skill, which is why skill is reported beside every outlook.

What the verification shows. Across the 47 counties, the Full Trajectory criterion with eight analogues has a mean OND modal hit rate of 43 percent against 33 percent by chance and positive RPSS in 20 counties; the Projected Season criterion reaches positive RPSS in 25. For MAM the mean RPSS is negative under every criterion and the hit rate is at chance level. The outlook should therefore be read as a historical frequency for MAM everywhere, and for OND in counties whose badge shows No skill.

Governing Mathematical Formulations:

Current ocean state. Lead-in values are the latest published RONI season and latest CPC ERSSTv6 DMI month; s is the target season located by label in each plume.

$$\mathbf{x}_0 = \left(\mathrm{RONI}_0^{\mathrm{lead}},\ \mathrm{DMI}_0^{\mathrm{lead}},\ \mathrm{RONI}_0^{\mathrm{peak}},\ \mathrm{DMI}_0^{\mathrm{peak}}\right), \quad \mathrm{RONI}_0^{\mathrm{peak}} = \mathrm{median}_m\left(\mathrm{Niño3.4}_{m,s}^{\mathrm{IRI}}\right) - 0.38, \quad \mathrm{DMI}_0^{\mathrm{peak}} =$$

Standardized analogue distance. Full Trajectory w = (0.5, 0.5); Lead-in Observed w = (1, 0); Projected Season w = (0, 1). Sigmas are 1991–2020 sample standard deviations for the target season.

$$D_i = \sqrt{w_{\mathrm{lead}}\left[\left(\frac{\Delta\mathrm{RONI}_i^{\mathrm{lead}}}{\sigma_{\mathrm{RONI}}^{\mathrm{lead}}}\right)^2 + \left(\frac{\Delta\mathrm{DMI}_i^{\mathrm{lead}}}{\sigma_{\mathrm{DMI}}^{\mathrm{lead}}}\right)^2\right] + w_{\mathrm{peak}}\left[\left(\frac{\Delta\mathrm{RONI}_i^{\mathrm{peak}}}{\sigma_{\mathrm{RONI}}^{\mathrm{peak}}}\right)^2 + \left(\frac{\Delta\mathrm{DMI}_i^{\mathrm{peak}}}{\sigma_{\mathrm{DMI}}^{\mathrm{peak}}}\right)^2\right]}$$

Analogue set and tercile probabilities. The eight nearest candidate years; displayed as rounded whole percentages, which need not sum to exactly 100.

$$\mathcal{A} = \underset{|S|=8}{\arg\min} \sum_{i \in S} D_i, \qquad \hat{p}_k = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \mathbb{I}(T_i = k), \quad k \in \{\mathrm{Drier}, \mathrm{Near\ normal}, \mathrm{Wetter}\}$$

Modal category. Ties are shown as no single modal category.

$$\hat{k} = \arg\max_k \hat{p}_k \quad \text{(undefined when two or three categories tie)}$$

Tercile bounds shown in millimetres. Hyndman and Fan Type 7 empirical quantiles of the county's seasonal totals.

$$t_1 = \hat{Q}_{1991\text{–}2020}\left(\tfrac{1}{3}\right), \qquad t_2 = \hat{Q}_{1991\text{–}2020}\left(\tfrac{2}{3}\right), \qquad \text{shown only if } N_{1991\text{–}2020} \geq 10$$

6.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	Equatorial Pacific and tropical Indian Ocean basin indices	Analogue candidates 1981 to 2025; current-season plumes
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	0.05 deg (~5.5 km, CHIRPS v3 native) aggregated to county mean	1981 to 2025; terciles from 1991–2020

Served Parquet Files in Platform: `enso_outlook_base.parquet`, `enso_hindcast_skill.parquet`, `enso_drivers_seasonal.parquet`, `iri_forecast_plume.json`, `iod_forecast_plume.json`

6.2.4 Methodological Assumptions & Analytical Limitations

- Eight analogues give probabilities in steps of 12.5 percent. A shift of one year between categories moves a probability by 12.5 points, so differences of that size carry no information on their own.
- Probabilities are shown whatever the hindcast skill. For MAM, and for OND in counties whose badge shows No skill, the bar is a historical frequency, not a forecast, and should not be used to depart from climatological planning on its own.
- The live lead-in coordinates differ from the historical ones. Historical candidates use the JAS (for OND) or DJF (for MAM) RONI and a three-month HadISST DMI mean, whereas the live state uses the most recently published RONI season and the latest single month of the CPC ERSSTv6 DMI. Early in the lead-in window the two can refer to different months.
- The in-season coordinates come from model plumes and inherit their errors. The IRI plume is converted to RONI space with a fixed 0.38 °C offset, which differs from the time-varying conversion used in the plume figure; the SINTEX-F IOD plume is used without adjustment.
- The hindcast skill shown in the strip uses observed in-season ocean values, so for the Full Trajectory and Projected Season criteria it is an upper bound on real-time skill (see Table 2.3).
- Candidate years are drawn from 1981 to 2025 and are not detrended. A warming Indian Ocean and the drying of the Long Rains mean that early analogue years may represent a different background climate.
- If the hindcast record for the exact criterion is missing, the card falls back to another record for the same county and season, and several display fields have fixed placeholder values when data are absent; the badge should be read only when a county-specific record is present.
- This outlook is not an official forecast. Operational decisions should follow the Kenya Meteorological Department (KMD) seasonal forecast and county advisories.

6.2.5 Controlled Vocabulary Keywords

`Analogue Year` • `Terciles` • `Conditional Probability` • `RONI` • `DMI` • `IRI Plume` • `SINTEX-F` • `RPSS` • `LOOCV` • `Modal Hit Rate` • `HSS` • `BSS` • `Forecast Skill` • `KMD` • `OND` • `MAM`

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6.3 Figure 2.2: Historical Analogue Scenario Multi-Hazard Diagnostics

Section in Platform: Section 2 | **Target Identifier:** `fig_4_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *What multi-hazard impacts occurred when this specific analogue ocean condition was observed in the past, and how did compound climate shocks propagate through crop production, rangeland pasture, and market purchasing power?*

6.3.1 Scientific Purpose & Context

Quantify and communicate Historical Analogue Scenario Multi-Hazard Diagnostics across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Historical multi-driver analog identification matching joint Pacific RONI/ONI and Indian Ocean DMI phases to past seasons with multi-hazard impacts.

6.3.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{Distance}(y) = \sqrt{w_1(z_{\text{RONI},y} - z_{\text{target}})^2 + w_2(z_{\text{DMI},y} - z_{\text{target}})^2}$$

6.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `driver_indices.parquet`, `chirps_county_monthly.parquet`

6.3.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.3.5 Controlled Vocabulary Keywords

`Analogue Year` • `Joint Ocean Phase` • `RONI` • `DMI`

6.4 Figure 5.1: Physical Teleconnection Mechanics: Pacific-to-Kenya Teleconnection

Section in Platform: Section 5.1 | **Target Identifier:** `fig_5_1` | **Output Type:** Figure

> **Policy / Scientific Question:** *What physical atmospheric wave dynamics connect equatorial Pacific warming to East African rainfall?*

6.4.1 Scientific Purpose & Context

Quantify and communicate Physical Teleconnection Mechanics: Pacific-to-Kenya Teleconnection across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Physical teleconnection mechanics: Walker circulation displacement, Indian Ocean dipole zonal wind reversals, and upper-level divergence modulating East African convective rainfall.

6.4.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

6.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `driver_indices.parquet`

6.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.4.5 Controlled Vocabulary Keywords

`Walker Circulation` • `ENSO`

6.5 Table 2.1: Two-Season Compound Agricultural & Livelihood Outcome Matrix

Section in Platform: Section 2 | **Target Identifier:** `table_2_1` | **Output Type:** Table

> **Policy / Scientific Question:** *How did consecutive MAM and OND seasons compound agricultural and livelihood shocks in the selected analogue year?*

6.5.1 Scientific Purpose & Context

Quantify and communicate Two-Season Compound Agricultural & Livelihood Outcome Matrix across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.5.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Two-season compound agricultural impact table pairing OND and MAM analog seasons with observed crop and rangeland yields.

6.5.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

6.5.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024
`harveststat-crop-production`	HarvestStat Africa - Harmonized Subnatio	County (ADM1), 47 counties	1965 to 2024 (harvest years; seasonal gap 2002 to 2014)

Served Parquet Files in Platform: `harveststat_county_production.parquet`, `driver_indices.parquet`

6.5.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.5.5 Controlled Vocabulary Keywords

`Double Failure` • `Analogue Year` • `HarvestStat`

6.6 Table 2.2: Official KMD Seasonal Sectoral Advisory Matrix

Section in Platform: Section 2 | **Target Identifier:** `table_2_2` | **Output Type:** Table

> **Policy / Scientific Question:** *What anticipatory actions are recommended across agriculture, livestock, water, and disaster management sectors?*

6.6.1 Scientific Purpose & Context

Quantify and communicate Official KMD Seasonal Sectoral Advisory Matrix across Kenyan counties to inform anticipatory action and climate adaptation planning.

6.6.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Tabular sectoral guidelines cross-referencing KMD forecast advisories across Agriculture, Water, Livestock, and Disaster Risk Reduction sectors.

6.6.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

6.6.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kmd-official-advisories`	kmd-official-advisories	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `kmd_cap_county_active.parquet`

6.6.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

6.6.5 Controlled Vocabulary Keywords

`KMD • AA • Anticipatory Action`

6.7 Table 2.3: Forecast Verification, Categorical Skill Assessment & Official Advisory Directory

Section in Platform: Section 2 | **Target Identifier:** `table_2_3` | **Output Type:** Table

> **Policy / Scientific Question:** *How much predictive skill does equatorial ocean state carry for county seasonal rainfall terciles when tested honestly against the historical record, and where should county planners obtain the official operational forecasts and warnings that the Explorer's outlook complements?*

6.7.1 Scientific Purpose & Context

A probabilistic seasonal outlook is only as useful as its verified track record. Section 2 of the Explorer issues an empirical analogue outlook for the coming season: it finds the historical years whose equatorial Pacific and Indian Ocean state most resembles the present, reads off the rainfall terciles those years produced in the selected county, and reports the resulting tercile frequencies as forecast probabilities. Without verification such an outlook is an appeal to plausibility. This card documents the verification framework that accompanies every outlook: a 45-year leave-one-out cross-validated (LOOCV) hindcast that scores the analogue method against the observed CHIRPS v3 terciles for every county, both seasons, every analogue-selection criterion and both ensemble sizes, using the Ranked Probability Skill Score (RPSS) as the headline metric with modal hit rate, Heidke Skill Score, Brier Skill Scores for the dry and wet terciles and severe false-alarm rates alongside. The skill badge, the plain-language skill statement and the Long Rains planning notice shown beside each outlook are all driven by this table.

The second purpose is institutional. Table 2.3 itself is the directory of official forecast, early warning and advisory products for Kenya: the Kenya Meteorological Department (KMD) seasonal outlooks and Common Alerting Protocol warnings, the IGAD Climate Prediction and Applications Centre (ICPAC) Greater Horn of Africa Climate Outlook Forum consensus, the National Drought Management Authority (NDMA) county drought early warning bulletins, the Kenya Food Security Steering Group (KFSSG) assessments and the Kenya Red Cross Society early action protocols. The Explorer's analogue outlook is a complement to these products, intended to make the historical base rate visible; it is not an official forecast, and the hindcast skill reported here applies to the Explorer's analogue method only. The directory routes planners to the mandated sources for operational decisions.

Verification also delivers the platform's most important piece of epistemic humility. The hindcast shows that ocean state carries modest, county-dependent skill for the Short Rains (OND) and no usable skill for the Long Rains (MAM). The Explorer still displays the MAM analogue frequencies, but every MAM outlook carries a fixed notice that they are a historical frequency guide rather than a forecast.

6.7.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: Observed terciles from the 1991–2020 WMO standard normal; hindcast years 1981 to 2025 (N = 45); distance standardization from 1991–2020

Forecast object. For a target season (OND or MAM) and a county, the analogue outlook is a probability vector over K = 3 ordered rainfall categories, Below normal (Dry), Near normal and Above normal (Wet), defined by the empirical 1991–2020 terciles of the county's CHIRPS seasonal total. Candidate years are characterised by four ocean-state coordinates: the Relative Oceanic Niño Index (RONI) and the Dipole Mode Index (DMI), each averaged over a pre-season lead-in window and over the target season itself. Each coordinate difference between two years is standardized by that coordinate's 1991–2020 standard deviation, and a Euclidean distance combines them under one of three selectable criteria: Full Trajectory (equal weight on lead-in and concurrent coordinates), Lead-in Observed (lead-in coordinates only) and Projected Season (concurrent coordinates only). The k nearest years (k = 5 or k = 8) form the analogue ensemble, and the forecast probability of each category is the fraction of analogue years whose observed tercile fell in it.

Leave-one-out cross-validation. For every hindcast year T in 1981 to 2025, year T is withheld, the remaining 44 years are ranked by distance from T, the k nearest form the ensemble, and the resulting probability vector is scored against the observed tercile of year T. The test year never contributes to its own forecast. Scores are averaged over the 45 test years to give one record per county, season, criterion and k; the live outlook always uses the eight nearest analogue years, so the notebook displays the k = 8 record for the user's selected criterion (if that record is missing it falls back to any record for the same county and season).

Ranked Probability Score. For each test year the forecast and observation vectors are accumulated over the ordered categories, P_m = sum of forecast probabilities up to category m and O_m = sum of the observation indicator up to m. Because the cumulative sums reach 1 at m = K, only the first K − 1 = 2 terms carry information, and the RPS is the sum of squared differences of the cumulative vectors over those terms. The reference forecast is the equal-odds climatology p_clim = (1/3, 1/3, 1/3), whose RPS is computed against the same observation each year. The RPSS is one minus the ratio of the mean forecast RPS to the mean climatological RPS over the 45 years; RPSS = 1 is a perfect deterministic forecast, 0 matches climatology and negative values are worse than climatology.

Categorical metrics. The modal forecast category is the category with the largest analogue count. When two or three categories tie, the hit is credited fractionally (1/2 or 1/3) and the case is recorded in a "Tied" row of the 3 × 3 contingency table. The modal hit rate is the mean credited hit over the 45 years; the hit-rate gain is the hit rate minus the 33.3 percent expected from an uninformed three-way choice; the Heidke Skill Score rescales the hit count by its climatological expectation. Brier Skill Scores are computed separately for the binary events "Dry occurs" and "Wet occurs" against the climatological Brier score of (1/3 − o)^2. The severe false-alarm rate for wet is the share of modal Wet forecasts that verified Dry, and symmetrically for dry.

Skill classes. Records are labelled High (RPSS 0.20), Moderate (0.10 RPSS < 0.20), Marginal (0 RPSS < 0.10) and No skill (RPSS < 0). The class sets the colour of the skill

badge and selects the guidance text. It does not suppress the probabilities: tercile percentages are displayed for every county and season whatever the class, and the user is expected to read them together with the badge and, for MAM, the standing planning notice.

6.7.2.1 Methodological Rationale

Why RPSS is the headline metric. The three rainfall categories are ordered, and the cost of a forecast error grows with distance: issuing a confident Above-normal outlook when Below-normal occurs is a worse failure than issuing Near-normal. The Brier score treats each category as a separate yes/no event and is blind to this ordering, and the area under the relative operating characteristic (ROC) curve is defined for one binary event at a time and measures discrimination only, ignoring whether the stated probabilities are calibrated. The Ranked Probability Score (Epstein 1969; Murphy 1971) is the multi-category generalisation of the Brier score computed on cumulative probabilities, so a forecast that places its mass two categories from the outcome is penalised more than one that places it one category away. It is a strictly proper scoring rule, so it cannot be gamed by hedging, and it rewards both reliability and resolution (Wilks 2011; Jolliffe and Stephenson 2012). Expressing it as a skill score against equal-odds climatology answers the question a planner actually asks: does conditioning on ocean state beat assuming that each tercile is equally likely?

Why leave-one-out. Ocean-rainfall relationships estimated and evaluated on the same years overstate skill, because the test year's own outcome has been used to estimate the conditional frequencies. Withholding each year in turn is the only cross-validation scheme that leaves 44 training years for every test year in a record this short, and it is the standard protocol for small-sample seasonal hindcasts. It removes in-sample optimism in the analogue selection; it does not remove information shared between adjacent years through persistence and trend (see limitations).

What the hindcast shows. For OND with the Projected Season criterion and k = 8, the county-average RPSS is +0.02, 25 of 47 counties have positive RPSS, the best counties (Marsabit, Mandera, Vihiga) reach +0.30 to +0.32, the mean modal hit rate is 43 percent against the 33.3 percent baseline and the mean Heidke Skill Score is +0.15. The Full Trajectory criterion at k = 8 is similar (20 of 47 positive, mean hit 43 percent). Skill is concentrated in the eastern and northeastern counties, where the Short Rains respond to the combined ENSO and IOD signal, and is weakest in the lake basin and the western highlands. For MAM every criterion and ensemble size yields a negative county-average RPSS (between −0.10 and −0.22), at most two counties exceed zero, and the modal hit rate is indistinguishable from chance (33 to 35 percent). This is the boreal spring predictability barrier: ENSO events decay or transition during MAM, the Pacific signal that organises the Short Rains is absent, and Long Rains variability is governed by regional circulation (the Western-V pattern, Indian Ocean moisture flux, the Turkana jet and intraseasonal activity) that the two-index analogue does not capture. Negative RPSS values arise because an analogue ensemble of five or eight years is sharp: it assigns probabilities in steps of 0.2 or 0.125 and is penalised relative to the flat climatology whenever that sharpness is misplaced. The platform treats these results as a finding to be communicated, not a defect to be tuned away.

Governing Mathematical Formulations:

Standardized analogue distance between candidate years. Delta denotes the difference between years i and j; sigma values are 1991–2020 standard deviations. Full Trajectory uses w_lead = w_conc = 0.5, Lead-in Observed uses (1, 0), Projected Season uses (0, 1).

$$D_{i,j} = \sqrt{w_{\text{lead}} \left(\frac{(\Delta\text{RONI}_{\text{lead}})^2}{\sigma^2_{\text{RONI,lead}}} + \frac{(\Delta\text{DMI}_{\text{lead}})^2}{\sigma^2_{\text{DMI,lead}}} \right) + w_{\text{conc}} \left(\frac{(\Delta\text{RONI}_{\text{conc}})^2}{\sigma^2_{\text{RONI,conc}}} + \frac{(\Delta\text{DMI}_{\text{conc}})^2}{\sigma^2_{\text{DMI,conc}}} \right)}$$

Leave-one-out analogue forecast probabilities. Categories k are Dry, Near and Wet from the county's 1991–2020 empirical terciles; the test year never enters its own ensemble.

$$p^{(-T)}_{T,k} = \frac{1}{|\mathcal{A}_T|} \sum_{i \in \mathcal{A}_T} \mathbb{I}(\text{Terc}_i = k), \qquad \mathcal{A}_T = \text{the } k \text{ nearest years to } T \text{ among } \{1981, \ldots, 2025\} \backslash \{T\}$$

Cumulative forecast and observation vectors. Categories are ordered Dry < Near < Wet; the third cumulative term is identically 1 and drops out.

$$P_m = \sum_{k=1}^{m} p_k, \qquad O_m = \sum_{k=1}^{m} o_k, \qquad m \in \{1, 2\}, \quad o_k = \mathbb{I}(\text{observed category} = k)$$

Ranked Probability Score and climatological reference. K = 3; the reference forecast is the equal-odds climatology (1/3, 1/3, 1/3).

$$\text{RPS}_T = \sum_{m=1}^{K-1} (P_m - O_m)^2, \qquad \text{RPS}^{\text{clim}}_T = \left(\tfrac{1}{3} - O_1\right)^2 + \left(\tfrac{2}{3} - O_2\right)^2$$

Ranked Probability Skill Score. RPSS = 1 is perfect, 0 equals climatology, negative is worse than climatology; the ratio of means is used, not the mean of ratios.

$$\text{RPSS} = 1 - \frac{\frac{1}{N}\sum_{T=1}^{N} \text{RPS}_T}{\frac{1}{N}\sum_{T=1}^{N} \text{RPS}^{\text{clim}}_T}, \qquad N = 45$$

Leave-one-out modal hit rate with tie splitting. A tie among two or three modal categories credits 1/2 or 1/3 of a hit; the gain is measured against the 33.3 percent uninformed baseline.

$$\text{HR}_{\text{modal}} = \frac{1}{N} \sum_{T=1}^{N} \frac{\mathbb{I}\left(\text{Terc}_T \in \arg\max_k p^{(-T)}_{T,k}\right)}{\left|\arg\max_k p^{(-T)}_{T,k}\right|}, \qquad \text{Gain} = \text{HR}_{\text{modal}} - \tfrac{1}{3}$$

Heidke Skill Score and Brier Skill Scores. H is the credited hit count; BSS is computed separately for the drought (Dry) and flood-relevant (Wet) terciles.

$$\mathrm{HSS} = \frac{H - N/3}{N - N/3}, \qquad \mathrm{BSS}_c = 1 - \frac{\frac{1}{N}\sum_T \left(p_{T,c} - o_{T,c}\right)^2}{\frac{1}{N}\sum_T \left(\frac{1}{3} - o_{T,c}\right)^2}, \quad c \in \{\mathrm{Dry}, \mathrm{Wet}\}$$

Severe false-alarm rates. Share of modal Wet (Dry) forecasts that verified in the opposite extreme tercile, taken from the 3×3 contingency table.

$$\mathrm{FAR}^{\mathrm{severe}}_{\mathrm{Wet}} = \frac{n(\hat{c} = \mathrm{Wet},\ c = \mathrm{Dry})}{n(\hat{c} = \mathrm{Wet})}, \qquad \mathrm{FAR}^{\mathrm{severe}}_{\mathrm{Dry}} = \frac{n(\hat{c} = \mathrm{Dry},\ c = \mathrm{Wet})}{n(\hat{c} = \mathrm{Dry})}$$

Skill class used for the badge and guidance text. The Long Rains outlook falls in the No-skill class for almost every county; its probabilities are still shown, with a notice that they are a historical frequency guide rather than a forecast.

$$\mathrm{Class} = \begin{cases} \text{High} & \mathrm{RPSS} \geq 0.20 \\ \text{Moderate} & 0.10 \leq \mathrm{RPSS} < 0.20 \\ \text{Marginal} & 0 \leq \mathrm{RPSS} < 0.10 \\ \text{No skill} & \mathrm{RPSS} < 0 \end{cases}, \qquad \text{badge colour and guidance text set by Class}$$

6.7.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	Equatorial Pacific (Niño 3.4 relative to tropical mean) and tropical Indian Ocean basin indices	1950 to present; hindcast uses 1981 to 2025
`enso-hindcast-skill`	enso-hindcast-skill	County (ADM1), 47 counties plus Ilemi Triangle	45 hindcast years (1981 to 2025), OND and MAM
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	0.05 deg aggregated to county mean	1981 to present

Served Parquet Files in Platform: `enso_hindcast_skill.parquet`, `enso_state_probabilities.parq`
`enso_outlook_base.parquet`

6.7.4 Methodological Assumptions & Analytical Limitations

- The verification sample is 45 seasons per county. Standard errors on a 45-year RPSS are large, and the difference between a Marginal and a Moderate class for one county is within sampling noise; the robust findings are the seasonal contrast (OND modest, MAM none) and the regional pattern, not individual county rankings.
- Leave-one-out removes the test year but not the information shared between neighbouring years through ENSO and IOD persistence, multi-year rainfall regimes or the long-term trend. The fixed 1991–2020 terciles also mean that 15 of the 45 hindcast years lie outside the baseline, and trends over 1981 to 2025 are not removed. Both effects make the hindcast a mildly optimistic estimate of real-time skill.
- In the hindcast the concurrent-season ocean coordinates are the observed values. In real time the Projected Season and Full Trajectory criteria use model plume forecasts of those coordinates, which carry their own error, so the hindcast for those criteria is a perfect-prognosis upper bound. The live outlook also translates the IRI Niño 3.4 plume median into RONI space with a fixed −0.38 °C offset. Even the Lead-in Observed criterion does not reproduce real-time conditions exactly: historical candidates use the JAS (for OND) or DJF (for MAM) RONI and a three-month HadISST DMI mean, whereas the live lead-in uses the most recently published RONI season and the latest single month of the CPC ERSSTv6 DMI.
- The equal-odds climatology is the conventional but weakest-informed reference. A trend-aware or persistence-based reference would be harder to beat and would lower the reported RPSS for OND further.
- Ensembles of five or eight analogues give forecast probabilities in coarse steps (multiples of 0.2 or 0.125). Sharpness is therefore imposed by design rather than learned from the data, and negative RPSS values partly reflect this coarseness.
- Observed terciles are county means of CHIRPS, which carries larger absolute uncertainty in the sparsely gauged northern and eastern counties; tercile ranks are more robust than totals but are not immune to it.
- Ties in the modal category are credited fractionally. This is a neutral convention, but it means the hit rate is not a count of unambiguous correct calls.
- The directory entries in Table 2.3 are institutional pointers and are maintained manually from the publishing institutions' portals; they are not themselves verified products, and the hindcast skill reported here does not apply to KMD, ICPAC or NDMA products, which have their own verification regimes.

6.7.5 Controlled Vocabulary Keywords

`RPSS` • `RPS` • `LOOCV` • `Modal Hit Rate` • `HSS` • `BSS` • `Forecast Skill` • `RONI` • `DMI` • `KMD` • `NDMA` • `KFSSG` • `AA` • `Terciles`

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7 Precipitation Climatology, Empirical Quantiles & Regimes

8 Precipitation Climatology, Empirical Quantiles & Regimes

This chapter defines the non-parametric empirical quantile baselines ($P_{33}, P_{67}, P_{10}, P_{90}$) over the 1991–2020 WMO standard normal period. It provides mathematical rationale for treating extreme positive rainfall skewness without Gaussian distortion, and formalizes independent seasonal standardization for consecutive sequence classification.

8.1 Figure 3.1: Multi-Decadal Precipitation Climatology & Empirical Tercile Bounds

Section in Platform: Section 3 | **Target Identifier:** `fig_2_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *How unusual was seasonal rainfall or drought across the historical satellite record, and where do this season's totals sit against the empirical tercile and extreme-decile bounds of the 1991–2020 normal?*

8.1.1 Scientific Purpose & Context

Kenya's rainfall is bimodal. The Intertropical Convergence Zone (ITCZ) crosses the equator twice a year, and each passage drives a rainy season: the March to May Long Rains (MAM) on the northward crossing and the October to December Short Rains (OND) on the southward crossing. The two seasons are not interchangeable. MAM delivers the larger share of annual rainfall in the western and central highlands and underpins the main maize harvest, whereas OND dominates the eastern and northern arid and semi-arid lands (ASAL) and is the season most strongly modulated by the El Niño–Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD). Between the two sits the long dry season, during which pastoral systems depend entirely on the pasture and water recharge delivered by the preceding rains.

Figure 3.1 places every season since 1981 for the selected county against this climatology. Bars show the county-mean seasonal total from the Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS v3), the 1991–2020 WMO standard normal mean is drawn as a reference line, and four dashed horizontal rules mark the empirical tercile bounds (P33 and P67) and extreme deciles (P10 and P90) of the same 30-year baseline. A strip beneath the x-axis colours each year by the concurrent ocean driver phase (ENSO and IOD for OND; the Western-V index for MAM) so that the reader can see whether an anomalous season

coincided with a known teleconnection state. The anomaly view re-expresses the bars as departures from the normal in millimetres and in percent, and the SPEI-3 option substitutes the three-month Standardized Precipitation-Evapotranspiration Index so that evaporative demand is accounted for.

The figure exists because the rainfall regime, not the annual mean, determines livelihood risk. Pastoral and marginal rainfed systems in the ASAL counties operate close to the lower tercile in most years. A season that falls at or below P33 removes the margin that herds and household food stocks rely on, and a season at or below P10 is the kind of event that triggers the National Drought Management Authority (NDMA) alarm phases and humanitarian scale-up. Expressing each season as a position within the empirical distribution of its own county and season gives county planners, the Kenya Meteorological Department (KMD) and humanitarian partners a single, comparable language for how unusual a season was.

8.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal (N = 30 seasons per county and season)

Data chain. CHIRPS v3.0 monthly precipitation at 0.05 degree resolution is summed per pixel into the twelve rolling three-month windows (JFM to DJF) and averaged over the pixels inside each county polygon (`value_mean`; the within-county pixel standard deviation is retained as `value_sd`). For the selected county, season and variable the figure queries one value per year from 1981 onward. The climatological baseline is the subset of years 1991 to 2020 inclusive, giving N = 30 seasonal totals per county and season.

Empirical quantile estimator. All tercile and decile bounds are sample quantiles computed with the Hyndman and Fan (1996) Type 7 definition, the default in R (`quantile(type = 7)`), NumPy (`method = "linear"`), DuckDB (`quantile_cont`) and D3 (`d3.quantile`, used in the browser). The N baseline values are sorted ascending, X(1) <= X(2) <= … <= X(N). For a probability p the estimator sets h = (N - 1)p + 1, takes the integer part j = floor(h) and the fractional part gamma = h - j, and interpolates linearly between the j-th and (j+1)-th order statistics. The estimator is continuous in p, uses only order statistics, and makes no assumption about the shape of the distribution. With N = 30 the four bounds used by the figure fall at h = 3.9 (P10, between the 3rd and 4th smallest years), h = 10.67 (P33), h = 20.33 (P67) and h = 27.1 (P90). Bounds are rounded to whole millimetres for display only; classification uses the unrounded values.

Classification. Each season is assigned one of five regime labels by comparison with the bounds: Much drier (R <= P10), Drier (P10 < R <= P33), Near normal (P33 < R < P67), Wetter (P67 <= R < P90) and Much wetter (R >= P90). The three-way tercile collapse used elsewhere in the platform (Drier / Near normal / Wetter) is the same rule with the decile bounds dropped. The anomaly view reports R - mean(1991–2020) in millimetres and as a percentage of the normal; no standardization by the standard deviation is applied to rainfall in this figure.

SPEI-3 option. When the user selects SPEI-3 the bars show the index value directly. SPEI is already a standardized quantity, produced upstream by fitting a log-logistic distribution to

the three-month climatic water balance (precipitation minus reference evapotranspiration) and mapping it to a standard normal variate, so no further tercile computation is applied; the conventional drought categories of the index are used instead.

8.1.2.1 Methodological Rationale

Why empirical quantiles rather than Gaussian bounds. The tercile convention used by many seasonal forecasting centres, mu plus or minus 0.43 sigma, is exact only when seasonal rainfall is normally distributed, because Phi(-0.43) = 0.333. East African seasonal rainfall is not normally distributed. Rainfall is bounded below by zero and generated by a small number of convective events, so the distribution has a long right tail. Computed from the CHIRPS county series over 1991–2020, the sample skewness coefficient of OND totals has a median of 1.23 across Kenya's 47 counties (interquartile range 0.95 to 1.65), exceeds 1.0 in 33 counties and reaches 3.0 in Isiolo and Wajir; no county has negative OND skew. MAM is less skewed (median 0.82) but still positive in 45 of 48 units. Coefficients of variation reach 0.90 for OND in the northeastern counties.

Consequences of imposing symmetry. For a right-skewed variable the mean exceeds the median and the standard deviation is inflated by the tail, so both symmetric bounds are displaced upward relative to the true tercile cut points. The lower bound mu - 0.43 sigma sits above the empirical P33 (median displacement +4 percent of P33 for OND across counties, up to +31 percent in the most skewed counties), so more than one third of baseline years fall below it: the Gaussian rule labels a median 36.7 percent of 1991–2020 OND seasons as dry where by construction P33 labels exactly one third. The error is in the direction of over-diagnosing drought and is largest precisely in the ASAL counties where drought response is most costly. The upper bound mu + 0.43 sigma is displaced further still (median +16 percent above the empirical P67 for OND, +52 percent in Wajir), so only about 23 percent of baseline years exceed it instead of one third: the symmetric rule under-counts wet seasons and overstates the rarity of the convective deluges that drive flood and Rift Valley fever risk. At the extremes the Gaussian rule becomes physically meaningless: mu - 1.28 sigma, the parametric P10, is negative whenever the coefficient of variation exceeds 0.78, which occurs in the northeastern counties, so no observed season could ever be classified as "much drier".

Illustration with a lognormal model. If seasonal totals follow a lognormal law with coefficient of variation 0.5, the true tercile bounds are 0.73 mu and 1.10 mu and the true P10 is 0.49 mu, whereas the Gaussian rule gives 0.79 mu, 1.22 mu and 0.36 mu. The mismatch is a property of skewness, not of sampling noise, and does not shrink as the record lengthens. The empirical Type 7 estimator avoids it by never leaving the observed order statistics, at the cost of larger sampling variance for the decile bounds (see limitations). Because the same estimator is applied in the browser, in the DuckDB queries and in the Python build scripts, tercile labels are reproducible across the platform.

Governing Mathematical Formulations:

Hyndman and Fan Type 7 sample quantile. X(1) <= … <= X(N) are the sorted 1991–2020 seasonal totals for one county and season (N = 30); the estimator interpolates linearly between adjacent order statistics.

$$\hat{Q}(p) = (1-\gamma)\, X_{(j)} + \gamma\, X_{(j+1)}, \qquad h = (N-1)\,p + 1, \quad j = \lfloor h \rfloor, \quad \gamma = h - j$$

Empirical tercile bounds over the 1991–2020 normal. Drier than normal is R <= T33, wetter than normal is R >= T67; by construction each tercile holds one third of the baseline years.

$$T_{33} = \hat{Q}_{1991\text{–}2020}\left(\tfrac{1}{3}\right), \qquad T_{67} = \hat{Q}_{1991\text{–}2020}\left(\tfrac{2}{3}\right)$$

Extreme decile bounds. Much drier is R <= D10 and much wetter is R >= D90; with N = 30 these bounds rest on the 3rd/4th and 27th/28th order statistics.

$$D_{10} = \hat{Q}_{1991\text{–}2020}(0.10), \qquad D_{90} = \hat{Q}_{1991\text{–}2020}(0.90)$$

Five-class regime assignment. R is the county-mean seasonal total; the three-way tercile labels used elsewhere drop the decile splits.

$$\text{Regime}(R) = \begin{cases} \text{Much drier} & R \le D_{10} \\ \text{Drier} & D_{10} < R \le T_{33} \\ \text{Near normal} & T_{33} < R < T_{67} \\ \text{Wetter} & T_{67} \le R < D_{90} \\ \text{Much wetter} & R \ge D_{90} \end{cases}$$

Sample skewness coefficient (Fisher-Pearson). Positive values indicate a long right tail; OND county series have median gamma1 = 1.23 over 1991–2020, which is why symmetric Gaussian bounds are rejected.

$$\gamma_1 = \frac{\frac{1}{N}\sum_{i=1}^{N}\left(R_i - \bar{R}\right)^3}{\left[\frac{1}{N}\sum_{i=1}^{N}\left(R_i - \bar{R}\right)^2\right]^{3/2}}$$

Gaussian tercile bounds (shown for contrast, not used). Both symmetric bounds are displaced upward under positive skew, over-diagnosing dry seasons and under-counting wet ones.

$\mu \pm 0.43\,\sigma$ since $\Phi(-0.43) \approx 0.333$; $\mu - 0.43\sigma > T_{33}$ and $\mu + 0.43\sigma > T_{67}$ whenever $\gamma_1 > 0$

Anomaly view. Departures are expressed in millimetres and percent of the normal; rainfall is never divided by its standard deviation in this figure.

$$A_y = R_y - \bar{R}_{1991\text{–}2020}, \qquad A_y^{\%} = 100\,\frac{R_y - \bar{R}_{1991\text{–}2020}}{\bar{R}_{1991\text{–}2020}}$$

8.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	0.05 deg (~5.5 km, CHIRPS v3 native) aggregated to county mean	1981 to present (rolling 3-month seasons labelled by end year)

Served Parquet Files in Platform: `chirps_county.parquet`, `chirps_county_monthly.parquet`

8.1.4 Methodological Assumptions & Analytical Limitations

- CHIRPS blends thermal infrared cold cloud duration estimates with station observations. Station density in northern and northeastern Kenya is sparse, so county means for the arid rangelands rely more heavily on the satellite component and on the CHPclim climatology than county means in the highlands; absolute totals carry larger uncertainty there than the rank-based tercile labels do.
- The 0.05 degree (about 5.5 km) grid is averaged to a single county value. Strong orographic gradients inside large counties (for example the Marsabit and Nyambene massifs against the surrounding lowlands) are collapsed into one number; the within-county pixel standard deviation is stored but not used in classification.
- Seasons are fixed calendar-month windows (MAM, OND and the other ten rolling windows). Late onset or early cessation that shifts rain across a window boundary is not captured, and the seasonal total says nothing about the dry-spell structure within the season.
- The baseline holds only 30 values per county and season. Tercile bounds estimated from 30 observations have appreciable sampling error, and the decile bounds rest on only three observations in each tail; small differences between a season and a bound should not be over-interpreted.
- The 1991–2020 normal is treated as stationary. The observed drying of MAM since the 1980s alongside a wetting tendency in OND means that the baseline mixes regimes; tercile frequencies computed over the full 1981 to present record will not be exactly one third.
- Seasons are labelled by the calendar year in which the window ends (DJF 2020 spans December 2019 to February 2020). Windows the record cannot cover in full are absent.
- SPEI-3 is standardized upstream with a fitted log-logistic distribution rather than with the empirical quantiles used for rainfall, so the two variables are not category-for-category comparable.
- County aggregations rely on official polygon boundaries from the UN OCHA Common Operational Datasets (COD-AB); historical boundary changes are not back-cast.

8.1.5 Controlled Vocabulary Keywords

`CHIRPS v3` • `WMO Normal` • `Empirical Quantiles` • `Hyndman-Fan Type 7` • `Positive Skewness` • `Terciles` • `SPEI-3` • `MAM` • `OND` • `COD-AB`

8.1.6 References

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8.2 Figure 3.1B: Consecutive Seasonal Sequences & Multi-Season Drought Persistence

Section in Platform: Section 3 | **Target Identifier:** `fig_2_1b` | **Output Type:** Chart

> **Policy / Scientific Question:** *Did this county's two rainy seasons fail back to back, how often has that happened before, how long did unbroken dry runs last, and what happened to maize yields when they did?*

8.2.1 Scientific Purpose & Context

Drought impacts in Kenya compound across seasons rather than arriving in one. A single below-normal season stresses pasture and delays planting, but households, herds and shallow aquifers usually carry enough buffer to reach the next rains. Two consecutive failed seasons

remove that buffer: dry-season grazing reserves are exhausted before the next rains, groundwater and pan recharge is skipped, breeding females are sold or die so the herd loses its reproductive core, seed stocks are eaten, and food prices rise while livestock prices collapse. The 2020 to 2023 drought, in which five successive seasons from OND 2020 to OND 2022 fell below normal across the ASAL counties, left more than four million people in Integrated Food Security Phase Classification (IPC) Phase 3 or worse by early 2023 according to the Kenya Food Security Steering Group short rains assessment. Operationally this is why NDMA drought phases and humanitarian early action protocols key on the sequence of seasons rather than on the latest one alone.

Figure 3.1B makes the sequence explicit. It pairs each production year Y with the two seasons that feed it, the Short Rains of the preceding calendar year (OND, Y−1) and the Long Rains of the same year (MAM, Y), and classifies the pair as Dry/Dry, Dry/Normal, Dry/Wet, Normal/Dry and so on. The OND(Y−1) → MAM(Y) pairing is used because those two seasons bracket the long dry season and jointly determine the pasture, water and planting conditions that the Y harvest and the Y livestock year depend on. A domino strip shows the pair state per production year, bars show the unbroken run length of consecutive dry seasons at the close of each year, and an outcome overlay shows county maize yield from HarvestStat and KNBS production so that the reader can see what happened to production in double-failure years relative to all years. Summary cards report the number and share of double failures in the record, the longest dry run with its total rainfall deficit, and the mean yield shortfall in double-failure years.

The analysis is descriptive and historical. It does not forecast the next season; its purpose is to give planners the empirical base rate of consecutive failure for their county, so that a dry OND can be read against how often a dry MAM has followed, and so that run-length thresholds in contingency plans are grounded in the county's own record.

8.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal, applied separately to OND and to MAM (full-record baseline available as a toggle)

Inputs. For the selected county the module queries the county-mean seasonal total for every OND and MAM season from 1981 onward. The baseline toggle selects either the 1991–2020 WMO standard normal (default) or the full record; the formulation below is written for the default.

Independent seasonal standardization. The sample mean and sample standard deviation ($N - 1$ denominator) are computed separately for OND and for MAM from the baseline years, giving two distinct parameter pairs (mu_OND, sigma_OND) and (mu_MAM, sigma_MAM). Each season's total is converted to a z-score using the parameters of its own season only. The two seasons are never pooled: OND and MAM differ in mean by a factor that ranges from well under one in the western highlands to well over two in the northeast, and their standard deviations differ correspondingly, so a pooled z would encode the seasonal cycle rather than the anomaly (see rationale).

State classification. A season is Dry when z < -0.5, Wet when z $> +0.5$ and Normal otherwise. The boundary values z $= \pm 0.5$ exactly fall in the Normal class. Seasons with no data are flagged missing and excluded from pairs. The anomaly in millimetres and in percent of the seasonal mean is carried alongside the z-score for display.

Chronological run length. All seasons are ordered MAM(1981), OND(1981), MAM(1982), OND(1982), and so on. Walking this sequence, a counter increments on every Dry season and resets to zero on any Normal, Wet or missing season; the counter value stored on each Dry season is its run length, the number of consecutive Dry seasons ending with it. The longest run in the record, the seasons it spans and its cumulative rainfall deficit relative to the sum of the seasonal normals are reported.

Production-year pairing. For each production year Y from the second year of the selected range, the pair is (State(OND, Y−1), State(MAM, Y)). A double failure is the Dry/Dry pair. The run length attached to year Y is the run at MAM(Y) if MAM(Y) is Dry, otherwise the run at OND(Y−1) if that season is Dry, otherwise zero; a run that was alive at OND(Y−1) but broken by a non-Dry MAM(Y) is flagged as broken. The ENSO phase and standardized Relative Oceanic Niño Index (RONI) for OND(Y−1), the IOD phase for OND(Y−1), and the RONI and Western-V phase for MAM(Y) are attached for the ocean-driver overlay.

Outcome overlay. County maize yield (t/ha, HarvestStat annual season) and production (HarvestStat and KNBS NAPR) are joined on harvest year Y. The summary compares the mean yield in double-failure years with the mean over all years with yield data and reports the percentage shortfall.

8.2.2.1 Methodological Rationale

Why the seasons must be standardized independently. Suppose the two seasons were pooled and a single mean and standard deviation computed from all 60 baseline season totals. Writing mu_O, mu_M for the seasonal means and sigma_O, sigma_M for the seasonal standard deviations, the pooled mean is (mu_O + mu_M)/2 and the pooled variance is the average of the two within-season variances plus a between-season term (mu_O − mu_M)^2/4. A season that is exactly normal for its own climatology, R = mu_s, would receive a pooled z-score of ±(mu_O − mu_M)/(2 sigma_pool), which is non-zero whenever the seasons differ in mean. In a county where OND averages 200 mm and MAM 450 mm, every ordinary OND would be classified Dry and every ordinary MAM Wet, and the "sequence" would simply reproduce the annual cycle. The between-season term also inflates sigma_pool, so genuine anomalies within either season would be compressed toward zero and under-detected. Standardizing each season against its own 1991–2020 moments removes both effects: a z of −0.5 means the same thing, half a within-season standard deviation below that season's own normal, whether it occurs in OND or MAM, which is the comparability a sequence analysis requires.

Why z-scores rather than empirical terciles here. Elsewhere in the platform rainfall regimes are defined by empirical quantiles because seasonal rainfall is positively skewed. This module is the one place where rainfall is expressed as a z-score, and it is done deliberately: the sequence and run-length logic needs a single anomaly scale that is comparable across two seasons with different distributions, and a symmetric threshold of half a standard deviation

is the same convention used for the ocean driver indices (RONI, DMI) that are overlaid on the strip. The consequence of the skew is accepted and reported rather than hidden. Under positive skew the share of seasons below z = −0.5 is somewhat higher than the 31 percent a Gaussian would give, and it varies by county with the degree of skew, so the module reports double-failure and run-length frequencies as empirical counts (X of N production years) rather than as nominal probabilities, and it never converts the z-scores to tercile labels.

Why the production-year pairing. Crop and livestock outcomes in year Y are driven by OND(Y−1) and MAM(Y) jointly: OND(Y−1) sets the dry-season pasture and water reserve and the residual soil moisture that MAM(Y) planting starts from, and MAM(Y) is the main cropping season in most counties. Pairing OND(Y) with MAM(Y) would split the dry season across two production years and would align the Y harvest with rains that fall after it.

Governing Mathematical Formulations:

Independent seasonal moments over the baseline. Each season has its own mean and sample standard deviation; the two seasons are never pooled.

$$\mu_s = \frac{1}{N_s}\sum_{y\in B} R_{s,y}, \qquad \sigma_s = \sqrt{\frac{1}{N_s-1}\sum_{y\in B}\left(R_{s,y}-\mu_s\right)^2}, \qquad s \in \{\mathrm{OND}, \mathrm{MAM}\},\ B = 1991\text{–}2020$$

Independent seasonal z-scores. A z of −0.5 is half a within-season standard deviation below that season's own 1991–2020 normal, in either season.

$$z_{\mathrm{OND},y} = \frac{R_{\mathrm{OND},y}-\mu_{\mathrm{OND}}}{\sigma_{\mathrm{OND}}}, \qquad z_{\mathrm{MAM},y} = \frac{R_{\mathrm{MAM},y}-\mu_{\mathrm{MAM}}}{\sigma_{\mathrm{MAM}}}$$

Pooled standardization bias (why pooling is rejected). A perfectly normal season would receive a non-zero pooled z-score, and the between-season term compresses true anomalies.

$$\sigma^2_{\mathrm{pool}} = \frac{\sigma^2_{\mathrm{OND}}+\sigma^2_{\mathrm{MAM}}}{2} + \frac{\left(\mu_{\mathrm{OND}}-\mu_{\mathrm{MAM}}\right)^2}{4}, \qquad z_{\mathrm{pool}}\big|_{R=\mu_s} = \pm\frac{\mu_{\mathrm{OND}}-\mu_{\mathrm{MAM}}}{2\,\sigma_{\mathrm{pool}}} \neq 0$$

Seasonal state assignment. Strict inequalities; the boundary values are Normal. Missing seasons are excluded from pairs and reset the run counter.

$$\mathrm{State}(s,y) = \begin{cases} \mathrm{Dry} & z_{s,y} < -0.5 \\ \mathrm{Wet} & z_{s,y} > +0.5 \\ \mathrm{Normal} & -0.5 \le z_{s,y} \le +0.5 \end{cases}$$

Chronological dry-run length. The run length is the number of consecutive Dry seasons ending at t; any non-Dry season resets it to zero.

$$L_t = (L_{t-1} + 1)\ \mathbb{I}(z_t < -0.5)\,, \qquad L_0 = 0, \qquad t \text{ indexes } \ldots, \mathrm{MAM}_y, \mathrm{OND}_y, \mathrm{MAM}_{y+1}, \ldots$$

Production-year pair and double failure. Production year Y is fed by the preceding Short Rains and the same-year Long Rains; the nine possible pair states collapse to the four highlighted sequences Dry/Dry, Dry/Wet, Wet/Dry and Wet/Wet plus the Normal-containing pairs.

$$\mathcal{P}_Y = (\mathrm{State}(\mathrm{OND}, Y-1),\ \mathrm{State}(\mathrm{MAM}, Y))\,, \qquad \mathrm{DF}_Y = \mathbb{I}(\mathcal{P}_Y = (\mathrm{Dry}, \mathrm{Dry}))$$

Empirical double-failure frequency and yield shortfall. Frequencies are reported as counts of production years, and the yield comparison uses HarvestStat county maize yield in double-failure years against all years with data.

$$f_{\mathrm{DF}} = \frac{\sum_Y \mathrm{DF}_Y}{\#\{Y : \mathcal{P}_Y \text{ complete}\}}, \qquad \Delta Y^{\%}_{\mathrm{DF}} = 100\,\frac{\overline{Y}_{\text{maize}|\mathrm{DF}} - \overline{Y}_{\text{maize}}}{\overline{Y}_{\text{maize}}}$$

8.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	0.05 deg (~5.5 km, CHIRPS v3 native) aggregated to county mean	1981 to present; pairs from production year 1982
`harveststat-crop-production`	HarvestStat Africa - Harmonized Subnatio	County (ADM1)	1965 to 2024 (seasonal gap 2002 to 2014)
`knbs-napr`	KNBS National Agricultural Production Re	County (ADM1)	2019 to 2024

Served Parquet Files in Platform: `chirps_county_monthly.parquet`, `chirps_county.parquet`

8.2.4 Methodological Assumptions & Analytical Limitations

- Seasons are fixed calendar windows. A Short Rains season that starts late and runs into January, or a Long Rains season that fails in April but recovers in June, is scored on the window total only, so sequence states can differ from the agronomic experience of the season.

- The z-score threshold of ±0.5 is applied to a positively skewed variable. The share of seasons classed Dry is therefore not exactly the Gaussian 31 percent and differs between counties and between OND and MAM; comparisons across counties should use the empirical counts shown, not the nominal threshold.
- Pasture and water respond to rainfall with a lag, and herd condition integrates over several seasons. A Dry/Dry pair is a rainfall statement; the livelihood consequence depends on the state the system entered the pair in, which the figure shows only through the preceding run length.
- Consecutive seasons are treated as separate observations for classification, but OND and the following MAM are not statistically independent (they share ENSO and IOD persistence and soil moisture memory). The module does not model this dependence; the double-failure frequency it reports is the observed joint frequency, not a product of marginals.
- The 1991–2020 baseline gives 30 values per season; the default z-scores for the 1980s and for years after 2020 are measured against a baseline they do not belong to. The full-record toggle trades this for a baseline that drifts as years are added.
- The outcome overlay joins maize yield on harvest year. HarvestStat county series are incomplete for 2002 to 2014 and KNBS NAPR starts in 2019, so the yield-shortfall statistic rests on a subset of double-failure years and on maize only; pastoral outcomes (mortality, livestock prices) are not in the overlay.
- Yields are not detrended in this module, so the shortfall in early double-failure years is measured against a long-run mean that includes later, higher-yielding years.
- County aggregations rely on official polygon boundaries from the UN OCHA Common Operational Datasets (COD-AB).

8.2.5 Controlled Vocabulary Keywords

`Seasonal z-score` • `Consecutive Sequence` • `Double Failure` • `Dry Run` • `WMO Normal` • `OND` • `MAM` • `HarvestStat` • `KNBS NAPR` • `IPC` • `RONI`

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8.3 Figure 3.2: Empirical Contingency Probability Matrix: Rainfall Terciles Conditioned on Ocean Driver Phase

Section in Platform: Section 3 | **Target Identifier:** `fig_2_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *For this county, how often has each rainy season been wetter, near normal or drier than the 1991–2020 normal when a given ocean driver phase was in place, and how far does that conditional frequency move away from the one-in-three climatological base rate?*

8.3.1 Scientific Purpose & Context

Seasonal forecasts for Kenya are expressed as shifts in the odds of the three rainfall terciles, but county planners rarely see the historical evidence behind those shifts for their own county. Figure 3.2 supplies it. For every year in the record it places the county's seasonal rainfall in a tercile of its own 1991–2020 distribution and the season's ocean driver in a phase class, then cross-tabulates the two. The result is the empirical conditional frequency of a wetter, near-normal or drier season under each driver phase, shown as one stacked bar per phase against a dashed reference line at the one-in-three climatological base rate.

The two rainy seasons are shown side by side because they respond to different drivers. For the Short Rains (OND) the user can condition on the Pacific El Niño–Southern Oscillation, measured by the Relative Oceanic Niño Index (RONI), or on the Indian Ocean Dipole, measured by the Dipole Mode Index (DMI). For the Long Rains (MAM) the default driver is the Western-V sea surface temperature gradient in the western Pacific, with RONI as an alternative. Phases can be shown as three classes (positive, neutral, negative) or five (strong and moderate on each side of neutral), and as shares or as raw counts of years.

The figure turns a teleconnection into a plain statement a contingency planner can use, such as "in this county, seven of the last thirteen El Niño Short Rains seasons were in the wettest third". It also shows where the signal is weak. Across Kenya's 47 counties, OND seasons that coincided with El Niño fell in the wetter tercile 57 percent of the time when pooled, against 33 percent expected by chance, and at least half the time in 41 counties. The same conditioning gives no consistent shift for MAM. The Explorer surfaces both results so that county plans weight ENSO strongly for OND and treat it as uninformative for MAM.

8.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: Tercile bounds and driver standardization from the 1991–2020 WMO standard normal; matrix counts over the user-selected range (default 1981 to 2025)

Rainfall terciles. Terciles are assigned offline by the outlook builder. For each county and season, the 1991–2020 seasonal totals (county-mean CHIRPS v3 PTOT) form the baseline, and a county is skipped if fewer than 20 baseline years are available. The tercile bounds are the empirical 33.3rd and 66.7th percentiles computed with linear interpolation between order statistics (NumPy's default method, equivalent to the Hyndman and Fan Type 7 estimator used elsewhere in the platform). Every year from 1981 to 2025 is then classed Drier if its total is at or below the lower bound, Wetter if it is strictly above the upper bound, and Near normal otherwise. Only the tercile bounds are used here; the decile bounds of Figure 3.1 are not.

Driver phase. For each season the selected driver is reduced to one value per year and standardized against its own 1991–2020 mean and sample standard deviation. For ENSO the value is the CPC RONI averaged over the concurrent season. For the IOD and the Western-V index the value is the mean of the available monthly values over October to December or March to May respectively (DMI from HadISST, falling back month by month to ERSST). The standardized value z is binned with symmetric thresholds: in three-class mode, positive phase if z $+0.5$, negative phase if z -0.5, and neutral otherwise; in five-class mode, strong positive if z $+1.5$, moderate positive if $+0.5$ z $< +1.5$, neutral if $-0.5 <$ z $< +0.5$, moderate negative if $-1.5 <$ z -0.5, and strong negative if z -1.5. Phase labels are El Niño and La Niña for RONI, positive and negative IOD for DMI, and high and low Western-V.

Contingency matrix. For the selected county and the years in the chosen range that have both a tercile and a driver value, the matrix counts years n_ij in driver class i and tercile j. Each row is normalised by its total to give the conditional frequency of each tercile within that phase, and the bars are drawn as these row shares (or as raw counts). Rows with fewer than five years are flagged in amber as small samples. No significance test or confidence interval is applied; the only reference is the 33.3 percent climatological base rate.

Insight summary. A text panel pools all positive-phase classes, reports the most frequent tercile and its share, and gives the mean and standard deviation of the seasonal rainfall anomaly in percent of the 1991–2020 mean for those years.

8.3.2.1 Methodological Rationale

Why a contingency table rather than a correlation. East African rainfall responds to ocean forcing asymmetrically. In OND, El Niño years shift the distribution strongly toward wet outcomes, while La Niña years produce a weaker and less consistent dry shift. A Pearson correlation between a driver index and seasonal rainfall compresses this into one number that is dominated by a few extreme years, such as 1997 and 2019. A contingency table keeps the asymmetry visible: each phase gets its own conditional distribution, and each cell is a count of real seasons that a planner can verify against memory.

Why terciles of each county's own distribution. Seasonal rainfall in Kenya is strongly positively skewed and its level varies by an order of magnitude between counties, so a common millimetre threshold would be meaningless and a Gaussian tercile rule would misclassify dry seasons. Empirical terciles of each county's own 1991–2020 record make every county's categories equiprobable by construction, so a conditional frequency above one third is directly interpretable as a shift in the odds.

Why standardized driver thresholds. Standardizing each driver against its own 1991–2020 distribution puts RONI, DMI and Western-V on a common scale, so the same ±0.5 and ±1.5 standard deviation thresholds give comparable phase frequencies across drivers and seasons. The threshold is applied to the standardized value rather than to the raw index, so it differs from the conventional raw thresholds (0.5 °C for RONI, 0.4 °C for DMI) used for phase labels elsewhere in the Explorer. Because the 1991–2020 standard deviation of OND RONI is about 1.12, the OND El Niño class here requires roughly 0.56 °C, close to the conventional rule; for MAM the standard deviation is about 0.64, so a MAM El Niño class begins near 0.32 °C.

What the data show. Pooled over all 47 counties and 1981 to 2025, OND seasons in the El Niño class (13 years) were Wetter 57 percent, Near normal 25 percent and Drier 18 percent of the time; at the county level the share Wetter ranges from 31 to 69 percent, with a median of 54 percent, and reaches at least one half in 41 counties. OND La Niña seasons (18 years) were Drier 39 percent of the time, and only 10 counties reach a one-half Drier share, which confirms the asymmetry. For MAM, El Niño seasons were Wetter only 30 percent of the time and La Niña seasons Drier 37 percent, with almost no county above one half in either case: the Long Rains carry no consistent ENSO signal, in agreement with the hindcast skill reported in Table 2.3.

Governing Mathematical Formulations:

Empirical tercile bounds and class assignment. Bounds use linear interpolation between order statistics of the county's 1991–2020 seasonal totals; counties with fewer than 20 baseline years are excluded.

$$t_1 = \hat{Q}_{1991\text{–}2020}\left(\tfrac{1}{3}\right), \quad t_2 = \hat{Q}_{1991\text{–}2020}\left(\tfrac{2}{3}\right), \qquad T_y = \begin{cases} \text{Drier} & R_y \le t_1 \\ \text{Wetter} & R_y > t_2 \\ \text{Near normal} & \text{otherwise} \end{cases}$$

Standardized driver value. s is the sample standard deviation (N − 1); M_y is the set of available months in the season, so a season with missing months is averaged over the months present.

$$z_y = \frac{x_y - \bar{x}_{1991\text{–}2020}}{s_{1991\text{–}2020}}, \qquad x_y = \begin{cases} \overline{\text{RONI}}_{\text{season},y} & \text{ENSO} \\ \frac{1}{|M_y|}\sum_{m\in M_y} \text{DMI}_{m,y} & \text{IOD (OND)} \\ \frac{1}{|M_y|}\sum_{m\in M_y} \text{WV}_{m,y} & \text{Western-V (MAM)} \end{cases}$$

Three-class and five-class phase assignment. Thresholds are in standard deviation units of the 1991–2020 driver distribution, not in raw index units.

$$\Phi_3(z) = \begin{cases} + & z \geq 0.5 \\ - & z \leq -0.5 \\ 0 & \text{otherwise} \end{cases}, \qquad \Phi_5(z) = \begin{cases} \text{strong+} & z \geq 1.5 \\ \text{moderate+} & 0.5 \leq z < 1.5 \\ 0 & -0.5 < z < 0.5 \\ \text{moderate}- & -1.5 < z \leq -0.5 \\ \text{strong}- & z \leq -1.5 \end{cases}$$

Row-normalised conditional tercile frequency. Each bar is one driver class; its segments sum to one. The dashed reference is the climatological base rate of 1/3.

$$\hat{P}(T = j \mid \Phi = i) = \frac{n_{ij}}{n_{i\cdot}}, \qquad n_{ij} = \#\{y \in [y_0, y_1] : \Phi(z_y) = i,\ T_y = j\}, \qquad \text{flag if } n_{i\cdot} < 5$$

Shift from the climatological base rate (interpretation). Not displayed numerically; read as the distance of each bar segment from the dashed line. With n_i. years the sampling standard error of a share near 1/3 is about sqrt(2/(9 n_i.)), roughly 0.13 for 13 years.

$$\Delta_{ij} = \hat{P}(T = j \mid \Phi = i) - \tfrac{1}{3}$$

Seasonal rainfall anomaly (insight panel). Reported as mean and standard deviation over the pooled positive-phase years.

$$A_y^{\%} = 100\,\frac{R_y - \bar{R}_{1991\text{–}2020}}{\bar{R}_{1991\text{–}2020}}$$

8.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	0.05 deg (~5.5 km, CHIRPS v3 native) aggregated to county mean	1981 to 2025
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	Equatorial Pacific, tropical Indian Ocean and western Pacific basin indices	1950 to present; 1981 to 2025 used

Served Parquet Files in Platform: `enso_outlook_base.parquet`, `driver_indices.parquet`, `enso_drivers_seasonal.parquet`

8.3.4 Methodological Assumptions & Analytical Limitations

- Sample sizes are small. Over 1981 to 2025 there are only 12 to 19 seasons in each three-class phase per season, and as few as two to four in the strong classes of five-class mode. A single season moves a share by six to eight percentage points, and the sampling standard error of a share near one third is about 13 percentage points for 13 years. No significance test or confidence interval is shown, so modest departures from the dashed line should not be read as a signal.
- Counts are pooled over the whole selected range, including the 15 years outside the 1991–2020 baseline. Trends in rainfall or in driver behaviour across 1981 to 2025, including the drying of the Long Rains and the warming of the Indian Ocean, are not removed.
- The driver phase uses the concurrent season. This describes the historical association between rainfall and the ocean state during the season, not predictability from information available before the season starts; forecast skill is assessed separately in Table 2.3.
- Phase thresholds are applied to standardized values, so the effective raw threshold depends on the 1991–2020 variability of each index and season (about 0.56 °C for OND RONI and 0.32 °C for MAM RONI). Phase classes here therefore differ slightly from the raw-threshold phase labels used in other figures.
- Seasonal driver means for the IOD and Western-V are taken over whichever months are available, with no completeness check, so a season with a missing month is represented by fewer than three months.
- Tercile boundaries treat a total exactly equal to the upper bound as Near normal, whereas Figure 3.1 classes such a total as Wetter. The difference affects at most a single boundary year per county.
- ENSO, the IOD and the Western-V gradient are not independent; positive IOD events frequently co-occur with El Niño. The figure conditions on one driver at a time and cannot separate their contributions.
- County rainfall is a county-mean CHIRPS v3 value; uncertainty is larger in the sparsely gauged northern and eastern counties.
- County aggregations rely on official polygon boundaries from the UN OCHA Common Operational Datasets (COD-AB).

8.3.5 Controlled Vocabulary Keywords

`Contingency Matrix` • `Conditional Probability` • `Joint Ocean Phase` • `Terciles` • `Empirical Quantiles` • `RONI` • `DMI` • `Western-V / WV` • `WMO Normal` • `CHIRPS v3` • `OND` • `MAM`

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8.4 Figure 3.3: Oceanic Driver Teleconnection Relationships (OLS Regression)

Section in Platform: Section 3 | **Target Identifier:** `fig_2_3` | **Output Type:** Chart

> **Policy / Scientific Question:** *What is the continuous sensitivity of county precipitation to Pacific and Indian Ocean warming?*

8.4.1 Scientific Purpose & Context

Quantify and communicate Oceanic Driver Teleconnection Relationships (OLS Regression) across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Continuous scatter correlation between standardized oceanic driver indices and seasonal rainfall anomalies with ordinary least squares (OLS) regression line and confidence bands.

8.4.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$r = \frac{\sum(z_{\text{driver}} - \bar{z})(R - \bar{R})}{\sqrt{\sum(z_{\text{driver}} - \bar{z})^2 \sum(R - \bar{R})^2}}$$

8.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county_monthly.parquet`, `driver_indices.parquet`

8.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.4.5 Controlled Vocabulary Keywords

`Pearson r • OLS Regression • Teleconnection Gradient`

8.5 Figure 3.4: Multi-Decadal Gridded Field History & Spatial Patterns

Section in Platform: Section 3 | **Target Identifier:** `fig_2_4` | **Output Type:** Map

> **Policy / Scientific Question:** *Where within the county did historical rainfall, vegetation, or drought anomalies concentrate?*

8.5.1 Scientific Purpose & Context

Quantify and communicate Multi-Decadal Gridded Field History & Spatial Patterns across Kenyan counties to inform anticipatory action and climate adaptation planning.

8.5.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Multi-decadal gridded rainfall raster analysis evaluating spatial gradients, elevation effects, and rainshadow patterns across subcounty zones.

8.5.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

8.5.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`africa-precipitation-monthly-seasonal`	Monthly and Seasonal Precipitation from	africa — 0.05 deg (~5.5 km, CHIRPS native)	1981-01 to present (open-ended)

Served Parquet Files in Platform: `chirps_county.parquet`

8.5.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

8.5.5 Controlled Vocabulary Keywords

`CHIRPS v3`

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9 Satellite Earth Observation, Hazards & Exposure

10 Satellite Earth Observation, Hazards & Exposure

This chapter covers Earth observation satellite telemetry for rapid flood inundation and rangeland vegetative health. It formalizes Sentinel-1 Synthetic Aperture Radar (SAR) backscatter thresholding from the Copernicus Global Flood Monitoring (GFM) service, hydrodynamic return period modeling (JRC), and MODIS 250m dekadal NDVI vegetation greenness anomalies.

10.1 Figure 3.5: Subcounty Flood Hazard & Infrastructure Exposure Breakdown

Section in Platform: Section 3 | **Target Identifier:** `fig_2_5` | **Output Type:** Map

> **Policy / Scientific Question:** *Which subcounties, populations, and public infrastructure assets face extreme flood inundation?*

10.1.1 Scientific Purpose & Context

Quantify and communicate Subcounty Flood Hazard & Infrastructure Exposure Breakdown across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Subcounty flood hazard profiling pairing Copernicus GFM Sentinel-1 SAR observed surface water extent with JRC flood return periods (1-in-10 to 1-in-100 year models).

10.1.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{Exposure Rate} = \frac{\text{Pop}_{\text{inundated}}}{\text{Pop}_{\text{total}}} \times 100\%$$

10.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-flood-gfm`	Kenya Observed Flood Extent (Copernicus	kenya — 0.001 deg (~111 m); aggregated from 20 m Sentinel-1 tiles (a coarse pixel is flooded if any 20 m sub-pixel flooded)	2018-01 to present (open-ended)
`eastafrica-flood-jrc`	JRC / GloFAS Return-Period River Flood H	world, kenya — 3 arc-seconds (~90 m)	Static baseline (return periods 10 to 500 years, CEMS-GLOFAS v2.1.2)
`kenya-facilities-hotosm`	Kenya Health and Education Facilities (H	kenya — Facility features (points, some building polygons)	2026-08
`kenya-roads-osm`	Kenya Classified Road Network (OpenStree	kenya	2026-08

Served Parquet Files in Platform: `exposure_gfm_seasonal.parquet`, `exposure_jrc_rp.parquet`

10.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.1.5 Controlled Vocabulary Keywords

`Copernicus GFM • SAR • JRC Flood RP`

10.2 Figure 4.3: Continuous MODIS Dekadal NDVI Pasture Biomass Dynamics

Section in Platform: Section 4 | **Target Identifier:** `fig_3_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *When did historical rangeland vegetation collapse into severe pasture deficits?*

10.2.1 Scientific Purpose & Context

Quantify and communicate Continuous MODIS Dekadal NDVI Pasture Biomass Dynamics across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Continuous MODIS 250m 16-day NDVI vegetation anomaly z-scores tracking pasture biomass, forage desiccation, and post-shock rangeland green-up.

10.2.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$z_{\mathrm{NDVI},t} = \frac{\mathrm{NDVI}_t - \mathrm{Median}_t(2000\text{–}2020)}{\sigma_t(2000\text{–}2020)}$$

10.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`eastafrica-ndvi-modis`	MODIS NDVI Seasonal Vegetation Condition	eastern-africa, kenya — ~250 m (0.0021 deg, MOD13Q1 native)	2000 to 2025
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `ndvi_county.parquet`, `driver_indices.parquet`

10.2.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.2.5 Controlled Vocabulary Keywords

`MODIS NDVI • ASAL`

10.3 Figure 4.5: Subcounty Sentinel-1 SAR Flood Inundation & Asset Exposure

Section in Platform: Section 4 | **Target Identifier:** `fig_3_3` | **Output Type:** Map

> **Policy / Scientific Question:** *What was the observed satellite footprint of extreme flood events and what assets were exposed?*

10.3.1 Scientific Purpose & Context

Quantify and communicate Subcounty Sentinel-1 SAR Flood Inundation & Asset Exposure across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Spatial intersection model evaluating physical asset exposure to observed flood inundation: population headcount, cropland area, road network kilometers, and health/education facilities.

10.3.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$E_{\text{asset}} = \int_{\Omega_{\text{inundated}}} D_{\text{asset}}(x, y)\, dxdy$$

10.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-flood-exposure-intersect`	Kenya Flood x Exposure, Per-Sub-County T	kenya — IEBC sub-counties (290 admin-2 units)	2018 to 2025
`kenya-flood-gfm`	Kenya Observed Flood Extent (Copernicus	kenya — 0.001 deg (~111 m); aggregated from 20 m Sentinel-1 tiles (a coarse pixel is flooded if any 20 m sub-pixel flooded)	2018-01 to present (open-ended)
`kenya-population-worldpop`	WorldPop Constrained Gridded Population	kenya — ~100 m (3 arc-seconds)	2020
`kenya-facilities-hotosm`	Kenya Health and Education Facilities (H	kenya — Facility features (points, some building polygons)	2026-08
`kenya-roads-osm`	Kenya Classified Road Network (OpenStree	kenya	2026-08

Served Parquet Files in Platform: `exposure_totals.parquet`, `exposure_vop.parquet`

10.3.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.3.5 Controlled Vocabulary Keywords

`KPLC` • `Copernicus GFM` • `HOT-OSM`

10.4 Figure 4.6: Humanitarian Emergency Chronology & Situation Reports

Section in Platform: Section 4 | **Target Identifier:** `fig_3_4` | **Output Type:** Chart

> **Policy / Scientific Question:** *What humanitarian crisis reports were filed by UN OCHA and government agencies across historical disasters?*

10.4.1 Scientific Purpose & Context

Quantify and communicate Humanitarian Emergency Chronology & Situation Reports across Kenyan counties to inform anticipatory action and climate adaptation planning.

10.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Humanitarian crisis impact chronology compiling UN OCHA ReliefWeb disaster dispatches and NDMA monthly drought early warning bulletins across alarm phases.

10.4.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

10.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`reliefweb`	reliefweb	County (ADM1)	Historical multi-decadal
`ndma-bulletin-index`	ndma-bulletin-index	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `reliefweb_county.parquet`, `ndma_bulletin_index.parquet`

10.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

10.4.5 Controlled Vocabulary Keywords

`ReliefWeb / GLIDE • NDMA • Drought Phase`

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11 Agricultural Production & Economic Valuation

12 Agricultural Production & Economic Valuation

This chapter formalizes the agricultural production anomaly functions, historical crop yield detrending, and the county economic capital valuation engine. It documents KNBS National Agriculture Production Report (NAPR) extractions, HarvestStat historical yields, and the subcounty spatial weight apportionment model under Beta review governance.

12.1 Figure 1.1: County Agricultural & Livestock Production Time Series (2019–2024)

Section in Platform: Section 1.1 | **Target Identifier:** `fig_1_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *How has crop and livestock production varied over recent statutory reporting years?*

12.1.1 Scientific Purpose & Context

Quantify and communicate County Agricultural & Livestock Production Time Series (2019–2024) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.1.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Official agricultural and pastoral asset inventories from KNBS National Agriculture Production Report (2024/2025). County share computed relative to national 47-county totals.

12.1.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$\text{Share}_c = \frac{Y_c}{\sum_{i=1}^{47} Y_i} \times 100\%$$

12.1.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024

Served Parquet Files in Platform: `knbs_napr_county_production.parquet`, `knbs_napr_livestock.parquet`, `knbs_napr_livestock_products.parquet`

12.1.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.1.5 Controlled Vocabulary Keywords

`KNBS NAPR`

12.2 Figure 1.2: Subnational Agricultural Capital Valuation & Subcounty Spatial Apportionment

Section in Platform: Section 1.2 | **Target Identifier:** `fig_1_2` | **Output Type:** Chart

> **Policy / Scientific Question:** *What agricultural and livestock capital is exposed to climate extremes in the county, how much of it is standing herd wealth as opposed to annual harvest and offtake value, and how is it distributed across the county's constituencies?*

12.2.1 Scientific Purpose & Context

Climate risk screening needs a value at risk. A drought that kills a fifth of a county's cattle and a drought that cuts its maize harvest by a fifth are different events: the first destroys capital that takes years of breeding to rebuild, the second destroys one year's income. Global gridded value-of-production surfaces, including the Atlas-wide MapSPAM crop and Gridded Livestock of the World (GLW4) livestock layers expressed in constant 2015 international dollars, conflate these two things. They value livestock by annual output (meat offtake and milk), so a pastoral county's wealth is represented by a small flow number while the herd itself, the asset that drought actually erodes, does not appear. In the earliest version of this figure that approach valued Marsabit's cattle at about 31 million US dollars and omitted camels entirely, which no pastoral planner would recognise.

Figure 1.2 therefore rebuilds the county portfolio from Kenya's statutory statistics and separates two quantities explicitly. Standing stock is the replacement value of the live herd: head counts from the 2019 Kenya Population and Housing Census and the Kenya National Bureau of Statistics (KNBS) National Agricultural Production Report (NAPR), valued at county sale prices with a capital-share adjustment for herd structure. Annual flow is the value of what the county produces in a year: crop harvests at farm-gate prices, livestock offtake and milk, and official livestock products (eggs, honey, hides, poultry meat). Both are reported in constant 2021 Kenyan shillings with a fixed 2021 conversion to US dollars, so that changes between years reflect physical quantities, not price inflation or exchange-rate movements. A portfolio chart shows the composition by commodity, and a constituency view apportions every county total across the 290 parliamentary constituencies (IEBC sub-counties) with mass-preserving spatial weights.

The output is a screening estimate of exposure, built for comparing counties and sub-counties and for sizing contingency planning. It is published under Beta review governance: livestock prices in the arid and semi-arid counties are annual medians from National Drought Management Authority (NDMA) county drought early warning bulletins used under research fair dealing while formal licensing is sought, so the asset tables are shown for expert review, are not released as open data and carry no bulk download.

12.2.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: Valuation base year 2021 (constant 2021 KES; USD at the 2021 annual average rate); crops 2019 to 2024, livestock and products 2021

Tier 1, quantities. Crop production by county and year comes from the KNBS NAPR 2023/24 and 2024/25 editions (deterministic PDF table extraction with additivity gates), supplemented by Agriculture and Food Authority (AFA) county returns for paddy rice and by Horticultural Crops Directorate county values for vegetables and fruits in 2021. Livestock head counts by county and species (cattle, camels, goats, sheep, donkeys, pigs, poultry) come from the 2019 Kenya Population and Housing Census (KPHC) Volume IV household livestock tables and the NAPR livestock population tables for 2021; the dairy share of the cattle herd is taken from the KPHC exotic-dairy count and, in highland counties only, scaled so that the national dairy herd matches the NAPR total. Livestock product quantities (milk, eggs, honey, hides, poultry meat) are the KNBS 2021 county figures.

Tier 2, prices. The base year is 2021 and all prices are 2021 Kenyan shillings (KES) at farm-gate or primary-market level. Crop prices follow a documented preference order: AFA Yearbook 2021 farm-gate prices first (coffee on a cherry-equivalent basis, tea at the Kenya Tea Development Agency green-leaf rate), then KNBS Economic Survey Table 7.4 average gross prices to farmers, then MapSPAM 2020 FAOSTAT-derived producer prices as the last resort. Live-animal sale prices in the 23 arid and semi-arid counties are 2019 to 2021 annual medians of the NDMA county drought early warning bulletins; elsewhere they are 2021 Kenya Agricultural Market Information System (KAMIS) medians and literature values for dairy cattle, donkeys, pigs and poultry. Milk is priced at county milk prices. Every parameter and price carries source identifiers resolvable in a sha256-verified source registry, and the build fails if any identifier is missing.

Tier 3, livestock parameters. Standing stock for cattle, goats and sheep is head count times a capital share of 0.75 times the sale price; the capital share reflects that a herd's immatures and breeding females are not all saleable at the quoted price of a mature animal. Camels are valued at the NDMA sale price times an empirical herd-structure ratio (1.39 in Marsabit, 0.99 in Isiolo, 1.17 nationally). Annual flow is head count times an annual offtake rate times sale price (cattle 0.12 in arid and semi-arid zones and 0.08 in the highlands; camels 0.04; goats 0.16; sheep 0.14) plus milk litres times milk price, where cattle milk is dairy head times 800 L per head per year plus non-dairy head times a zone coefficient (55 L in arid, 100 L in semi-arid and highland zones), and camel milk is 180 L per head per year. Zone coefficients derive from the Behnke and Muthami (2011) audit of Kenyan livestock contribution to GDP, cross-checked against the primary sources, with goat milk from Field (1985). Counties are assigned to arid, semi-arid or highland zones from the 2015 Ending Drought Emergencies county list.

Tier 4, currency. Flows and stocks are converted to US dollars at the KNBS Economic Survey 2021 annual average of 109.65 KES per USD. The rate is fixed to the base year so that the dollar series is a constant-price series; a current rate is available only as an explicit user override. The legacy constant 2015 international dollar layers from MapSPAM and GLW4 are retained for comparison with the Atlas-wide exposure product but do not enter the county totals.

Tier 5, subcounty apportionment. Each county total for a commodity is distributed across its IEBC parliamentary constituencies in proportion to a spatial prior specific to that commodity. For crops and horticulture the prior is MapSPAM 2020 v1r2 physical area for the matching crop (or crop group), summed from the 5 arc-minute grid cells whose centroids fall inside each constituency. For livestock and livestock products the prior is the KPHC 2019 household livestock count for the species (or species group) by constituency, obtained by crosswalking census sub-counties to IEBC constituencies with the GLW4 density surface as the reference layer. Where a prior sums to zero within a county (12 percent of crop rows, under 1 percent of livestock rows) the constituency land-area share is used instead and the row is flagged as a fallback. Shares are gated to sum to one within each county and commodity to within 1e-12, and the apportioned values are gated to re-aggregate exactly to the county totals.

12.2.2.1 Methodological Rationale

Why constant base-year prices. The purpose of the figure is to expose physical production and herd capital to climate hazard, so the valuation must not move when prices move. Kenyan farm-gate prices and the shilling both shifted sharply over the data window (the shilling went from 109.65 per US dollar in 2021 to 139.85 in 2023, and KNBS revised the 2022 maize price upward by 35 percent between editions). Valuing every year's quantities at the same 2021 prices and the same 2021 exchange rate means that a change in the series is a change in tonnes or head, which is what a drought or flood alters. The 2021 base year was chosen because it is the latest year for which KNBS county livestock populations, the Economic Survey price tables, the AFA yearbook and the NDMA bulletin price medians all overlap.

Why stock and flow are separated. Drought mortality and distress sales erode the herd, which is a capital stock; a poor harvest erodes one year's output, which is a flow. Reporting only flow, as the gridded international-dollar layers do, makes pastoral counties look poor and under-states what a multi-season drought puts at risk; reporting only stock makes crop counties look poor and ignores the income shock. The capital-share parameter corrects a second conflation: multiplying every head by the sale price of a mature animal overstates herd value because a herd is mostly females and immatures, so the 0.75 share, derived from empirical Kenyan herd demographic structures, brings the stock value to a replacement basis. Camels receive a species-specific ratio because their herd structure and price structure differ from those of cattle and small stock.

Why mass-preserving dasymetric apportionment. County totals are statutory; constituency totals are not published. The honest way to produce a sub-county figure is to keep the county total fixed and distribute it with the best available spatial evidence of where the commodity is, which is the pycnophylactic (mass-preserving) property of dasymetric mapping. The priors are chosen to be commodity-specific so that maize follows maize area and camels follow camel counts, not population or land area. MapSPAM and GLW4 enter here only as spatial shape, never as level, so their known biases in absolute value (which motivated the rebuild) do not propagate into the totals.

Why Beta review governance. The NDMA bulletin prices are copyright NDMA with no open licence; they are used as derived annual medians under research fair dealing while

permission is sought. Until that permission exists the assets cannot be redistributed, so the module carries a Beta Review Edition banner, provides no bulk download, and is excluded from the open Atlas data store and the Comprehensive Data Hub. The values are screening estimates for exposure comparison and contingency sizing; they are not calibrated for use as the basis of fiscal transfers, insurance payouts or compensation without further validation against county-level market and herd surveys.

Governing Mathematical Formulations:

County crop flow (gross value of production at constant 2021 farm-gate prices). Q is county production in tonnes from KNBS NAPR (plus AFA rice and HCD horticulture); the price for each crop is taken from the first available source in the preference order.

$$F_{c,y}^{\text{crop}} = \sum_{i=1}^{I} Q_{c,i,y}\, P_i^{2021}, \qquad P_i^{2021} \in \{\text{AFA farm-gate} \succ \text{KNBS ES Table 7.4} \succ \text{FAOSTAT via MapSPAM}\}$$

County livestock standing stock (herd replacement capital). H is head count (KPHC 2019 and NAPR 2021), kappa is the capital share or camel herd-structure ratio, and the sale price is the NDMA 2019 to 2021 median in ASAL counties or the KAMIS 2021 median elsewhere.

$$S_c = \sum_j H_{c,j}\, \kappa_j\, P_{c,j}^{\text{sale}}, \qquad \kappa_j = \begin{cases} 0.75 & j \in \{\text{cattle}, \text{goats}, \text{sheep}\} \\ \rho_c^{\text{camel}} & j = \text{camels} \\ 1 & \text{otherwise} \end{cases}$$

County livestock annual flow (offtake plus milk). r is the annual offtake rate by species and zone z(c); M is milk litres with the dairy herd at 800 L per head and the non-dairy herd at the zone coefficient (55 L arid, 100 L semi-arid and highland); camel milk is 180 L per head.

$$F_c^{\text{live}} = \sum_j H_{c,j}\, r_{j,z(c)}\, P_{c,j}^{\text{sale}} + \sum_j M_{c,j}\, P_c^{\text{milk}}, \qquad M_{c,\text{cattle}} = H_c^{\text{dairy}} \cdot 800 + \left(H_{c,\text{cattle}} - H_c^{\text{dairy}}\right) \cdot m_{z(c)}$$

County livestock product flow and portfolio totals. Products (eggs, honey, hides, poultry meat) use KNBS 2021 county quantities at the KNBS 2021 national unit value; the dollar conversion is fixed at the 2021 annual average rate.

$$F_c^{\text{prod}} = \sum_g Q_{c,g}\, \bar{P}_g^{\text{KE}}, \qquad F_c = F_c^{\text{crop}} + F_c^{\text{live}} + F_c^{\text{prod}}, \qquad V^{\text{USD}} = \frac{V^{\text{KES}}}{109.65}$$

Subcounty apportionment shares (mass-preserving dasymetric weights). Weights are commodity-specific; where the prior sums to zero within a county the constituency land-area share is substituted and the row flagged as a fallback.

$$w_{sc,i} = \frac{A_{sc,i}}{\sum_{k \in c} A_{k,i}} \text{ (crops: MapSPAM physical area)}, \qquad w_{sc,j} = \frac{H^{\mathrm{KPHC}}_{sc,j}}{\sum_{k \in c} H^{\mathrm{KPHC}}_{k,j}} \text{ (livestock: census head)},$$

Apportioned constituency values and county-sum identity. The build gates that apportioned values re-aggregate exactly to the statutory county totals.

$$S_{sc} = \sum_j w_{sc,j} S_{c,j}, \qquad F_{sc} = \sum_i w_{sc,i} F_{c,i} + \sum_j w_{sc,j} F_{c,j}, \qquad \sum_{sc \in c} S_{sc} = S_c, \quad \sum_{sc \in c} F_{sc} = F_c$$

12.2.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`knbs-napr`	KNBS National Agricultural Production Re	County (ADM1), 47 counties	2019 to 2024 (crops); 2021 (livestock and products, baseline year)
`kenya-population-knbs-census`	KNBS 2019 Census Population Counts by Co	Census sub-county, crosswalked to 290 IEBC constituencies	2019
`crop-vop-intld15`	MapSPAM 2020 Adaptation Atlas, Sub-Sahar	5 arc-minute (~10 km) grid, summed to IEBC constituencies	2020
`livestock-vop`	Gridded Livestock of the World (GLW4) Va	1 km (0.008333 deg)	2020

Served Parquet Files in Platform: `economic_valuation_baseline.parquet`, `economic_valuation_subcounty_weights.parquet`, `economic_valuation_prices.json`

12.2.4 Methodological Assumptions & Analytical Limitations

- Beta Review Edition governance: livestock prices for the 23 ASAL counties derive from NDMA county drought early warning bulletins used under research fair dealing pending formal NDMA licensing. The asset tables are provided for expert review only, carry no bulk download, and are not published to the open Atlas data store or the Comprehensive Data Hub.
- The values are macro-level exposure screening estimates. They are suitable for comparing counties and constituencies and for sizing contingency plans; they are not calibrated for direct use in fiscal transfers, index insurance payouts or compensation schedules without validation against county market and herd surveys.

- Livestock head counts are 2019 census household counts and 2021 NAPR populations. Pastoral herds are mobile and partly enumerated away from the household, and herd sizes changed substantially through the 2020 to 2023 drought; the 2021 baseline predates most of that mortality.
- The capital share (0.75), camel herd-structure ratios, offtake rates and milk coefficients are zone-level parameters from audited literature, not county surveys. Each carries a documented range in the parameter file; the point estimates are used in the figure and the ranges are not propagated as uncertainty bands.
- Milk coefficients rest on Behnke and Muthami (2011) field audits and on Field (1985) for small stock, with the dairy-herd scaling applied in highland counties only; dry-county dairy herds are deliberately left unscaled because national scaling inflated them 1.3 to 13 times.
- Crops are valued for 2019 to 2024 but livestock and products for 2021 only, so the portfolio is a 2021 snapshot and the crop time series should not be read as a trend in total assets.
- Crop prices mix three sources with different bases (farm-gate, gross price to farmers, FAOSTAT producer price). Horticulture enters as county-reported values rather than as quantity times price.
- MapSPAM 2020 physical area is itself a modelled allocation of national and sub-national statistics to a 10 km grid, so crop apportionment inherits its allocation errors; in heterogeneous pastoral ecotones the cropland prior and the census livestock prior both carry classification error, and 12 percent of crop rows fall back to land-area shares.
- The KPHC 2019 census sub-counties do not coincide with IEBC constituencies; the crosswalk introduces boundary error in counties where the two geographies diverge.
- Fisheries, forestry, draught power, manure and the non-market services of livestock are outside the valuation; the flow is a gross value, not value added, and input costs are not deducted.
- County aggregations rely on official polygon boundaries (IEBC constituencies via UN OCHA COD-AB); historical boundary changes are not back-cast.

12.2.5 Controlled Vocabulary Keywords

`VOP` • `Herd Capital` • `Subcounty Weights` • `Spatial Apportionment` • `Beta Review Governance` • `KNBS NAPR` • `KNBS` • `NDMA` • `KAMIS` • `FAO / FAOSTAT` • `GLW / Livestock Density` • `Cropland Mask` • `ASAL`

12.2.6 References

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12.3 Figure 4.1: Dual-Season Ocean Drivers vs. Agricultural & Livestock Output

Section in Platform: Section 4 | **Target Identifier:** `fig_3_1` | **Output Type:** Chart

> **Policy / Scientific Question:** *In the years covered by official county statistics, how did crop production, livestock numbers and livestock product value move relative to their typical level, and what were the two rainy seasons and the ocean drivers doing in the seasons that fed each year?*

12.3.1 Scientific Purpose & Context

Annual agricultural output in Kenya is fed by two rainy seasons, and harvests lag the rains that grew them. Figure 4.1 lines up the official county statistics from the Kenya National Bureau of Statistics (KNBS) National Agricultural Production Report (NAPR) with the seasons that fed each year, so a planner can see whether a weak year followed a failed season, a double failure, or neither.

The upper panel plots the selected crops, livestock species or livestock products for the county as an index relative to each item's median over the surveyed years (100 = typical), or as raw values. The lower strips show, for each year, the seasonal rainfall anomaly in percent of the 1991–2020 mean and the seasonal ocean driver values (Relative Oceanic Niño Index,

Dipole Mode Index and the Western-V index) against their phase thresholds, with an optional vegetation (NDVI) overlay. For crops the year is a harvest year, and it is paired with the preceding Short Rains (OND of the previous calendar year) and the concurrent Long Rains (MAM). Livestock and product years are paired with the same calendar year's MAM and OND, with lagged values available in the table view.

The figure is a descriptive, record-linking tool. The KNBS county series currently span only 2019 to 2024 for crops and 2021 to 2023 for livestock and products, so it supports reading individual recent years against their seasons, for example the 2020 to 2022 multi-season drought, rather than estimating how output responds to ocean drivers in general. Longer-run crop responses are examined with HarvestStat county series elsewhere in Section 4.

12.3.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: Output indexed to the median of the surveyed years; rainfall anomalies against the 1991–2020 WMO standard normal

Output series. For the selected county the figure assembles three groups from the KNBS NAPR extractions: crop production in tonnes (with paddy rice from Agriculture and Food Authority county returns), livestock head counts by species, and livestock product value in Kenyan shillings, excluding values flagged as suspect during extraction. The user chooses a group and up to a few items (by default the two leading food crops). Only positive values are plotted. The years shown are the years with data for the selected items: currently 2019 to 2024 for crops and 2021 to 2023 for livestock and products.

Median index. Each item's typical level is the median of its positive values over the surveyed years. In the default view each value is plotted as an index, 100 times the value divided by that median, so that items with very different units share one axis. The table view reports the percentage deviation from the median and flags deviations of at least +15 percent and at most −20 percent. No trend is fitted or removed.

Season pairing. For crops, year Y is a harvest year paired with OND of year Y−1 and MAM of year Y, following the national convention that Short Rains crops are harvested in the following calendar year. For livestock and products, year Y is paired with MAM and OND of the same year, and the table adds the preceding seasons as lagged columns.

Seasonal climate strips. Rainfall is shown as the percentage anomaly of the county-mean CHIRPS v3 seasonal total against its 1991–2020 mean; SPEI-3 is available as an alternative. The ocean drivers are the CPC RONI for the season (falling back to a seasonal mean of monthly values when the seasonal record is missing), the mean of the available monthly HadISST DMI values over the season (ERSST where HadISST is missing), and the mean of the standardized Western-V index over March to May. Each driver is drawn against its phase thresholds: ±0.5 °C for RONI, ±0.4 °C for DMI and ±0.5 standard deviations for Western-V. The bars are coloured by season, not by phase.

12.3.2.1 Methodological Rationale

Why a median index rather than a detrended anomaly. Detrending needs a series long enough to estimate a trend separately from the shocks of interest. The KNBS county series are four to six years long, and a linear trend fitted to them would absorb a drought year as readily as a real trend in area or productivity. The median of the surveyed years is the most robust typical level available for so short a record: it is not pulled by one extreme year, and it makes items in tonnes, head and shillings comparable on one axis. Long-run teleconnection effects on yield are better examined with the multi-decadal HarvestStat county series shown elsewhere in Section 4, which are long enough to separate trend from shocks.

Why explicit season pairing. Placing an annual harvest beside same-year seasonal shading invites a misreading, because the Short Rains of year Y are harvested in year Y+1. The figure therefore pairs each crop harvest year with the preceding OND and the concurrent MAM, labels the axis as harvest year, and draws the two seasons as separate bars, so the lag is part of the visual grammar rather than a footnote. Livestock numbers and products are census and survey-year quantities with no single harvest date, so they are paired with the same year's seasons and the lagged seasons are reported alongside.

Why raw driver thresholds here. The strips are a reading aid placed against the physical index values that county planners encounter in bulletins and outlooks, so they use the conventional raw thresholds for each index rather than the standardized classes used for the contingency analysis in Figure 3.2.

Governing Mathematical Formulations:

Median index of output. V is production (t), head count or product value (KSh) for item j; the set of years is the item's surveyed years in the county.

$$I_{j,Y} = 100\,\frac{V_{j,Y}}{\tilde{V}_j}, \qquad \tilde{V}_j = \text{median}\{V_{j,y} : y \in \mathcal{Y}_j,\ V_{j,y} > 0\}$$

Deviation from the median (table view). Asymmetric flag thresholds highlight shortfalls more readily than surpluses.

$$\delta_{j,Y} = 100\,\frac{V_{j,Y} - \tilde{V}_j}{\tilde{V}_j}, \qquad \text{flag: } \delta \geq +15\% \text{ or } \delta \leq -20\%$$

Season pairing by output type. Crop years are harvest years; Short Rains crops are harvested in the following calendar year.

crops: $Y \leftrightarrow (\text{OND}_{Y-1},\ \text{MAM}_Y)$, livestock and products: $Y \leftrightarrow (\text{MAM}_Y,\ \text{OND}_Y)$ with lagged $(\text{MAM}_Y$

Seasonal rainfall anomaly. County-mean CHIRPS v3 seasonal totals.

$$A^{\%}_{s,y} = 100\,\frac{R_{s,y} - \bar{R}_{s,1991\text{–}2020}}{\bar{R}_{s,1991\text{–}2020}}, \qquad s \in \{\mathrm{MAM}, \mathrm{OND}\}$$

Driver phase reference lines. Seasonal means of the indices drawn against conventional raw thresholds.

$$\mathrm{RONI} \gtrless \pm 0.5^{\circ}\mathrm{C}, \qquad \mathrm{DMI} \gtrless \pm 0.4^{\circ}\mathrm{C}, \qquad \text{Western-V} \gtrless \pm 0.5\,\sigma$$

12.3.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
knbs-napr	KNBS National Agricultural Production Re	County (ADM1), 47 counties	Crops 2019 to 2024; livestock and products 2021 to 2023
africa-precipitation-monthly-seasonal	Monthly and Seasonal Precipitation from	0.05 deg (~5.5 km, CHIRPS v3 native) aggregated to county mean	1981 to 2025; anomalies against 1991–2020
africa-spei-drought	SPEI Drought Index, 3- and 12-month (Afr	County (ADM1)	1981 to 2025
eastafrica-ndvi-modis	MODIS NDVI Seasonal Vegetation Condition	County (ADM1) aggregate of 250 m MODIS NDVI	Surveyed years only
enso-driver-indices	Ocean Climate Teleconnection Drivers (Pa	Equatorial Pacific, tropical Indian Ocean and western Pacific basin indices	Surveyed years and preceding season

Served Parquet Files in Platform: knbs_napr_county_production.parquet, afa_rice.parquet, knbs_napr_livestock.parquet, knbs_napr_livestock_products.parquet, chirps_county.parquet, ndvi_county.parquet, driver_indices.parquet, enso_outlook_base.parque

12.3.4 Methodological Assumptions & Analytical Limitations

- The record is very short. Crops have at most six surveyed years and livestock and products three, so the figure cannot estimate a response to ocean drivers; it only places individual years beside their seasons. No correlation, regression or significance test is computed.
- The median of four to six years is itself uncertain and is pulled toward whatever conditions dominated the surveyed years; for 2019 to 2024 that period includes the 2020 to 2022 drought, so the "typical" level may be lower than a longer-run normal.

- Output is not detrended, and changes in planted area, herd management or KNBS survey coverage between editions are not separated from weather effects.
- Crop production mixes MAM-grown and OND-grown harvests in one annual figure, so a single harvest year reflects both seasons in unknown proportions that vary by county and crop.
- Livestock head counts and product values are stock and survey quantities whose response to drought is lagged and cumulative; pairing them with same-year seasons understates the effect of earlier seasons, which the lagged table columns only partly address.
- Values flagged as suspect in the NAPR product tables are excluded, and blanks are not treated as zeros, so some items have gaps.
- The optional NDVI overlay is drawn only for surveyed years, so the preceding Short Rains value is often unavailable for the first year shown.
- Driver strips use seasonal means of whatever monthly values are available, without a completeness check, and the Western-V index is defined only for MAM.
- County aggregations rely on official polygon boundaries from the UN OCHA Common Operational Datasets (COD-AB).

12.3.5 Controlled Vocabulary Keywords

`KNBS NAPR • KNBS • AFA • CHIRPS v3 • SPEI-3 • MODIS NDVI • RONI • DMI • Western-V / WV • WMO Normal • OND • MAM • Double Failure`

12.3.6 References

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12.4 Figure 4.2: Empirical Ocean Driver Phase Response vs Mean Output

Section in Platform: Section 4 | **Target Identifier:** `fig_3_1b` | **Output Type:** Chart

> **Policy / Scientific Question:** *What is the historical average production anomaly during El Niño, La Niña, or IOD phases?*

12.4.1 Scientific Purpose & Context

Quantify and communicate Empirical Ocean Driver Phase Response vs Mean Output across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.4.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Empirical phase response distribution of agricultural production showing distribution spreads, interquartile ranges, and outlier seasons.

12.4.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

12.4.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`knbs-napr`	KNBS National Agricultural Production Re	kenya	2019 to 2024
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `knbs_napr_county_production.parquet`, `driver_indices.parquet`

12.4.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.4.5 Controlled Vocabulary Keywords

`KNBS NAPR`

12.5 Figure 4.2B: Multi-Decadal Seasonal Crop Harvests & Climate Teleconnections (HarvestStat Africa)

Section in Platform: Section 4 | **Target Identifier:** `fig_3_1c` | **Output Type:** Chart

> **Policy / Scientific Question:** *How did subnational seasonal crop yields and production track historical rainfall terciles across 1990–2024?*

12.5.1 Scientific Purpose & Context

Quantify and communicate Multi-Decadal Seasonal Crop Harvests & Climate Teleconnections (HarvestStat Africa) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.5.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Long-term historical crop yield time-series (1982–2020 HarvestStat) detrended using linear and polynomial models to evaluate teleconnection persistence over four decades.

12.5.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$Y_{\text{detrended}} = Y_t - (a \cdot t + b) + \bar{Y}$$

12.5.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`harveststat-crop-production`	HarvestStat Africa - Harmonized Subnatio	County (ADM1), 47 counties	1965 to 2024 (harvest years; seasonal gap 2002 to 2014)
`enso-driver-indices`	Ocean Climate Teleconnection Drivers (Pa	global-oceans, equatorial-pacific, indian-ocean	1950-01 to present (open-ended)

Served Parquet Files in Platform: `harveststat_county_production.parquet`, `driver_indices.parquet`

12.5.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.5.5 Controlled Vocabulary Keywords

`HarvestStat • Yield Anomaly`

12.6 Figure 4.4: Pastoral Market Prices & Terms of Trade Monthly Purchasing Power

Section in Platform: Section 4 | **Target Identifier:** `fig_3_2b` | **Output Type:** Chart

> **Policy / Scientific Question:** *How severely did food prices surge and pastoral purchasing power collapse during climate shocks?*

12.6.1 Scientific Purpose & Context

Quantify and communicate Pastoral Market Prices & Terms of Trade Monthly Purchasing Power across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.6.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

Pastoral terms-of-trade (ToT) computed as the ratio of live medium-sized goat (shoat) price to 1 kg dry maize grain. Spikes and collapses indexed against ENSO shock windows.

12.6.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

Governing Mathematical Formulations:

$$\mathrm{ToT}_m = \frac{P_{\mathrm{goat},m}}{P_{\mathrm{maize},m}} \quad \text{(kg maize per goat)}$$

12.6.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`kenya-market-prices-fews`	Kenya Retail and Wholesale Market Prices	kenya	2000 to present (open-ended)
`kenya-market-prices-kamis`	Kenya Live-Animal Market Prices by Grade	kenya	2021-05-24 to present (open-ended)

Served Parquet Files in Platform: `market_prices.parquet`, `market_prices_kamis.parquet`

12.6.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.6.5 Controlled Vocabulary Keywords

`ToT` • `KAMIS`

12.7 Figure 4.4B: Regional Cross-Border Food Trade Flows & Market Shock Absorber (FEWS NET XBT)

Section in Platform: Section 4 | **Target Identifier:** `fig_3_2c` | **Output Type:** Chart

> **Policy / Scientific Question:** *How do regional food trade flows from Uganda, Tanzania, and Ethiopia buffer domestic grain supply during rainfall shocks?*

12.7.1 Scientific Purpose & Context

Quantify and communicate Regional Cross-Border Food Trade Flows & Market Shock Absorber (FEWS NET XBT) across Kenyan counties to inform anticipatory action and climate adaptation planning.

12.7.2 Mathematical Formulation & Algorithmic Approach

Baseline Reference Period: 1991–2020 WMO standard normal

East Africa cross-border formal and informal grain trade flows (FEWS NET XBT). Evaluates supply buffering from Uganda and Tanzania during regional drought shocks.

12.7.2.1 Methodological Rationale

Applies non-parametric or standardized metrics aligned with the platform conventions of empirical quantile partitioning and independent seasonal standardization to avoid distributional distortion.

12.7.3 Underlying Data Assets & Inputs

Provenance Key	Dataset Title	Spatial Resolution	Temporal Span
`xbt-trade`	xbt-trade	County (ADM1)	Historical multi-decadal

Served Parquet Files in Platform: `xbt_trade.parquet`

12.7.4 Methodological Assumptions & Analytical Limitations

- County aggregations rely on official polygon boundaries from UN OCHA COD-AB standard.
- Sub-seasonal temporal variations within rolling seasons are smoothed over seasonal totals.
- Data latency reflects standard publication cadences of underlying satellite and administrative surveys.

12.7.5 Controlled Vocabulary Keywords

`XBT Trade`

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13 Master Dataset Catalogue & Data Provenance

14 Master Dataset Catalogue & Data Provenance

This chapter documents the 23 formal data assets serving the Kenya ENSO Explorer platform. Each entry conforms to the Comprehensive Data Hub (CDH) Schema v0.3.0, recording upstream producers, spatial-temporal coverages, scientific caveats, and open-access licensing terms.

14.1 `enso-driver-indices`: Ocean Climate Teleconnection Drivers (Pacific RONI, Indian Ocean DMI & Western-V)

Category: Climate | **Publishing State:** `authored` | **License:** Open Data

14.1.1 Summary

Harmonised multi-decadal monthly and seasonal time series of primary equatorial ocean-atmosphere climate teleconnection drivers governing Kenyan rainfall: NOAA CPC Relative Oceanic Niño Index (RONI) for Pacific ENSO state; NOAA CPC / HadISST Dipole Mode Index (DMI) for Indian Ocean Dipole state; and Funk et al. Western-V SST index for MAM Long Rains teleconnections. Standardised and compiled into analysis-ready parquet tables (1950 to present).

- **Temporal Coverage:** 1950-01 to present (open-ended)
- **Spatial Resolution:** global-oceans, equatorial-pacific, indian-ocean

14.1.2 Data Producers & Custodians

- **NOAA Climate Prediction Center** - https://www.cpc.ncep.noaa.gov/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.1.3 Scientific Caveats & Guidance

[!WARNING] Attribution: “Data provided by NOAA Climate Prediction Center (CPC) and compiled for the African Agriculture Adaptation Atlas (AAAA).” RELATIVE OCEANIC NINO INDEX (RONI) OVER RAW NINO 3.4. CPC computes RONI by subtracting the tropical-mean (20S-20N) SST anomaly from the Nino 3.4 anomaly. This detrending removes secular tropical ocean warming (~+0.6 degC since 1950) that otherwise causes historical cool baselines to register as artificial permanent El Nino conditions. Always use RONI for multi-decadal analogue selection. INDIAN OCEAN DIPOLE (IOD) DOMINANCE IN SHORT RAINS. The DMI is the primary physical forcing of Kenyan rainfall variability during the October-December (OND) Short Rains. Warm ENSO and positive IOD co-occur frequently but can decouple; off-diagonal states (El Nino + negative IOD) produce subdued anomalies. SEASONAL ASYMMETRY AND LONG RAINS (MAM) DECOUPLING. Pacific ENSO indices exhibit weak correlation with Kenyan rainfall during the March-May (MAM) Long Rains (r ~ 0.08). The Western-V index captures western Pacific warm pool gradients that govern convective subsidence over the Horn of Africa, providing independent predictive skill where ENSO fails. Z-SCORE STANDARDIZATION REQUIREMENT. Pacific SST anomalies (~0.5-2.5 degC) and regional rainfall anomalies exhibit a 10.6:1 variance distortion ratio. Multi-driver analogue selection must use z-score standardisation across all driver dimensions to ensure equal physical weighting.

14.2 `africa-precipitation-monthly-seasonal`: Monthly and Seasonal Precipitation from CHIRPS v3 (Africa)

Category: Climate | **Publishing State:** `authored` | **License:** Open Data

14.2.1 Summary

Total precipitation (PTOT) aggregated from CHIRPS v3.0 and delivered as per-pixel monthly Cloud Optimized GeoTIFFs and per-year rolling 3-month (seasonal) sums over Africa, 1981 to present. This is the rainfall backbone of the KE-ENSO Explorer’s seasonal/ENSO analysis: the twelve seasonal windows (JFM, FMA, …, DJF) are the tri-month sums the notebook compares against ENSO phase and against the drought (SPEI), vegetation (NDVI) and flood layers. A derived aggregation product, not a redistribution of the CHIRPS daily archive.

- **Temporal Coverage:** 1981-01 to present (open-ended)
- **Spatial Resolution:** africa — 0.05 deg (~5.5 km, CHIRPS native)

14.2.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Climate Hazards Center, UC Santa Barbara** - https://www.chc.ucsb.edu/data/chirps

14.2.3 Scientific Caveats & Guidance

> [!WARNING] Source CHIRPS is public domain ("no restrictions on use"); these reprocessed COGs are released CC-BY-4.0 with attribution to the Climate Hazards Center, UC Santa Barbara. Seasonal windows are labelled by the calendar year in which the window ends (DJF_2020 = Dec 2019 to Feb 2020). Edge windows that the record cannot cover in full are absent (NDJ and DJF 1981; windows ending after the latest monthly file), so not every season x year combination of the template exists.

14.3 `africa-spei-drought`: SPEI Drought Index, 3- and 12-month (Africa)

Category: Climate | **Publishing State:** `authored` | **License:** Open Data

14.3.1 Summary

Standardized Precipitation-Evapotranspiration Index at 3-month (SPEI-03) and 12-month (SPEI-12) accumulation, as per-pixel monthly Cloud Optimized GeoTIFFs over Africa, 1981 to present. Built from CHIRPS v3 precipitation and a potential-evapotranspiration climate water balance fitted with a log-logistic distribution; negative values are drier than the reference climatology. The drought layer of the KE-ENSO Explorer: the short accumulation tracks seasonal/agricultural drought, the long accumulation hydrological drought.

- **Temporal Coverage:** 1981-01 to present (open-ended)
- **Spatial Resolution:** africa — 0.05 deg (~5.5 km, CHIRPS grid)

14.3.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.3.3 Scientific Caveats & Guidance

[!WARNING] SPEI is a z-score anomaly against the fitted reference distribution, not an absolute deficit. Arid-cell -Inf values (zero-precipitation tails of the log-logistic fit) are clamped to NoData.

14.4 `eastafrica-flood-jrc`: JRC / GloFAS Return-Period River Flood Hazard (Global)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.4.1 Summary

Modelled global riverine flood hazard from the JRC / GloFAS Global Flood Maps (CEMS-GLOFAS v2.1.2): inundation depth for seven return periods, 10 to 500 years, at 3 arc-seconds, as a static baseline climatology. Published as Cloud Optimized GeoTIFFs with internal overviews on Source Cooperative, which is the canonical read location. The Adaptation Atlas additionally holds a provisional Kenya crop of the same data. Within the KE-ENSO Explorer this is the MODELLED flood-prone baseline complementing the OBSERVED Copernicus GFM extent (record kenya-flood-gfm): JRC answers "how flood-prone at magnitude X", GFM answers "where and when it actually flooded".

- **Temporal Coverage:** None
- **Spatial Resolution:** world, kenya — 3 arc-seconds (~90 m)

14.4.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Source Cooperative / Radiant Earth Foundation** - https://source.coop/nlebovits/jrc-glofas
- **European Commission Joint Research Centre (JRC)** - https://data.jrc.ec.europa.eu/

14.4.3 Scientific Caveats & Guidance

[!WARNING] FEDERATED TO THE CLOUD-OPTIMIZED UPSTREAM (2026-09-18, following review on cdh-catalog#32). The full global product is already published as COGs with internal overviews under CC-BY-4.0 on Source Cooperative, at the same seven return periods and the same 3 arc-second resolution as the Atlas crop. The canonical asset in this record is therefore the global collection, not our subset. The Kenya crop is retained as a provisional second asset because

the KE-ENSO Explorer currently reads it, but it carries no information the global collection lacks and the performance argument for keeping it is untested. Measuring that is the open task; retiring the duplicate is the likely outcome. Static modelled hazard (return-period depths); not an observed time series and not scenario-projected, so no temporal coverage is declared. Pair with the GFM observed extent for the "modelled vs observed" flood story.

14.5 `kenya-flood-gfm`: Kenya Observed Flood Extent (Copernicus GFM, Sentinel-1)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.5.1 Summary

Observed flood extent for Kenya from the Copernicus Emergency Management Service Global Flood Monitoring (GFM) product, which classifies flooding from every Sentinel-1 SAR acquisition (ensemble of the DLR, TU Wien and LIST algorithms; permanent and reference water excluded). Delivered as ~111 m aggregates built directly from the 20 m SAR tiles: a monthly flood-occurrence layer with its valid-observation count, a rolling 3-month (seasonal) layer aligned to the KE-ENSO Explorer precipitation seasons, and a full-record history (occurrence frequency and ever-flooded footprint), 2018 to present. Replaces the MODIS Global Flood Database (ended 2018) with a cloud-penetrating, continuously updated SAR record. The 20 m per-acquisition archive is not materialised here (16,000+ acquisitions); it can be re-pulled from the EODC STAC on demand.

- **Temporal Coverage:** 2018-01 to present (open-ended)
- **Spatial Resolution:** kenya — 0.001 deg (~111 m); aggregated from 20 m Sentinel-1 tiles (a coarse pixel is flooded if any 20 m sub-pixel flooded)

14.5.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Copernicus Emergency Management Service (CEMS)** - https://global-flood.emergency.copernicus.eu/

14.5.3 Scientific Caveats & Guidance

[!WARNING] Attribution required: "Contains modified Copernicus Emergency Management Service information [year]". Pixel coding of the flooded layers: 0 = not flooded, 1 = flooded, 255 = not observed (SAR gap or excluded) - 255 is absence of observation, NOT dry; the nobs layer gives the count of valid observations so no-flood is distinguishable from no-observation. Seasonal windows are named by the initials of their months and labelled by the year of their FIRST month (DJF_2018 = Dec 2018 to Feb 2019), so NDJ and DJF exist for 2018 to 2024 only. This differs from the CHIRPS seasonal record (africa-precipitation-monthly-seasonal), which labels NDJ and DJF by the year the window ends; align on those two windows with care. A ~7-year record measures recent observed flood occurrence, not return-period magnitude.

14.6 `eastafrica-ndvi-modis`: MODIS NDVI Seasonal Vegetation Condition (East Africa)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.6.1 Summary

Normalized Difference Vegetation Index from MODIS MOD13Q1 (250 m, 16-day composites), averaged over the two East African rainy seasons - the long rains (MAM, March to May) and the short rains (OND, October to December) - for every year from 2000 to 2025, over an East Africa window centred on Kenya. Written as Cloud Optimized GeoTIFFs. The vegetation-condition layer of the KE-ENSO Explorer: greenness response to seasonal rainfall and ENSO phase.

- **Temporal Coverage:** 2000 to 2025
- **Spatial Resolution:** eastern-africa, kenya — ~250 m (0.0021 deg, MOD13Q1 native)

14.6.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **NASA LP DAAC / USGS EROS** - https://lpdaac.usgs.gov/products/mod13q1v061/

14.6.3 Scientific Caveats & Guidance

[!WARNING] Source MODIS data are free and unrestricted; these aggregated COGs are released CC-BY-4.0 with attribution to NASA LP DAAC. NDVI ranges roughly -0.2 to 1.0 (higher = greener). Only the MAM and OND windows are published (no other seasons, no annual mean). Each file covers the full East Africa window (~275 MB); read with a bbox/overview to avoid whole-file downloads.

14.7 `eastafrica-wrsi-fews`: FEWS NET Water Requirement Satisfaction Index, Cropland and Rangeland (Kenya)

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.7.1 Summary

End-of-season Water Requirement Satisfaction Index (WRSI) for cropland (maize) and rangeland growing seasons from USGS FEWS NET, for the long rains (MAM) and short rains (OND) of every year from 2003, cropped to Kenya and written as Cloud Optimized GeoTIFFs. WRSI is the ratio of seasonal water supplied to crop or pasture water demand (0 to 100 percent); low values flag agricultural water deficit. The crop and rangeland monitoring layer of the KE-ENSO Explorer, on the same CHIRPS v3 precipitation backbone as the rainfall and SPEI layers.

- **Temporal Coverage:** 2003 to present (open-ended)
- **Spatial Resolution:** kenya — 0.1 deg (~11 km, FEWS NET WRSI grid)

14.7.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **USGS FEWS NET** - https://earlywarning.usgs.gov/fews

14.7.3 Scientific Caveats & Guidance

[!WARNING] Cropland and rangeland use different water-demand parameterisations and growing-season definitions (FEWS NET region codes / end-of-season dekads), so the two products are not interchangeable. The {crop} x {season} cells map to FEWS NET East Africa region codes as follows (FEWS NET zip-product instructions, Table 1, 2025-02-03): rangeland OND = e1 (Sep-Jan), rangeland MAM = e2 (Feb-Jul), cropland OND = et (maize, Oct-Feb), cropland MAM = ee (maize, Mar-Nov). Not every season x year combination of the template exists: the most recent season is only published once FEWS NET releases its

end-of-season product. PROVENANCE: objects published between 2026-08-18 and 2026-09-20 had the cropland and rangeland labels swapped (e1/e2 served as cropland, ek/et as rangeland); every object was replaced in place on 2026-09-20 and copies fetched before that date should be discarded.

14.8 `kenya-flood-exposure-intersect`: Kenya Flood x Exposure, Per-Sub-County Tables

Category: Satellite | **Publishing State:** `authored` | **License:** Open Data

14.8.1 Summary

Pre-computed flood x exposure statistics for Kenya's 290 IEBC sub-counties (admin 2), as three small Parquet tables the KE-ENSO Explorer reads directly (DuckDB-WASM) instead of intersecting heavy rasters and vectors in the browser. Two hazard products - Copernicus GFM observed flood (per season x year, 2018 to 2025) and JRC modelled return-period flood hazard - are each crossed with population (WorldPop constrained 100 m grid, levelled to the official KNBS county totals), classified roads and the electricity transmission grid (length exposed) and health and school facilities (count exposed); a third table holds the per-sub-county denominators. Tables carry no geometry: join on adm2_pcode to the boundary record kenya-admin-codab.

- **Temporal Coverage:** 2018 to 2025
- **Spatial Resolution:** kenya — IEBC sub-counties (290 admin-2 units)

14.8.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.8.3 Scientific Caveats & Guidance

[!WARNING] Derived aggregate statistics; no input geometry is redistributed. Please attribute the inputs: Copernicus GFM and JRC Global Flood Maps (Copernicus / European Commission), WorldPop constrained population (CC-BY-4.0), OpenStreetMap roads and HOTOSM facilities ((c) OpenStreetMap contributors, ODbL), KPLC grid (CC0-1.0, energydata.info), IEBC COD-AB boundaries (OCHA, CC-BY-3.0-IGO). In the GFM table, observed_pct is the SAR-observed share of the sub-county in that season: low values mean sparse observation, not absence of flooding. Road and grid kilometres are cell-resolution approximations (in-cell line length within flooded cells). Every table also carries adm1_pcode, adm1_name and adm2_name label columns. POPULATION

METHOD (issue #28), stated in full. Headcounts are NOT raw gridded pixel sums. Every published sub-county population is built as three explicit factors, each published as its own column:

[!WARNING] pop_adm2 = pop_adm2_grid x pop_scale_census x pop_growth_county

[!WARNING] (1) pop_adm2_grid - the 100 m gridded pixel sum, which supplies the SHARE of the county's people

[!WARNING] living in that sub-county, and nothing else.

[!WARNING] (2) pop_scale_census = census county total / gridded county total - replaces the LEVEL of the

[!WARNING] gridded surface with the enumerated KNBS 2019 census, county by county. The gridded surfaces

[!WARNING] run about 17 per cent above the census (WorldPop constrained ~55.2 M, GRID3 ~55.9 M, census

[!WARNING] 47,564,296), so this factor is ~0.855 nationally but spans 0.324-1.394 across individual

[!WARNING] counties (measured on the published grid, 2026-09-17): WorldPop exceeds the census by up to

[!WARNING] threefold in some counties and falls below it in others. Any aggregate therefore moves by a

[!WARNING] weighted mix of county factors, not by the national one - flood-exposed population sits mostly

[!WARNING] in low-factor counties, so the GFM exposed total fell about 35 per cent where the national

[!WARNING] denominator fell 14 per cent. No time component.

[!WARNING] (3) pop_growth_county = target county total / census county total - the county's proportional

[!WARNING] CHANGE OVER TIME, from the KNBS Vol XVI projections (record

[!WARNING] kenya-population-knbs-projections). Exactly 1.0 when the denominator is the census itself.

[!WARNING] County is the only level where any of this is defensible: COD-AB adm2 is 290 IEBC constituencies while the census reports 345 KNBS sub-counties, and only 183 names match (record kenya-population-knbs-census); KNBS does not project below county at all. FIXED-SHARE ASSUMPTION. Each sub-county's share of its county is held constant at the gridded 2020 distribution, so every sub-county in a county moves by exactly that county's percentage change. Sub-county-specific growth is NOT represented: a fast-urbanising sub-county and a depopulating one in the same county grow at the same rate here. Growth does vary between counties - the 2025 factors span 1.03 to 1.15 - but not within one. YEAR MATCHING (the published default from 2026-09-17). The GFM table records OBSERVED floods and carries a year, so a single fixed denominator is wrong in one direction or the other: the 2019 census undercounts a 2024 flood by roughly 9 per cent, and a 2025 projection overcounts a 2018 one by a similar margin. Each GFM row is therefore levelled against ITS OWN year's county population, and pop_year records which year was used. Years the KNBS projections cover (2020 onward) use them; 2018 and 2019 fall back to the enumerated census, because nothing official projects backwards from the 2020 base and inventing a figure would be worse than stating the fallback. Consequence for consumers: pop_source VARIES BY ROW in the GFM table - "knbs-census-2019" for 2018-2019 and "knbs-projection-" thereafter - so do not read one value and assume it applies to the file. The JRC table is a return-period hazard with no event year, and the totals table is a static denominator, so both take a single stated reference year (pop_year, same on every row). TWO DEFINITIONS OF THE GROWTH FACTOR, recorded in pop_method:

[!WARNING] county-growth-from- - pop_growth_county = projection(year) / projection(base year,

[!WARNING] 2020). The enumerated census remains the level and KNBS supplies only the shape of change;

[!WARNING] national 2025 lands near 51.96 M because the ~2.6 per cent step KNBS puts between census night

[!WARNING] (August 2019) and its own 2020 base is deliberately not imported.

[!WARNING] county-level - pop_growth_county = projection(year) / census(2019), so the county total IS the

[!WARNING] published official projection and national 2025 is 53,330,964 - the figure a Kenyan

[!WARNING] counterpart would quote - at the cost of superseding the census anchor with the KNBS base-level

[!WARNING] revision.

[!WARNING] pop_source names the population vintage (knbs-census-2019, or knbs-projection-); pop_grid_source names the surface supplying the share. Nothing is discarded: the raw gridded figures remain in pop_exposed_grid and pop_total_grid. pop_pct is identical under every option - the factors cancel.

14.9 `kenya-population-knbs-census`: KNBS 2019 Census Population Counts by County and Sub-county (Kenya)

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.9.1 Summary

Enumerated population from the 2019 Kenya Population and Housing Census (KNBS), as admin-level tables: national, the 47 counties keyed on IEBC COD-AB adm1_pcode, the 345 KNBS sub-county rows in the Bureau's own sub-county universe, and a county x 5-year age group x sex table. The official denominator for anything the KE-ENSO Explorer presents to a Kenyan audience: the census total is 47,564,296, where the two gridded surfaces the Explorer also carries (kenya-population-worldpop 55.2 M, kenya-population-grid3 55.9 M) sit about 17 per cent higher.

- **Temporal Coverage:** 2019
- **Spatial Resolution:** kenya — counties (47 admin-1 units); a sub-county table is included in the KNBS universe

14.9.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/

14.9.3 Scientific Caveats & Guidance

[!WARNING] LICENCE EVIDENCE. Verified 2026-09-18 against the HDX API: the dataset "Kenya: Population Per County and sub county from Census Report", published by KNBS's OWN HDX organisation (Kenya National Bureau of Statistics), carries license_id other-pd-nr, "Public Domain / No restrictions (CC0)", and its resource is the identical workbook this record ingests ("Distribution of Population by Sex, Number of Households, Land Area, Population Density and Sub County.xlsx"). That is an explicit act of dedication by the rights holder over this content, so CC0-1.0 is asserted here rather than a LicenseRef. Note that KNBS census VOLUMES carry a general "All rights reserved" notice; it does not govern this table, which KNBS itself released without restriction. Attribution

is given as a courtesy though CC0 does not require it. The supporting evidence gathered earlier still stands: Evidence for the licence, recorded because knbs.or.ke itself carries no terms page (the terms-and-conditions URL 404s and the footer asserts “All Rights Reserved”), which is misleading: (1) KNBS operates an Open License Agreement covering its Open Data Platform, granting a worldwide, royalty-free, non-exclusive licence to use, copy, modify, publish, adapt, distribute and create derivative works for commercial and non-commercial purposes, subject to attribution. (2) KNBS’s own account on the Humanitarian Data Exchange published the SAME sub-county table used here - “Distribution of Population by Sex, Number of Households, Land Area, Population Density and Sub County” - under Public Domain / No restrictions (https://data.humdata.org/dataset/kenya-population-per-county-from-census-report-2019). (3) The same census figures are redistributed by UNFPA under CC-BY-3.0-IGO (HDX cod-ps-ken). Required attribution, carried by any product derived from this record: “This product was adapted from the Statistical Office of Kenya’s information, which is licensed under the KNBS Open License Agreement.”

[!WARNING] ADMIN LEVELS DIFFER BY LEVEL. County joins one-for-one with IEBC COD-AB (47 units, adm1_pcode). Sub-county does NOT: COD-AB adm2 is 290 IEBC constituencies while the census reports 345 KNBS sub-county rows (a different universe - constituency splits such as Embakasi East/West/Central against sub-county Embakasi, plus 12 forest and national-park units such as Aberdare Forest and Mt Elgon Forest). Normalised in-county name matching resolves 183 of the 290 COD-AB units, so the adm2 table is published in the KNBS universe, name-keyed, with adm2_pcode_codab filled only where a name matches exactly one COD-AB unit inside the same county and codab_match recording how each row resolved. Use county totals plus a gridded within-county share for any adm2 denominator; do not join this table to COD-AB sub-counties on name. Male + Female (47,562,772) falls 1,524 short of the published Total because KNBS suppresses the intersex count below national level; the residual is carried as pop_intersex. The age-and-sex table comes from HDX cod-ps-ken (UNFPA, CC-BY-3.0-IGO), which redistributes the same census (Volume III) with adm1 p-codes attached. DO NOT SWITCH THE SOURCE TO THE HDX COPY of the sub-county table, despite its cleaner licence status: that copy is defective. It carries 346 rows to this source’s 345 because the Tharaka-Nithi COUNTY total (393,177) appears as if it were a sub-county, inflating the national figure to 47,957,473 against the published 47,564,296. This ingest uses the knbs.or.ke workbook, which reconciles exactly. (It also spells Kutulo as “Kotulo”; same value, harmless.)

14.10 `kenya-population-knbs-projections`: KNBS Official Population Projections by County, 2020-2045 (Kenya)

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.10.1 Summary

Kenya's official cohort-component population projections, from the 2019 KPHC Analytical Report on Population Projections (Volume XVI, KNBS 2022), tabulated from the report's Appendix 5 as county x year x 5-year age group x sex, keyed on IEBC COD-AB adm1_pcode. National trajectory 48.82 M (2020), 53.33 M (2025), 57.81 M (2030), 62.16 M (2035), 66.31 M (2040), 70.18 M (2045). Lets forward-framed flood and drought exposure use a population denominator for the year in question instead of a 2020 surface read as current.

- **Temporal Coverage:** 2020 to 2045
- **Spatial Resolution:** kenya — counties (47 admin-1 units)

14.10.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/

14.10.3 Scientific Caveats & Guidance

> [!WARNING] MIXED-STATUS RECORD - READ BEFORE REUSE (decision 2026-09-18). The source documents are KNBS publications, linked here and not hosted by the Atlas. Three DERIVED parquet tables were published to digital-atlas BEFORE the licence was correctly established; they reproduce KNBS figures, so they are NOT open data. They are documented here deliberately: leaving a public object undocumented is worse than describing its true status. They must not be treated as open data, and this record must NOT be submitted to cdh-catalog, until written permission exists from KNBS (info@knbs.or.ke / directorgeneral@knbs.or.ke). Permission request is deferred as of 2026-09-18, so this state persists by decision, not by oversight. HOW THE ATLAS USES THIS WITHOUT REDISTRIBUTING IT. The exposure chain applies projections as a dimensionless growth ratio on a census anchor, never as absolute published values: R/observational/_population_helpers.R, POP_METHOD=county-growth (the default at 7_zonal_exposure.R:138 and 7b_relevel_exposure_pop.R:64) computes projection_county(year) / projection_county(2020) and multiplies the CC0 census count by it, so no published artifact reproduces a KNBS projection. The alternative POP_METHOD=county-level reproduces the official county total exactly and must NOT be used for anything published. SOURCING - CORRECTED 2026-10-07. An earlier version of this note claimed the ratios are read from the Summary Report rather than Volume XVI. That was WRONG: it

recorded an intention from the 2026-09-18 decision that was never implemented. python/ingest_population_knbs_projections.py downloads exactly one PDF, Volume XVI (PDF_URL, one urlretrieve_checked call), and the Summary Report URL appears nowhere in python/ or R/. The published tables are their own evidence - they carry 2020-2035 annually plus 2040 and 2045, while the Summary Report carries only 2020-2025, 2030, 2035, 2040 and 2045, so 2026-2029 and 2031-2034 can only have come from Volume XVI. The ratios therefore derive from the rights-reserved document, and the ratio method alone is what keeps published artifacts clear of KNBS figures. Switching the source is not a one-line change: the live reference year is 2026, which the Summary Report does not contain, so it would mean constraining POP_REF_YEAR to covered years - moving the national denominator from 52,837,534 (2026) to 51,961,873 (2025). Tracked on issue #33.

[!WARNING] LICENCE - CORRECTED 2026-09-18, READ BEFORE REUSE OR REDISTRIBUTION. This record previously claimed the KNBS Open License Agreement. That was WRONG. Volume XVI was read directly on 2026-09-18 and its copyright page states: "(c) 2022 Kenya National Bureau of Statistics. All rights reserved. No part of this publication may be reproduced, distributed, or transmitted in any form or by any means … without the prior written permission of the Bureau." The KNBS Open License Agreement is scoped to data published on the KNBS Open Data Platform and does not reach a PDF published on knbs.or.ke, and a search of HDX on 2026-09-18 found NO licensed mirror of these projections anywhere - the UNFPA COD-PS product (cod-ps-ken, CC-BY-3.0-IGO) carries the 2019 census only and contains no projections, so there is no openly licensed substitute. CONSEQUENCE. Extracting these figures for analysis is ordinary research use. REDISTRIBUTING them, including as the derived parquet files this record points at, needs written permission from KNBS (info@knbs.or.ke / directorgeneral@knbs.or.ke, the addresses given in the report). Until that permission exists this record must NOT be submitted to cdh-catalog, and the published derivatives should be treated as pending clearance rather than as open data. Tracked in hazards_prototype issue 33. This constraint does NOT apply to the census counts - record kenya-population-knbs-census is CC0-1.0, released without restriction by KNBS's own HDX account.

[!WARNING] LICENCE EVIDENCE. Weaker footing than the census record, and the difference is worth stating. KNBS's Open License Agreement grants worldwide, royalty-free, non-exclusive rights to reuse, adapt, redistribute and create derivative works with attribution - but it is scoped to data published on the KNBS Open Data Platform, and Volume XVI is a PDF hosted on knbs.or.ke. Unlike the 2019 census tables, no licensed mirror of these projections exists anywhere (searched HDX, open.africa and the Kenya data portal, 2026-09-17): UNFPA's COD-PS covers the census, not Volume XVI. What is published here is a derived tabulation of figures printed in a public national statistical report, attributed, not a copy of the report. Confirm with KNBS before this record is contributed to cdh-catalog. Required attribution: "This product was adapted from the Statistical Office of Kenya's information, which is licensed under the

KNBS Open License Agreement."

[!WARNING] COUNTY ONLY. Volume XVI does not project below county - no sub-county name appears anywhere in the report - so this record stops at county. The KE-ENSO exposure tables (record kenya-flood-exposure-intersect) do carry sub-county figures for a projection year, and the way they get there is stated there in full: each sub-county's share of its county is held fixed at the gridded 2020 distribution and the whole county is moved by its own percentage change from this table (pop_growth_county), with the enumerated census supplying the level. That means no sub-county-specific growth is represented; growth varies between counties (2025 factors span 1.03 to 1.15) but is uniform within one. Anyone wanting genuine sub-county differentials needs a source that produces them - this one does not. YEAR COVERAGE IS UNEVEN. The report's annual panels cover 2020-2035; 2040 and 2045 appear only in the five-yearly milestone panel, so 2036-2039 and 2041-2044 are absent. TWO SOURCE DEFECTS, both handled and logged by the ingest rather than silently absorbed. (1) Busia 2030, age 80+, female reads 3,120 in the 2030-2035 panel against 5,799 in the two other panels that carry 2030; the printed row total (9,714) reconciles only with 5,799, so 5,799 is used. (2) Kiambu's 2026-2030 panel labels its open-ended top age group "80-84" where every other panel says "80+"; it is relabelled "80+", without which it would double-count against the overlapping panels. KNBS rounds each printed cell independently, so sums do not close exactly: age groups land within five people of the printed All Ages row, and the 47 counties sum to roughly 15 to 20 people below the national row in every year. The identical report is also hosted as …-Analytical-Report-on-Population-Projections-Vol-XVI.pdf (2024/05, 32.9 MB); its extracted text is identical to the 2023/09 upload used here.

14.11 `kenya-population-worldpop`: WorldPop Constrained Gridded Population 2020 (Kenya)

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.11.1 Summary

Gridded population count for Kenya, 2020, from WorldPop's constrained (top-down dasymetric) 100 m product, cropped to Kenya and written as a Cloud Optimized GeoTIFF. The primary population-exposure surface of the KE-ENSO Explorer flood and drought overlay; paired with the GRID3 / WOPR bottom-up surface (record kenya-population-grid3), which has the same grain but a different method.

- **Temporal Coverage:** 2020
- **Spatial Resolution:** kenya — ~100 m (3 arc-seconds)

14.11.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **WorldPop, University of Southampton** - https://www.worldpop.org/

14.11.3 Scientific Caveats & Guidance

[!WARNING] Constrained = top-down redistribution of national totals onto built-settlement extents. Use for spatial distribution, not as a census-anchored count. Compare with the GRID3 / WOPR bottom-up surface (record kenya-population-grid3). DISCREPANCY AGAINST THE CENSUS: the UN-adjusted national total, ~55.2 M, is about 17 per cent above the enumerated KNBS 2019 census figure of 47,564,296 (record kenya-population-knbs-census). Any absolute headcount taken straight off this surface is therefore ~17 per cent above what a Kenyan user will compare it against. The KE-ENSO exposure tables handle this by taking the LEVEL from KNBS at county and only the within-county SHARE from this grid (record kenya-flood-exposure-intersect); shares and percentages are unaffected either way. An earlier version of this record quoted ~55.9 M, which is the GRID3 total, not this one.

14.12 `kenya-roads-osm`: Kenya Classified Road Network (OpenStreetMap)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.12.1 Summary

Kenya's classified road network (motorway, trunk, primary, secondary and tertiary highways) extracted from the Geofabrik Kenya OpenStreetMap extract and delivered as GeoJSON. The road-access overlay of the KE-ENSO Explorer flood and drought exposure: the network that matters for access, not every track and footpath.

- **Temporal Coverage:** 2026-08
- **Spatial Resolution:** kenya

14.12.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **OpenStreetMap contributors (Geofabrik extract)** - https://www.openstreetmap.org/copyright

14.12.3 Scientific Caveats & Guidance

[!WARNING] ODbL: attribution "(c) OpenStreetMap contributors" and share-alike apply to redistribution of the data. Classified highways only (motorway to tertiary); footways, tracks and unclassified roads are excluded to keep the overlay light. Snapshot of the Geofabrik extract as of August 2026.

14.13 `kenya-facilities-hotosm`: Kenya Health and Education Facilities (HOTOSM)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.13.1 Summary

Health facilities and education facilities (schools) for Kenya from the Humanitarian OpenStreetMap Team exports on HDX (OpenStreetMap-derived), clipped to Kenya and delivered as two GeoJSON files. The critical-asset overlay of the KE-ENSO Explorer flood and drought exposure. HOTOSM is the open working substitute for the official registries (KMHFR health facilities, GIGA schools), whose APIs were not reachable when the layer was built.

- **Temporal Coverage:** 2026-08
- **Spatial Resolution:** kenya — Facility features (points, some building polygons)

14.13.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Humanitarian OpenStreetMap Team (via HDX)** - https://data.humdata.org/organization/hot

14.13.3 Scientific Caveats & Guidance

[!WARNING] ODbL: attribution "(c) OpenStreetMap contributors, via Humanitarian OpenStreetMap Team" and share-alike apply to redistribution of the data. Snapshot of the HDX export as of August 2026. OSM-derived completeness varies by region; this is not an authoritative facility registry.

14.14 `kenya-power-grid-kplc`: Kenya Electricity Transmission Grid (KPLC)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.14.1 Summary

Kenya's electricity transmission network - 11, 33, 66, 132 and 220 kV lines - from the Kenya Power and Lighting Company dataset on energydata.info, merged into one line layer (141,205 features) with a voltage attribute and delivered as GeoJSON. The 132 and 220 kV lines are the national backbone (highest-value at-risk assets). The power-infrastructure overlay of the KE-ENSO Explorer flood and drought exposure. The source's "Unidentified Grid Network" layer is excluded; gridfinder's modelled medium-voltage distribution grid is a deferred complement, not included here.

- **Temporal Coverage:** 2026-08
- **Spatial Resolution:** kenya

14.14.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **Kenya Power and Lighting Company (KPLC), via energydata.info** - https://energydata.info/dataset/kenya-kenya-electricity-network

14.14.3 Scientific Caveats & Guidance

> [!WARNING] Transmission lines only (11 / 33 / 66 / 132 / 220 kV), not the distribution or medium-voltage network. Snapshot of the energydata.info dataset as retrieved in August 2026. The S3 object is served as application/octet-stream; it parses as GeoJSON.

14.15 `kenya-admin-codab`: Kenya Administrative Boundaries, Counties and Sub-counties (IEBC COD-AB)

Category: Spatial | **Publishing State:** `authored` | **License:** Open Data

14.15.1 Summary

Official Kenya administrative boundaries - 47 counties (admin 1) and 290 sub-counties (admin 2) - from the OCHA Common Operational Dataset for Administrative Boundaries (COD-AB), source IEBC, with official p-codes, harmonised to EPSG:4326 and delivered as two GeoJSON files. The reporting-unit backbone for aggregating KE-ENSO Explorer flood and drought exposure (record kenya-flood-exposure-intersect joins on adm2_pcode). Preferred over GAUL for exposure because it carries p-codes and excludes disputed slivers.

- **Temporal Coverage:** 2019-10 to 2025-01
- **Spatial Resolution:** kenya — Admin 1 (47 counties) and admin 2 (290 sub-counties)

14.15.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** (Peter Steward)
- **OCHA / Independent Electoral and Boundaries Commission (IEBC), via HDX** - https://data.humdata.org/dataset/cod-ab-ken

14.15.3 Scientific Caveats & Guidance

> [!WARNING] Geometries were made valid and reprojected but not simplified or otherwise altered. Attribute columns follow the COD-AB convention (adm0/adm1/adm2 _pcode and _name, valid_on, area_sqkm).

14.16 `crop-vop-intld15`: MapSPAM 2020 Adaptation Atlas, Sub-Saharan Africa

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.16.1 Summary

Spatially disaggregated crop statistics for Sub-Saharan Africa: 42 crops across six production systems, harmonised to FAOSTAT country totals, as physical area, harvested area, production, yield and value of production. A regional adaptation of IFPRI's MapSPAM 2020 built for the Africa Agriculture Adaptation Atlas, supplementing the global release with national and sub-national sources. This is the crop-production basis of the Atlas exposure chain: every crop value-of- production layer, and through them the hazard-exposure product, is distributed across these production shares.

- **Temporal Coverage:** 2020
- **Spatial Resolution:** sub-saharan-africa

14.16.2 Data Producers & Custodians

- **Alliance of Bioversity International and CIAT** - https://alliancebioversityciat.org/
- **International Food Policy Research Institute (IFPRI)** - https://www.ifpri.org/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.16.3 Scientific Caveats & Guidance

[!WARNING] NOT THE SAME DATASET AS THE HUB'S `spam2020` RECORD, DESPITE THE NAME. That record describes IFPRI's GLOBAL MapSPAM 2020 v2r2, licensed CC-BY-SA-4.0 and published as zarr. This one is the Atlas's Sub-Saharan Africa adaptation, licensed CC-BY-4.0 and published as GeoTIFF. Different scope, different licence, different format, different DOI. Do not substitute one for the other, and note that the share-alike term on the global release does NOT reach the Atlas exposure chain, which consumes this product. THREE PROCESSING LEVELS, AND THEY ARE NOT INTERCHANGEABLE. `raw` holds the released CSVs, one file per variable and technology code (P_TA, P_TH, P_TI and so on). `analysis-ready` holds rasterised GeoTIFFs. `atlas-harmonized` holds the Atlas value-of-production layers derived from them. The VoP chain reads `analysis-ready/variable=prod_t` and writes into `atlas-harmonized`. FILE NAMING DIFFERS BETWEEN LEVELS. The irrigated system is `irrigated` under `analysis-ready` and `irr` under `atlas-harmonized`. Anything templating a path across both levels must special-case it; the token values below are given per level for that reason. CURRENCY BASIS IS A KNOWN TRAP. Several value-of-production variables coexist - vop_intld15, vop_usd2015, vop_nominal-usd15, vop_nominal-usd21 - and they are NOT interchangeable. A past mismatch between the crop and livestock currency bases produced roughly a sevenfold error in cattle exposure. Constant international dollars (vop_intld15) is the Atlas standard; check the livestock counterpart is on the same basis before combining. Tracked as hazards_prototype issue 30. THREE OBJECTS ARE PUBLISHED UNDER A MALFORMED KEY. Directly beneath `region=ssa/` there are three parquet objects whose path reads `processing=atlas-harmonizedvariable=<name>.parquet`, missing the separator between the two hive fields: crop-livestock_all, vop_intld15-2021 and vop_nominal-usd-2021. They are not reachable by a hive-partitioned scan and should be republished or removed rather than left as a trap for the next reader.

14.17 `livestock-vop`: Gridded Livestock of the World (GLW4) Value of Production (Adaptation Atlas SSA)

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.17.1 Summary

Spatial density distribution and economic value of production for major livestock species (cattle, sheep, goats, camels, pigs, chickens) across Sub-Saharan Africa. Based on FAO Gridded Livestock of the World v4 (GLW4 2020) downscaled to 1 km / 5-minute resolution, harmonised to FAOSTAT country production values and national livestock populations, expressed in constant 2015 international dollars (vop_intld15).

- **Temporal Coverage:** 2020
- **Spatial Resolution:** sub-saharan-africa, kenya — 1 km (0.008333 deg)

14.17.2 Data Producers & Custodians

- **Food and Agriculture Organization of the United Nations (FAO)** - https://www.fao.org/livestock-systems/global-distributions/en/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.17.3 Scientific Caveats & Guidance

> [!WARNING] Attribution: "Data provided by FAO Gridded Livestock of the World (GLW4) and compiled for the African Agriculture Adaptation Atlas (AAAA)." CURRENCY BASIS MUST REMAIN CONSTANT INTERNATIONAL DOLLARS (vop_intld15). Several value-of-production metrics coexist (nominal USD, constant 2015 international dollars). To avoid severe distortions when pairing livestock with MapSPAM crop VoP, both portfolios must be evaluated on the identical constant 2015 international dollar basis. Past unit mismatches produced ~7x errors in cattle exposure. DENSITY VS PIXEL COUNTS. GLW4-2020 layers represent per-km2 animal density (D-DA), not raw counts per grid cell. Spatial aggregation across county administrative polygons must weight by grid-cell land area (excluding open waterbodies). LIVESTOCK VULNERABILITY LAGS. Unlike annual field crops which fail within a single dry dekad, pastoral herds exhibit biological inertia: mortality, distress sales, and conception deficits lag rainfall anomalies by 6 to 12 months.

14.18 `knbs-napr`: KNBS National Agricultural Production Reports (Crops, Livestock & Products)

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.18.1 Summary

Audited subnational agricultural statistics from the Kenya National Bureau of Statistics (KNBS) National Agricultural Production Reports (NAPR 2023/24 and 2024/25 editions). Contains county-level annual area harvested (ha), production (t), and gross farm-gate value (KES) across 22 major food and cash commodities (maize, beans, sorghum, tea, coffee, potatoes, pyrethrum, wheat, dairy milk, beef), cross-verified via dual-engine deterministic parsing (PyMuPDF) and additivity gates.

- **Temporal Coverage:** 2019 to 2024
- **Spatial Resolution:** kenya

14.18.2 Data Producers & Custodians

- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.18.3 Scientific Caveats & Guidance

> [!WARNING] Attribution: "Data extracted deterministically from the Kenya National Bureau of Statistics (KNBS) National Agricultural Production Reports (2023/24, 2024/25), Republic of Kenya." PROJECT RULE D1 COMPLIANCE: ZERO LLM INFERENCE. Every number served from this dataset is parsed directly from published government PDF tables by deterministic code (_sources/napr_extract.py) and passed through an automated machine additivity gate. No language model reads, types, or imputes any numerical value. BLANK IS NOT ZERO. KNBS data is compiled from administrative expert-estimates. Missing county-years represent unobserved reporting gaps, never zero production. Records with missing data are preserved as null and never imputed as false zeroes. CROSS-EDITION REBASE (2025 EDITION PRIORITY). Where reporting years overlap between the 2023/24 and 2024/25 editions, the latest edition is prioritised because it incorporates KNBS retrospective prior-year revisions. ADMINISTRATIVE ESTIMATES VS CENSUS SURVEYS. NAPR figures represent annual ministerial and county directorate of agriculture administrative estimates, not full-enumeration agricultural censuses.

14.19 `harveststat-crop-production`: HarvestStat Africa - Harmonized Subnational Crop Statistics

Category: Agri | **Publishing State:** `authored` | **License:** Open Data

14.19.1 Summary

Cleaned and harmonised subnational crop production, harvested area and yield for Sub-Saharan Africa, assembled from FEWS NET and FAO sources: 574,204 records, 33 countries, 94 crop types, primarily 1980-2022. A federated record - the Adaptation Atlas cites this dataset but does not host or redistribute it.

- **Temporal Coverage:** None
- **Spatial Resolution:** None

14.19.2 Data Producers & Custodians

- **NASA Harvest / University of Maryland** - https://github.com/HarvestStat/HarvestStat-Africa
- **Famine Early Warning Systems Network (FEWS NET)** - https://fews.net/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.19.3 Scientific Caveats & Guidance

> [!WARNING] LICENCE CONFIRMED 2026-09-18. The Dryad landing page does not print a licence, which is why this record was initially held as a draft. The Dryad API settles it: the dataset record returns license = https://spdx.org/licenses/CC0-1.0.html, at version 12, last modified 2026-07-17. CC0 requires no attribution; the citation below is given anyway. Quantities are metric tonnes, areas hectares, yields metric tonnes per hectare. Coverage is jagged across countries, crops and years - check per-country completeness before treating any national series as continuous.

14.20 `kenya-market-prices-fews`: Kenya Retail and Wholesale Market Prices (FEWS NET Data Warehouse)

Category: Markets | **Publishing State:** `authored` | **License:** Open Data

14.20.1 Summary

Staple and livestock market prices for Kenya from the FEWS NET Data Warehouse market price facts: 42 counties by market by product, 12 products, retail and wholesale, in Kenyan shillings, 2000 to 2026. Pulled through the FDW API. The price basis behind the Kenya ENSO Explorer's market panel and behind the pastoral terms of trade shown there, which the Atlas derives rather than reads.

- **Temporal Coverage:** 2000 to present (open-ended)
- **Spatial Resolution:** kenya

14.20.2 Data Producers & Custodians

- **Famine Early Warning Systems Network (FEWS NET), USAID** - https://fews.net/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.20.3 Scientific Caveats & Guidance

[!WARNING] LICENCE IS A POLICY, NOT A CREATIVE COMMONS GRANT. FEWS NET provides data as a public service, managed under USAID's policy on sharing Agency-funded data (ADS 579). Attribution is required, in the manner indicated in the metadata accompanying the data, prominently on any visual that uses it, with a link to the FEWS NET website, and without claiming or implying that FEWS NET endorses the use. FEWS NET disclaims liability and makes no guarantee of accuracy or completeness. THE CLAUSE THAT MATTERS FOR REDISTRIBUTION. FEWS NET states that "some data are provided by third parties, and may not be redistributed or reused without the consent of the original data provider, or may be subject to additional terms and conditions", and instructs users to check the primary source's own policy. Market price series are commonly collected by national agencies rather than by FEWS NET itself, so this clause is live for this dataset: reading the prices for analysis is unencumbered, republishing the series as an Atlas data product needs the originating agency identified first. Nothing derived from this is published by the Atlas today. TERMS OF TRADE IS OURS, NOT THEIRS. The pastoral terms of trade shown in the Explorer is computed here as goat price divided by maize price. It is not an NDMA figure and it is not published by FEWS NET in that form. It is a ratio, so it moves with either leg - a fall can mean cheaper livestock or dearer grain, which have opposite implications - and it should never be read as a price index. NOT NDMA. Kenya's National Drought Management Authority publishes drought early warning bulletins carrying market prices and terms of trade, and an earlier version of the Atlas metadata wrongly named them as a source. No NDMA bulletin is fetched, parsed or mirrored anywhere. The NDMA bulletins remain a candidate source, held as a local draft record. COVERAGE IS UNEVEN. 42 of 47 counties appear, markets within a county vary, and a product is not quoted in every market in every month. Absence is a gap in market reporting, not a zero price.

14.21 `kenya-market-prices-kamis`: Kenya Live-Animal Market Prices by Grade and Sex (MoALD KAMIS)

Category: Markets | **Publishing State:** `draft` | **License:** Open Data

14.21.1 Summary

Market-day livestock price quotes for Kenya from the Kilimo AgriMarkets Market Information System (KAMIS) of the Ministry of Agriculture and Livestock Development (MoALD): cattle, sheep, goats and camels priced per head at named livestock markets in 45 counties, each quote carrying a breed (classification), a quality grade (1 to 3), the animal's sex and the volume on offer, from May 2021 to the present. Harvested from the portal's search export. The only verified open Kenyan source that prices camels and sheep and that records grade and sex, which the FEWS NET series behind the Kenya ENSO Explorer does not; it extends the explorer's pastoral price panel from goats and cattle to all four pastoral species at market-day frequency.

- **Temporal Coverage:** 2021-05-24 to present (open-ended)
- **Spatial Resolution:** kenya

14.21.2 Data Producers & Custodians

- **Ministry of Agriculture and Livestock Development (MoALD), Kenya** - https://kamis.kilimo.go.ke/
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.21.3 Scientific Caveats & Guidance

> [!WARNING] NO PUBLISHED LICENCE. The KAMIS portal (kamis.kilimo.go.ke) footer asserts "Copyright (c) KAMIS" and gives the Ministry's contact (info@kilimo.go.ke, +254-20-2718870, Cathedral Road, Nairobi); it carries no licence statement and no terms of use, and the About page credits the National Agricultural and Rural Inclusive Growth Project (NARIGP) for the system. A copyright line without a licence is a reservation, not a grant. The Atlas reads the quotes for analysis and displays derived monthly medians; the market-day quotes are not republished to S3 or to the Climate Data Hub and no bulk download is offered until MoALD confirms terms of reuse. HOW THE PORTAL PAGES, AND WHY EARLIER COUNTS WERE WRONG. The HTML results table pages by an offset in the URL path (/site/market_search/), not by the page= query parameter, so one wide request returns only the first per_page rows. The Excel export honours per_page and returns a whole window in one file. The harvester pulls one export per species-year and checks the row count against the portal's own pagination total; on 2026-10-23 that was 116,634 rows across the four species. WHOLESALE IS THE LIVESTOCK PRICE. 99.8 percent of

live-animal quotes sit in the portal's Wholesale column, which is the per-head price at the livestock market; the Retail column is essentially empty for animals. The explorer therefore serves these as wholesale prices, distinct from the FEWS NET retail goat and cattle series it already shows. THE 2021-05-24 BULK BACKFILL. The portal's history starts on 2021-05-24, and that single date carries 11,079 quotes, about one tenth of the whole archive. It is the upload date of a backfill, not a market day. Those rows are kept in the market-day file with the flag launch_day_bulk and are excluded from the monthly medians. KAMIS cannot reach the 2011 or 2017 drought analogues; the FEWS NET series remains the long-run record. DIRTY ROWS ARE FLAGGED, NOT DROPPED. A county literally named test (345 quotes), 3,203 exact duplicate submissions, 67 non-positive prices and 591 order-of-magnitude outliers against the same market-day median are all present in the market-day file with a qc_flag, and absent from the monthly medians. County spellings are canonicalised to the 47-county key (Tana-River to Tana River, Homa-bay to Homa Bay, Muranga to Murang'a); the portal spelling is kept in county_raw. COVERAGE IS UNEVEN AND THINNING. Reporting peaked in 2022 (about 40,000 quotes) and fell to a few thousand a year by 2025; camel quotes stop at 2025-11-24. Absence of a market-month is a gap in reporting, not a zero price. Grade and breed are enumerator judgements with no published grading standard, so grade comparisons across markets are indicative only. NO COORDINATES. The portal names the market and county only; the file is plain parquet, not GeoParquet. OBFUSCATED DELIVERY, BY CHOICE. As a courtesy to the Ministry while reuse terms are sought, the market-day file is published to the explorer only as an AES-256-GCM container (market_prices_kamis.parquet.enc) that the explorer decrypts in the browser; the monthly medians it displays are derived aggregates. This stops casual download and crawling. It is not access control: a static page must hold whatever decrypts it, so a determined reader can recover the bytes. The plain file is not committed to the repository and no bulk download is offered.

14.22 `kenya-food-insecurity-ipc`: Kenya Acute Food Insecurity Phase Classification (IPC / KFSSG)

Category: Markets | **Publishing State:** `authored` | **License:** Open Data

14.22.1 Summary

IPC Acute Food Insecurity phase classifications and population-in-phase tables for Kenya, produced twice a year by the Kenya Food Security Steering Group after the long-rains and short-rains assessments. A federated record - the Adaptation Atlas cites this analysis but does not host it.

- **Temporal Coverage:** None
- **Spatial Resolution:** None

14.22.2 Data Producers & Custodians

- **Integrated Food Security Phase Classification (IPC) Global Partnership** - https://www.ipcinfo.org/
- **Kenya Food Security Steering Group (KFSSG)**
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.22.3 Scientific Caveats & Guidance

> [!WARNING] LICENCE CONFIRMED 2026-09-18, AND IT CONSTRAINS DOWNSTREAM USE. IPC states that all IPC results are licensed CC BY-NC-SA 3.0 IGO. Two consequences that do not apply elsewhere in the Atlas stack: NON-COMMERCIAL, so IPC figures cannot travel into a commercially licensed or commercially used product; and SHARE-ALIKE, so any adapted work built on them must carry the same licence, which conflicts with the CC-BY-4.0 most Atlas products use. Citing IPC phases and linking out is unencumbered; folding them into a derived Atlas layer is not. The IPC Analysis Platform is hosted by FAO and governed by its terms of use. A PHASE IS A CLASSIFICATION, NOT A MEASUREMENT. Phases 1-5 come from an expert consensus process over converging evidence, so they are ordinal and are NOT comparable across analysis rounds in the way a measured index is: methodology, evidence base and analyst panel all change between rounds. Do not difference phases between rounds to derive a trend, and do not interpolate between analyses. Kenya analyses are conducted twice a year - long rains (July-August) and short rains (January-February) - and the IPC process is hosted within NDMA.

14.23 `kenya-county-gender-datasheets`: Kenya County Gender Data Sheets, 2025

Category: Demographics | **Publishing State:** `authored` | **License:** Open Data

14.23.1 Summary

Gender and social-inclusion indicators for all 47 Kenyan counties, published as one data sheet per county in October 2025. Indicators cover health and fertility, education, economic participation, political representation, disability and attitudes to intimate-partner violence, drawn largely from the 2022 Kenya Demographic and Health Survey, the 2019 census and

the 2022 general election. A federated record - the Adaptation Atlas reads these sheets but does not host or redistribute them.

- **Temporal Coverage:** None
- **Spatial Resolution:** kenya

14.23.2 Data Producers & Custodians

- **Kenya National Bureau of Statistics (KNBS)** - https://www.knbs.or.ke/
- **State Department for Gender and Affirmative Action, Ministry of Gender, Culture and Children Services**
- **Council of Governors**
- **UN Women (Women Count)** - https://data.unwomen.org/women-count
- **SDGs Kenya Forum**
- **Alliance of Bioversity International and CIAT** (Peter Steward)

14.23.3 Scientific Caveats & Guidance

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table. Treating the whole table as “2022” is wrong for some indicators. BLANK IS NOT ZERO. The extraction serves a series only where at least 40 of 47 counties agree with the modal national value; non-matching county rows are dropped individually. So a served series can be silently missing up to seven counties, and a missing county-series is a parse or gate drop - NEVER an observed zero. Only 10 of 35 series reach all 47 counties; the worst is households with an elderly member at 40/47. Per-county row counts run 22-35. COVERAGE IS ASYMMETRIC WITHIN AN INDICATOR. One code can have 47 counties for male and 46 for female. Do not compute a sex ratio or a gender gap without checking both sides are present for that county. CHART-ONLY INDICATORS ARE ABSENT BY METHOD. Antenatal-visit and vaccination charts and internet usage are dropped by the gate. Their absence reflects the extraction, not the source. LABELS ARE NOT DEFINITIONS. The canonical label is the longest string seen across the 47 sheets and several remain truncated mid-sentence. Do not lift them into variable descriptions. COLUMN-ORDER FRAGILITY. Values are read Kenya-then-County from the printed header. A future edition that flips that header would swap every value silently; the coverage gate catches it only if the national column also becomes inconsistent.

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15 Controlled Vocabulary & Scientific Glossary

16 Controlled Vocabulary & Scientific Glossary

This chapter provides standardized canonical definitions, abbreviations, mathematical notations, and institutional citations for all **84 domain terms** across the Kenya ENSO Explorer platform. It establishes the controlled terminology required for interoperable subnational climate risk analytics.

16.1 Administrative & Geography

16.1.1 Arid and Semi-Arid Lands (ASAL)

The 23 designated arid and semi-arid counties of Kenya comprising over 80% of the national landmass, characterized by low and erratic rainfall, fragile rangeland ecosystems, and predominantly pastoralist livelihoods.

Mathematical Formulation / Metric:

$$\text{County Count} = 23 \text{ designated ASAL counties}$$

Canonical Institutional / Literature Reference: *Kenya Ministry of Devolution and ASALs.*

Active in Platform Sections: `0.0`, `1.0`, `1.1`, `4.3`, `4.4`

16.1.2 Common Operational Datasets - Administrative Boundaries (COD-AB)

The authoritative United Nations humanitarian GIS administrative boundary layers (ADM0 National, ADM1 County, ADM2 Sub-county) endorsed by UN OCHA and the Kenya Survey Department.

Mathematical Formulation / Metric:

$$Vectorpolygonalboundarylayerswithstandardized P-codes$$

Canonical Institutional / Literature Reference: *UN OCHA Humanitarian Data Exchange (HDX).*

Active in Platform Sections: `1.0`, `1.1`, `3.4`, `6.1`

16.1.3 Humanitarian OpenStreetMap Team Facilities (HOT-OSM)

Community-mapped geospatial vector points of health clinics, hospitals, and educational institutions in Kenya, validated for humanitarian disaster exposure modeling.

Mathematical Formulation / Metric:

$$Geospatialpointfeaturelayer$$

Canonical Institutional / Literature Reference: *Humanitarian OpenStreetMap Team (HOT) Kenya Export.*

Active in Platform Sections: `3.5`, `4.5`, `6.1`

16.1.4 Kenya Power and Lighting Company Grid (KPLC)

National electrical transmission line infrastructure vector dataset used in the platform to evaluate power grid asset vulnerability to flood inundation.

Mathematical Formulation / Metric:

$$Vectorlinenetwork(kVtransmissionlines)$$

Canonical Institutional / Literature Reference: *Kenya Power and Lighting Company GIS Repository.*

Active in Platform Sections: `4.5`, `6.1`

16.2 Climatology & Seasons

16.2.1 Double Season Failure (Double Failure)

A production year in which both feeding seasons were Dry, that is a Dry/Dry consecutive sequence (OND of the preceding year and MAM of the same year both below -0.5 standard deviations of their own 1991–2020 climatology). Double failures remove the dry-season buffer of pasture, water and food stocks and are the historical signature of acute drought emergencies in Kenya.

Mathematical Formulation / Metric:

$$\mathrm{DF}_Y = \mathbb{I}\left(z_{\mathrm{OND},Y-1} < -0.5 \ \wedge \ z_{\mathrm{MAM},Y} < -0.5\right)$$

Canonical Institutional / Literature Reference: *Kenya ENSO Explorer methods monograph, Figure 3.1B.*

Active in Platform Sections: `3.1B`

16.2.2 Long Rains (MAM)

The primary agricultural rainfall season across western, central, and northern Kenya, spanning the months of March, April, and May, driven by the northward migration of the Intertropical Convergence Zone (ITCZ).

Mathematical Formulation / Metric:

$$R_{\text{MAM}} = \sum_{m \in \{\text{Mar, Apr, May}\}} R_m$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD) National Seasonal Calendar.*

Active in Platform Sections: `0.0`, `1.0`, `2.1`, `3.1`, `3.1B`, `4.1`

16.2.3 Pastoral Boreal Summer Window (JJAS / JAS)

The dry season across eastern and southern pastoral rangelands (June–September) and the primary rainfall season in the western highlands and Lake Victoria basin.

Mathematical Formulation / Metric:

$$R_{\text{JJAS}} = \sum_{m \in \{\text{Jun, Jul, Aug, Sep}\}} R_m$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD) Regional Climatology.*

Active in Platform Sections: `1.0`, `3.1`, `4.3`

16.2.4 Short Rains (OND)

The secondary agricultural and pastoral rainfall season spanning October, November, and December, exhibiting the highest teleconnection sensitivity and predictive skill linked to Pacific ENSO and Indian Ocean Dipole dynamics.

Mathematical Formulation / Metric:

$$R_{\text{OND}} = \sum_{m \in \{\text{Oct, Nov, Dec}\}} R_m$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD) National Seasonal Calendar.*

Active in Platform Sections: `0.0`, `1.0`, `2.1`, `3.1`, `3.1B`, `4.1`

16.2.5 WMO 1991–2020 Standard Normal Baseline (WMO Normal)

The canonical 30-year climatological reference period defined by the World Meteorological Organization for assessing contemporary climate anomalies and evaluating extreme frequencies.

Mathematical Formulation / Metric:

$$Calendar period 1991-01-01 through 2020-12-31 (30 complete historical years)$$

Canonical Institutional / Literature Reference: *WMO-No. 1203 (2017). WMO Guidelines on the Calculation of Climate Normals.*

Active in Platform Sections: `0.0`, `2.1`, `3.1`, `3.1B`, `3.2`, `6.1`

16.2.6 Warm Dry Season Window (DJF)

The warm dry interval spanning December, January, and February, marked by crop maturation and harvesting in agricultural counties and grazing pressure in pastoral zones.

Mathematical Formulation / Metric:

$$R_{\mathrm{DJF}} = R_{\mathrm{Dec}} + R_{\mathrm{Jan}} + R_{\mathrm{Feb}}$$

Canonical Institutional / Literature Reference: *Kenya Meteorological Department (KMD).*

Active in Platform Sections: `1.0`, `3.1`

16.3 Demography & Equity

16.3.1 Gender Equality and Social Inclusion (GESI)

A developmental framework and empirical indicator suite examining how gender headship, intra-household decision-making, land tenure documentation, and labor division influence differential vulnerability to climate extremes.

Mathematical Formulation / Metric:

$$Multidimensional indicator suite benchmarking county vs national medians$$

Canonical Institutional / Literature Reference: *KNBS County Gender Data Sheets (2025).*

Active in Platform Sections: `1.3`, `6.1`

16.4 Economic & Valuation

16.4.1 Gross Value of Production (VOP)

The aggregate physical output of agricultural crops and livestock commodities valued at constant standardized gate prices, providing an uninflated metric of total agricultural capital at risk from climate extremes.

Mathematical Formulation / Metric:

$$\mathrm{VOP}_c = \sum_{k=1}^{K} Q_{c,k} \times P_k^{\text{constant}}$$

Canonical Institutional / Literature Reference: *FAOSTAT Valuation Methodology / Beta Review Governance.*

Active in Platform Sections: `1.2`, `4.5`

16.4.2 Standing Herd Capital (Herd Capital)

The replacement value of the live herd in a county, computed as head count times a capital-share adjustment for herd structure times the county sale price (0.75 of the mature sale price for cattle, goats and sheep; a species-specific herd-structure ratio for camels). It is a stock, distinct from the annual flow of offtake and milk, and is the quantity that drought mortality and distress sales erode.

Mathematical Formulation / Metric:

$$S_c = \sum_j H_{c,j}\, \kappa_j\, P_{c,j}^{\text{sale}}$$

Canonical Institutional / Literature Reference: *Behnke, R., & Muthami, D. (2011). The contribution of livestock to the Kenyan economy. IGAD LPI Working Paper 03-11.*

Active in Platform Sections: `1.2`

16.5 Food Security & Vulnerability

16.5.1 Integrated Food Security Phase Classification (IPC)

A multi-partner standardized scale categorizing food insecurity severity into five distinct phases: Phase 1 (Minimal), Phase 2 (Stressed), Phase 3 (Crisis), Phase 4 (Emergency), and Phase 5 (Catastrophe/Famine).

Mathematical Formulation / Metric:

$$\text{Phase } \in \{1, 2, 3, 4, 5\}$$

Canonical Institutional / Literature Reference: *IPC Global Partners (2021). Integrated Food Security Phase Classification Technical Manual Version 3.1.*

Active in Platform Sections: `4.6`, `6.1`

16.5.2 Reduced Coping Strategy Index (rCSI)

A standardized proxy measure of household food stress based on a weighted 5-question survey assessing behavioral adaptations taken by households experiencing food shortages over the past 7 days.

Mathematical Formulation / Metric:

$$\text{rCSI} = \sum_{i=1}^{5} w_i \times f_i, \quad w \in \{1, 1, 1, 2, 3\}$$

Canonical Institutional / Literature Reference: *Maxwell, D., et al. (2008). The Coping Strategies Index Field Methods Manual. CARE / WFP.*

Active in Platform Sections: `4.6`, `6.1`

16.6 Institutional & Governance

16.6.1 Agriculture and Food Authority (AFA)

The state regulator overseeing development, regulation, and promotion of scheduled agricultural crops in Kenya (including sugar, tea, coffee, nuts, and cereals).

Mathematical Formulation / Metric:

$$\textit{Stateregulatoryauthority}$$

Canonical Institutional / Literature Reference: *Agriculture and Food Authority Act (2013).*

Active in Platform Sections: `4.1`, `6.1`

16.6.2 Anticipatory Action (AA)

Acting ahead of predicted hazardous climate events to prevent or reduce humanitarian impacts before disaster strikes, triggered by probabilistic meteorological forecasts.

Mathematical Formulation / Metric:

$$\text{Trigger: } P(\text{Extreme}) \geq \tau_{\text{action}}$$

Canonical Institutional / Literature Reference: *UN OCHA / Red Cross Climate Centre Anticipatory Action Framework.*

Active in Platform Sections: `0.0`, `2.2`

16.6.3 Beta Review Governance (Beta Review Governance)

The publication posture applied to the Section 1.2 economic valuation assets while formal licensing of the National Drought Management Authority bulletin prices is sought: values are shown for expert review as modeled screening estimates, carry a Beta Review Edition banner, provide no bulk download, and are excluded from the open Atlas data store and the Comprehensive Data Hub. The figures are not calibrated for direct use in fiscal transfers or compensation.

Canonical Institutional / Literature Reference: *National Drought Management Authority County Drought Early Warning Bulletins (2019–2021), used under research fair dealing.*

Active in Platform Sections: `1.2`

16.6.4 Common Alerting Protocol (CAP)

An international XML standard (OASIS CAP 1.2) for exchanging public warnings across media. The Kenya Meteorological Department publishes its severe weather warnings as a machine-readable CAP feed, which the Explorer reads to list current and recent advisories by county.

Canonical Institutional / Literature Reference: *OASIS (2010). Common Alerting Protocol Version 1.2. OASIS Standard.*

Active in Platform Sections: `2.3`

16.6.5 Comprehensive Data Hub Schema (CDH Schema)

The institutional metadata and data publishing specification (v0.3.0) governing dataset descriptions, citation formatting, temporal-spatial schemas, and open data licensing across the Adaptation Atlas.

Mathematical Formulation / Metric:

$$Metadatacontractspecificationv0.3.0$$

Canonical Institutional / Literature Reference: *Africa Agriculture Adaptation Atlas CDH Technical Specification.*

Active in Platform Sections: `6.1`

16.6.6 Disaster Risk Reduction (DRR)

Policy objective aimed at preventing new and reducing existing disaster risk and managing residual risk, strengthening resilience.

Mathematical Formulation / Metric:

$$SendaiFrameworkforDisasterRiskReduction2015-2030.$$

Canonical Institutional / Literature Reference: *UNDRR Sendai Framework.*

Active in Platform Sections: `0.0`, `2.2`, `4.6`

16.6.7 Famine Early Warning Systems Network (FEWS NET)

A leading provider of early warning and analysis on food insecurity created by USAID, providing data on markets, trade, and agro-climatology across East Africa.

Mathematical Formulation / Metric:

$$Globalearlywarningnetwork$$

Canonical Institutional / Literature Reference: *USAID FEWS NET.*

Active in Platform Sections: `4.1`, `4.2B`, `4.4`, `4.4B`, `6.1`

16.6.8 Food and Agriculture Organization (FAO / FAOSTAT)

The specialized agency of the United Nations leading international efforts to defeat hunger, maintain global agricultural statistical databases (FAOSTAT), and establish standardized commodity price concepts.

Mathematical Formulation / Metric:

$$UNSpecializedStatisticalAgency$$

Canonical Institutional / Literature Reference: *FAO Statistical Databases (FAOSTAT).*

Active in Platform Sections: `1.2`, `4.1`, `6.1`

16.6.9 Kenya Agricultural and Livestock Research Organization (KALRO)

The principal agricultural research institution in Kenya developing crop varieties, rangeland management protocols, and agro-weather advisories.

Mathematical Formulation / Metric:

$$Nationalagriculturalresearchauthority$$

Canonical Institutional / Literature Reference: *KALRO Act (2013).*

Active in Platform Sections: `4.1`, `6.1`

16.6.10 Kenya Food Security Steering Group (KFSSG)

The multi-agency technical coordination body led by NDMA and national line ministries that conducts biannual post-harvest food security assessments (Short Rains Assessment - SRA; Long Rains Assessment - LRA).

Mathematical Formulation / Metric:

$$Biannualconsensusassessments(SRA/LRA)$$

Canonical Institutional / Literature Reference: *National Drought Management Authority (NDMA).*

Active in Platform Sections: `0.0`, `4.6`, `6.1`

16.6.11 Kenya Meteorological Department (KMD)

The canonical national hydrometeorological service of Kenya responsible for national weather forecasting, seasonal climate outlooks, marine telemetry, and severe weather early warnings.

Mathematical Formulation / Metric:

$$Nationalhydrometauthority$$

Canonical Institutional / Literature Reference: *Ministry of Environment, Climate Change and Forestry.*

Active in Platform Sections: `0.0`, `2.1`, `2.2`, `2.3`, `6.1`

16.6.12 Kenya National Bureau of Statistics (KNBS)

The principal government agency responsible for collecting, analyzing, and disseminating official statistical information in Kenya, including the decennial Population and Housing Census and economic surveys.

Mathematical Formulation / Metric:

$$Nationalstatisticalauthority$$

Canonical Institutional / Literature Reference: *Statistics Act (2006).*

Active in Platform Sections: `1.0`, `1.1`, `1.2`, `1.3`, `4.1`, `6.1`

16.6.13 NDMA Drought Phase (Drought Phase)

The official drought early warning stage assigned to each arid and semi-arid county in the National Drought Management Authority monthly bulletins: Normal, Alert, Alarm, Emergency, and Recovery. Stages combine biophysical indicators (rainfall, vegetation condition, water) with livelihood indicators (livestock condition, prices, nutrition) and trigger contingency financing.

Canonical Institutional / Literature Reference: *National Drought Management Authority (NDMA). County Drought Early Warning Bulletins. Nairobi: NDMA.*

Active in Platform Sections: `3.4`, `4.6`

16.6.14 National Drought Management Authority (NDMA)

The statutory leadership agency in the Government of Kenya responsible for drought early warning, crisis coordination, resilience building, and operating the national early warning bulletin monitoring network.

Mathematical Formulation / Metric:

$$Monthly drought monitoring across 23 ASAL counties$$

Canonical Institutional / Literature Reference: *National Drought Management Authority Act (2016).*

Active in Platform Sections: `0.0`, `1.2`, `4.6`, `6.1`

16.6.15 Regional Centre for Mapping of Resources for Development (RCMRD)

An intergovernmental organization based in Nairobi, Kenya, providing geospatial and remote sensing support, disaster risk mapping, and Earth observation analytics across Eastern and Southern Africa.

Mathematical Formulation / Metric:

$$Intergovernmental Earth observation center$$

Canonical Institutional / Literature Reference: *RCMRD Institutional Charter.*

Active in Platform Sections: `3.5`, `6.1`

16.6.16 ReliefWeb Disaster Chronicle (ReliefWeb / GLIDE)

The UN OCHA humanitarian information portal tracking disaster situations and compiling emergency response dispatches indexed by Global Identifier Numbers (GLIDE).

Mathematical Formulation / Metric:

$$GLIDE Unique Disaster Identifier standard$$

Canonical Institutional / Literature Reference: *UN OCHA ReliefWeb Service.*

Active in Platform Sections: `4.6`, `6.1`

16.7 Markets & Early Warning

16.7.1 Cross-Border Food Trade (XBT Trade)

The formal and informal monitored cross-border trade of staple cereal grains and livestock across East African border corridors (e.g. Busia, Malaba, Moyale, Namanga, Mandera) tracked quarterly by FEWS NET.

Mathematical Formulation / Metric:

$$V_{\text{XBT}} = \sum \text{Tonnes}_{\text{cereal}} \text{ (Import / Export)}$$

Canonical Institutional / Literature Reference: *FEWS NET East Africa Cross-Border Trade Bulletin.*

Active in Platform Sections: `4.4B`, `6.1`

16.7.2 Kenya Agricultural Market Information System (KAMIS)

The national market price surveillance platform operated by the Ministry of Agriculture and Livestock Development tracking daily wholesale and retail food commodity prices across Kenya's urban and rural markets.

Mathematical Formulation / Metric:

$$\textit{Daily/weekly market surveillance time series}$$

Canonical Institutional / Literature Reference: *Ministry of Agriculture and Livestock Development (MoALD).*

Active in Platform Sections: `4.4`, `6.1`

16.7.3 Pastoral Terms of Trade (ToT)

The ratio of pastoral purchasing power, measured as the number of kilograms of local staple grain (dry maize) that a pastoral household can purchase through the sale of one medium-sized live goat (shoat).

Mathematical Formulation / Metric:

$$\text{ToT}_m = \frac{P_{\text{goat},m}}{P_{\text{maize},m}} \quad \text{(kg/goat)}$$

Canonical Institutional / Literature Reference: *National Drought Management Authority (NDMA) Monthly Early Warning Bulletins.*

Active in Platform Sections: `4.4`, `6.1`

16.8 Ocean & Climate Drivers

16.8.1 Columbia IRI Prediction Plume (IRI Plume)

A multi-model ensemble of dynamical and statistical seasonal forecast models predicting Niño 3.4 SST anomalies up to 9 months ahead, curated monthly by the International Research Institute for Climate and Society (IRI).

Mathematical Formulation / Metric:

$$\bar{z}_{\text{IRI}} = \frac{1}{M} \sum_{m=1}^{M} z_{\text{model}_m}$$

Canonical Institutional / Literature Reference: *Columbia Climate School International Research Institute for Climate and Society.*

Active in Platform Sections: `2.2`

16.8.2 Dipole Mode Index (DMI)

The quantitative measure of the Indian Ocean Dipole, defined as the difference in sea surface temperature anomaly between the western tropical Indian Ocean (50°E–70°E, 10°S–10°N) and the southeastern tropical Indian Ocean (90°E–110°E, 10°S–0°).

Mathematical Formulation / Metric:

$$\text{DMI} = \text{SSTA}(50^\circ\text{–}70^\circ\text{E}, 10^\circ\text{S–}10^\circ\text{N}) - \text{SSTA}(90^\circ\text{–}110^\circ\text{E}, 10^\circ\text{S–}0^\circ)$$

Canonical Institutional / Literature Reference: *JAMSTEC Application Laboratory / NOAA Physical Sciences Laboratory.*

Active in Platform Sections: `0.0`, `2.1`, `3.2`, `3.3`, `4.1`

16.8.3 El Niño–Southern Oscillation (ENSO)

A coupled ocean-atmosphere phenomenon in the tropical Pacific characterized by irregular fluctuations between warm (El Niño), cool (La Niña), and neutral sea surface temperature anomalies, exerting global atmospheric teleconnections.

Mathematical Formulation / Metric:

$$\text{SSTA}_{\text{Niño 3.4}} \geq +0.5^\circ\text{C (El Niño)}, \leq -0.5^\circ\text{C (La Niña)}$$

Canonical Institutional / Literature Reference: *Trenberth, K. E. (1997). The definition of El Niño. BAMS, 78(12), 2771-2778.*

Active in Platform Sections: `0.0`, `2.1`, `2.2`, `3.1`, `3.2`, `3.3`, `4.1`, `5.1`

16.8.4 Indian Ocean Dipole (IOD)

An ocean-atmosphere coupled climate mode in the equatorial Indian Ocean characterized by an anomalous sea surface temperature gradient between the western tropical Indian Ocean and the eastern tropical Indian Ocean.

Mathematical Formulation / Metric:

$$\mathrm{DMI} = \mathrm{SSTA}_{\text{Western IO}} - \mathrm{SSTA}_{\text{Eastern IO}}$$

Canonical Institutional / Literature Reference: *Saji, N. H., et al. (1999). A dipole mode in the tropical Indian Ocean. Nature, 401(6751), 360-363.*

Active in Platform Sections: `0.0`, `2.1`, `2.2`, `3.2`, `3.3`, `4.1`

16.8.5 Joint Ocean Phase (Joint Ocean Phase)

The combined classification of a season by its ENSO phase and its Indian Ocean Dipole phase. ENSO phase is El Niño when the seasonal-mean Relative Oceanic Niño Index exceeds +0.5 °C, La Niña below -0.5 °C, and Neutral otherwise; IOD phase is positive when the October to December mean Dipole Mode Index exceeds +0.4 °C, negative below -0.4 °C, and neutral otherwise.

Mathematical Formulation / Metric:

$$\mathrm{Phase} = \big(\mathrm{ENSO}(\overline{\mathrm{RONI}}),\ \mathrm{IOD}(\overline{\mathrm{DMI}})\big)\,, \quad \pm 0.5^{\circ}\mathrm{C},\ \pm 0.4^{\circ}\mathrm{C}$$

Canonical Institutional / Literature Reference: *Saji, N. H., Goswami, B. N., Vinayachandran, P. N., & Yamagata, T. (1999). A dipole mode in the tropical Indian Ocean. Nature, 401, 360-363.*

Active in Platform Sections: `2.1`, `2.2`, `3.1`, `4.2`

16.8.6 Oceanic Niño Index (ONI)

NOAA's primary operational index for tracking ENSO, computed as the 3-month running mean of ERSST.v5 SST anomalies in the Niño 3.4 region (5°N–5°S, 120°–170°W) based on 30-year centered base periods.

Mathematical Formulation / Metric:

$$\mathrm{ONI} = \frac{1}{3}\sum_{m=1}^{3} \mathrm{SSTA}_{\text{Niño 3.4},m}$$

Canonical Institutional / Literature Reference: *NOAA Climate Prediction Center (CPC) Operational ENSO Criteria.*

Active in Platform Sections: `2.1`, `3.2`, `3.3`

16.8.7 Relative Oceanic Niño Index (RONI)

The Oceanic Niño Index (Niño 3.4 SST anomaly) minus the mean tropical ocean SST anomaly (20°S–20°N). Measures relative warmth against tropical background warming to identify true atmospheric convective forcing in a warming climate.

Mathematical Formulation / Metric:

$$\mathrm{RONI} = \mathrm{SSTA}_{\text{Niño 3.4}} - \mathrm{SSTA}_{\text{Tropical (20°S–20°N)}}$$

Canonical Institutional / Literature Reference: *van Oldenborgh, G. J., Hendon, H., Stockdale, T., L'Heureux, M., Coughlan de Perez, E., Singh, R., & van Aalst, M. (2021). Defining El Niño indices in a warming climate. Environmental Research Letters, 16(4), 044003.*

Active in Platform Sections: `0.0`, `2.1`, `3.2`, `3.3`, `4.1`

16.8.8 SINTEX-F Forecast Plume (SINTEX-F)

The Scale Interaction Experiment-Frontier coupled ocean-atmosphere general circulation model developed by JAMSTEC, used operationally for 9-month lead forecasts of the Indian Ocean Dipole and ENSO.

Mathematical Formulation / Metric:

$$Ensemble mean of 9 coupled dynamical simulation members$$

Canonical Institutional / Literature Reference: *Luo, J. J., et al. (2005). Extended-range ENSO and IOD predictions using SINTEX-F. J. Climate.*

Active in Platform Sections: `2.2`

16.8.9 Sea Surface Temperature Anomaly (SSTA)

The deviation of ocean surface temperature from the long-term climatological mean for a given location and calendar month, typically referenced to a 30-year WMO normal baseline.

Mathematical Formulation / Metric:

$$\mathrm{SSTA}_{y,m} = \mathrm{SST}_{y,m} - \bar{\mathrm{SST}}_{m(1991\text{–}2020)}$$

Canonical Institutional / Literature Reference: *WMO Climatological Standard Normals Technical Document No. 1203.*

Active in Platform Sections: `2.1`, `3.3`

16.8.10 Walker Circulation (Walker Circulation)

The zonal overturning circulation of the tropical atmosphere, with rising air over the warm western Pacific and Maritime Continent and sinking air over the eastern Pacific, and a corresponding Indian Ocean cell. ENSO and the Indian Ocean Dipole shift these cells, changing subsidence and moisture convergence over East Africa and so modulating the Short Rains.

Canonical Institutional / Literature Reference: *Nicholson, S. E. (2017). Climate and climatic variability of rainfall over eastern Africa. Reviews of Geophysics, 55(3), 590-635.*

Active in Platform Sections: `5.1`

16.8.11 Western Pacific V-shape Pattern (Western-V / WV)

A recognized sea surface temperature and atmospheric circulation dipole in the Western North Pacific that modulates moisture flux into the Greater Horn of Africa, particularly relevant during boreal spring (MAM).

Mathematical Formulation / Metric:

$$\text{WV Index} = \text{Standardized } z\text{-score of WNP SST/Wind anomaly dipole}$$

Canonical Institutional / Literature Reference: *Funk, C., et al. (2023). Western Pacific driver mechanics modulating East African rainfall.*

Active in Platform Sections: `2.1`, `5.1`

16.9 Official Statistics

16.9.1 HarvestStat Subnational Crop Statistics (HarvestStat)

A harmonized, multi-decadal historical subnational crop production database developed by the FEWS NET Science Team and UC Santa Barbara, extending from 1982 through 2020 for Kenyan counties.

Mathematical Formulation / Metric:

$$Harmonized historical time series with quality flags$$

Canonical Institutional / Literature Reference: *FEWS NET Science Team & Climate Hazards Center.*

Active in Platform Sections: `4.2B`, `6.1`

16.9.2 National Agriculture Production Report (KNBS NAPR)

The official national agricultural survey published by the Kenya National Bureau of Statistics providing subnational, county-level production statistics across 47 counties for crops, livestock headcounts, and animal products.

Mathematical Formulation / Metric:

$$Officialstatestatisticalrelease(2024/2025editions)$$

Canonical Institutional / Literature Reference: *Kenya National Bureau of Statistics (KNBS).*

Active in Platform Sections: `1.1`, `4.1`, `6.1`

16.10 Remote Sensing & Hazards

16.10.1 Climate Hazards Center InfraRed Precipitation with Station Data (CHIRPS v3)

A quasi-global high-resolution precipitation dataset blending cold cloud duration (CCD) satellite infrared imagery with calibrated in-situ station observations, providing gridded rainfall at 0.05° spatial resolution from 1981 to present.

Mathematical Formulation / Metric:

$$R_{\text{CHIRPS}} = R_{\text{IR}} \times \frac{S_{\text{gauged}}}{S_{\text{climatology}}}$$

Canonical Institutional / Literature Reference: *Funk, C., et al. (2015). The climate hazards infrared precipitation with stations. Scientific Data, 2, 150066.*

Active in Platform Sections: `0.0`, `1.0`, `2.1`, `3.1`, `3.4`, `6.1`

16.10.2 Cold Cloud Duration (CCD)

The duration of time a satellite thermal infrared pixel registers temperatures below a convective threshold (typically -40°C), used by CHIRPS as a physical proxy for convective precipitation.

Mathematical Formulation / Metric:

$$\text{CCD} = \sum_{t} \mathbb{I}(T_b(t) < 233.15\,\text{K}) \times \Delta t$$

Canonical Institutional / Literature Reference: *Milford, J. R., & Dugdale, G. (1990). Estimation of rainfall using cold cloud duration. Int. J. Remote Sens.*

Active in Platform Sections: `3.1`, `3.4`

16.10.3 Copernicus Global Flood Monitoring (Copernicus GFM)

An operational Earth observation service providing near-real-time satellite Synthetic Aperture Radar (SAR) flood inundation extents derived from Sentinel-1 constellation radar acquisitions at 20m resolution, independent of cloud cover.

Mathematical Formulation / Metric:

$$\text{SAR Backscatter Threshold: } \sigma_0 < \tau_{\text{water}} \text{ (dB)}$$

Canonical Institutional / Literature Reference: *European Commission Joint Research Centre (JRC) Copernicus GFM Service.*

Active in Platform Sections: `1.0`, `3.5`, `4.5`, `6.1`

16.10.4 Cropland Land Cover Mask (Cropland Mask)

A 10m spatial resolution binary landcover mask derived from ESA WorldCover Sentinel-1/Sentinel-2 classification identifying cultivated agricultural parcels.

Mathematical Formulation / Metric:

$$M_{\text{crop}}(x, y) \in \{0, 1\}$$

Canonical Institutional / Literature Reference: *Zanaga, D., et al. (2021). ESA WorldCover 10m 2020 v100.*

Active in Platform Sections: `1.2`, `4.5`

16.10.5 Gridded Livestock of the World (GLW / Livestock Density)

Global high-resolution modeled spatial distribution grids of cattle, goats, sheep, and camels disaggregated to sub-kilometer pixels using random forest environmental regressions.

Mathematical Formulation / Metric:

$$D_{\text{animal}}(x, y) \text{ (heads/km}^2\text{)}$$

Canonical Institutional / Literature Reference: *Gilbert, M., et al. (2018). Global distribution data for cattle, buffaloes, horses, sheep, goats, pigs, chickens and ducks in 2010. Scientific Data.*

Active in Platform Sections: `1.2`, `4.5`

16.10.6 JRC Flood Return Period Hazard Layers (JRC Flood RP)

Hydrodynamic inundation depth maps modeled by the European Commission Joint Research Centre for statistical return intervals (1-in-10, 1-in-20, 1-in-50, 1-in-100 years) based on calibrated river discharge distributions.

Mathematical Formulation / Metric:

$$T_{\text{RP}} = \frac{1}{P(\text{Exceedance in any single year})}$$

Canonical Institutional / Literature Reference: *Dottori, F., et al. (2016). Development and evaluation of a framework for global flood hazard mapping. Advances in Water Resources.*

Active in Platform Sections: `3.5`, `4.5`, `6.1`

16.10.7 Normalized Difference Vegetation Index (MODIS NDVI)

A satellite remote sensing index quantifying photosynthetic biomass and vegetation greenness by computing the normalized ratio of surface reflectance in near-infrared (NIR) and visible red (RED) spectrums.

Mathematical Formulation / Metric:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{RED}}}{\rho_{\text{NIR}} + \rho_{\text{RED}}}$$

Canonical Institutional / Literature Reference: *Rouse, J. W., et al. (1974). Monitoring the vernal advancement of retrogradation of natural vegetation. NASA/GSFC.*

Active in Platform Sections: `4.3`, `6.1`

16.10.8 Standardized Precipitation Index (SPI)

A widely used drought index based on the probability of precipitation for any time scale, fitted to a gamma distribution and transformed to a normal distribution with mean zero and standard deviation one.

Mathematical Formulation / Metric:

$$\text{SPI} = \Phi^{-1}(G(P))$$

Canonical Institutional / Literature Reference: *McKee, T. B., et al. (1993). The relationship of drought frequency and duration to time scales. AMS.*

Active in Platform Sections: `3.1`, `6.1`

16.10.9 Standardized Precipitation-Evapotranspiration Index (SPEI-3)

A multiscalar drought index that combines precipitation and atmospheric evaporative demand (potential evapotranspiration) accumulated over a 3-month rolling window, fitted to a log-logistic probability distribution.

Mathematical Formulation / Metric:

$$D_i = P_i - \mathrm{PET}_i, \quad \mathrm{SPEI} = \Phi^{-1}(F(D))$$

Canonical Institutional / Literature Reference: *Vicente-Serrano, S. M., et al. (2010). A multiscalar drought index sensitive to global warming: SPEI. J. Climate.*

Active in Platform Sections: `3.1`, `6.1`

16.10.10 Synthetic Aperture Radar (SAR)

An active microwave imaging sensor that illuminates the Earth's surface with radar pulses, capable of penetrating cloud cover, dense smoke, and nighttime darkness to detect smooth open water surfaces via specular reflection.

Mathematical Formulation / Metric:

$$\sigma^0 \text{ (radar backscatter coefficient in decibels)}$$

Canonical Institutional / Literature Reference: *Sentinel-1 Technical Guide, European Space Agency (ESA).*

Active in Platform Sections: `1.0`, `3.5`, `4.5`

16.10.11 Water Requirement Satisfaction Index (WRSI)

A geospatial agro-climatological crop water balance model calculating the ratio of actual seasonal evapotranspiration to crop water requirements, used by FEWS NET to identify crop and pasture moisture stress.

Mathematical Formulation / Metric:

$$\mathrm{WRSI} = \frac{\sum_{t=1}^{N} \mathrm{ET}_a}{\sum_{t=1}^{N} \mathrm{ET}_c} \times 100\%$$

Canonical Institutional / Literature Reference: *Senay, G. B., & Verdin, J. (2003). Characterizing drought conditions in the Greater Horn of Africa using WRSI. Int. J. Remote Sens.*

Active in Platform Sections: `4.1`, `6.1`

16.11 Statistical Methods

16.11.1 Analogue Year (Analogue Year)

A historical season whose equatorial ocean state most closely resembles the current or forecast state, selected by the smallest standardized Euclidean distance in Relative Oceanic Niño Index (RONI) and Dipole Mode Index (DMI) space. The rainfall terciles observed in the k nearest analogue years (k = 5 or 8) give the empirical outlook probabilities.

Mathematical Formulation / Metric:

$$\mathcal{A} = \{k \text{ years with smallest } D_i\}, \quad D_i = \sqrt{\sum_d (\Delta x_{i,d}/\sigma_d)^2}$$

Canonical Institutional / Literature Reference: *Kenya ENSO Explorer methods monograph, Table 2.3 (forecast verification).*

Active in Platform Sections: `2.1`, `2.2`, `4.2`

16.11.2 Bivariate Joint Probability Distribution (Joint Probability)

The empirical probability of two climate variables concurrently occupying designated states (e.g., El Niño in the Pacific co-occurring with positive Indian Ocean Dipole).

Mathematical Formulation / Metric:

$$P(A \cap B) = \frac{n(A \cap B)}{N_{\text{total}}}$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences.*

Active in Platform Sections: `3.2`, `4.1`

16.11.3 Brier Skill Score (BSS)

The skill of a probability forecast for one binary event (for example, the Dry tercile occurs) relative to the climatological probability, computed as one minus the ratio of the mean squared forecast error to the mean squared climatological error. The Explorer reports BSS separately for the drought-relevant Dry tercile and the flood-relevant Wet tercile.

Mathematical Formulation / Metric:

$$\text{BSS}_c = 1 - \frac{\overline{(p_c - o_c)^2}}{\overline{(1/3 - o_c)^2}}$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences (3rd ed.). Academic Press.*

Active in Platform Sections: `2.2`

16.11.4 Conditional Tercile Probability (Conditional Probability)

The empirical probability that seasonal rainfall falls in the Drier, Near normal or Wetter tercile given a stated ocean condition, estimated as the relative frequency of each tercile among historical seasons that share that condition. It is compared with the unconditioned climatological probability of one third.

Mathematical Formulation / Metric:

$$\hat{P}(T = k \mid \text{phase}) = \frac{\#\{y : T_y = k,\ \text{phase}_y\}}{\#\{y : \text{phase}_y\}}$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences (3rd ed.). Academic Press.*

Active in Platform Sections: `2.1`, `2.2`

16.11.5 Consecutive Seasonal Sequence (Consecutive Sequence)

The ordered pair of rainfall states for the two seasons that feed one production year: the Short Rains of the preceding calendar year (OND, Y−1) followed by the Long Rains of the same year (MAM, Y). Each season is classified Dry, Normal or Wet from its own independently standardized z-score before pairing.

Mathematical Formulation / Metric:

$$\mathcal{P}_Y = (\text{State}(\text{OND}, Y-1),\ \text{State}(\text{MAM}, Y))$$

Canonical Institutional / Literature Reference: *Kenya ENSO Explorer methods monograph, Figure 3.1B.*

Active in Platform Sections: `3.1B`

16.11.6 Contingency Table (Contingency Matrix)

A cross-tabulation of counts between two categorical variables, used here to tabulate ocean phase against observed rainfall tercile and, in forecast verification, the predicted modal tercile against the observed tercile. Row-normalized cells give conditional frequencies.

Mathematical Formulation / Metric:

$$n_{ij} = \#\{y : A_y = i,\ B_y = j\}, \quad \hat{P}(B = j \mid A = i) = n_{ij}/n_{i\cdot}$$

Canonical Institutional / Literature Reference: *Jolliffe, I. T., & Stephenson, D. B. (Eds.). (2012). Forecast Verification: A Practitioner's Guide in Atmospheric Science (2nd ed.). Wiley.*

Active in Platform Sections: `2.1`, `2.2`

16.11.7 Detrended Yield Anomaly (Yield Anomaly)

The residual of a crop production or yield series from its fitted linear time trend, expressed in absolute units and as a percentage of the trend value. Removing the trend separates year-to-year climate shocks from long-run changes in area, technology and reporting.

Mathematical Formulation / Metric:

$$a_y = Y_y - (\hat{\beta}_0 + \hat{\beta}_1 y), \qquad a_y^{\%} = 100\, a_y / (\hat{\beta}_0 + \hat{\beta}_1 y)$$

Canonical Institutional / Literature Reference: *Lobell, D. B., & Field, C. B. (2007). Global scale climate-crop yield relationships and the impacts of recent warming. Environmental Research Letters, 2(1), 014002.*

Active in Platform Sections: `3.1`, `4.1`, `4.2B`

16.11.8 Dry-Season Run Length (Dry Run)

The number of consecutive Dry seasons (z below -0.5) ending with a given season when all MAM and OND seasons are walked in strict chronological order. Any Normal, Wet or missing season resets the count to zero. The longest run in a county record and its cumulative rainfall deficit summarise multi-season drought persistence.

Mathematical Formulation / Metric:

$$L_t = (L_{t-1} + 1)\, \mathbb{I}(z_t < -0.5), \quad L_0 = 0$$

Canonical Institutional / Literature Reference: *Kenya ENSO Explorer methods monograph, Figure 3.1B.*

Active in Platform Sections: `3.1B`

16.11.9 Empirical Confidence Interval (95% CI)

The range of statistical estimates within which the true population parameter is expected to fall with 95% probability, calculated via non-parametric bootstrapping.

Mathematical Formulation / Metric:

$$\mathrm{CI}_{95\%} = [\hat{Q}_{\mathrm{boot}}(0.025), \hat{Q}_{\mathrm{boot}}(0.975)]$$

Canonical Institutional / Literature Reference: *Efron, B., & Tibshirani, R. J. (1994). An Introduction to the Bootstrap.*

Active in Platform Sections: `2.2`, `3.3`

16.11.10 Forecast Skill (Forecast Skill)

The improvement of a forecast system over a reference forecast, measured here by the leave-one-out cross-validated Ranked Probability Skill Score against equal-odds climatology and labelled High (RPSS at least 0.20), Moderate (0.10 to 0.20), Marginal (0 to 0.10) or No skill (below 0). The class sets the colour of the skill badge and the guidance text beside each outlook; probabilities are displayed in every class, so a No-skill outlook should be read as a historical frequency only.

Mathematical Formulation / Metric:

$$\text{Class(RPSS)} \in \{\text{High} \geq 0.20,\ \text{Moderate} \geq 0.10,\ \text{Marginal} \geq 0,\ \text{No skill} < 0\}$$

Canonical Institutional / Literature Reference: *Jolliffe, I. T., & Stephenson, D. B. (Eds.). (2012). Forecast Verification: A Practitioner's Guide in Atmospheric Science (2nd ed.). Wiley.*

Active in Platform Sections: `2.2`, `2.3`

16.11.11 Heidke Skill Score (HSS)

A categorical skill score that rescales the proportion of correct modal forecasts by the proportion expected from an uninformed forecast. For three equiprobable terciles the expected proportion is one third; HSS is 1 for perfect categorical forecasts, 0 for chance and negative for worse than chance.

Mathematical Formulation / Metric:

$$\text{HSS} = \frac{H - N/3}{N - N/3}$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences (3rd ed.). Academic Press.*

Active in Platform Sections: `2.2`

16.11.12 Hyndman-Fan Type 7 Quantile Estimator (Hyndman-Fan Type 7)

The standard continuous empirical quantile estimation algorithm used in R, Python, and DuckDB where sample fractiles are linearly interpolated between adjacent sorted values.

Mathematical Formulation / Metric:

$$\widehat{Q}(p) = X_{(\lfloor h \rfloor)} + (h - \lfloor h \rfloor)(X_{(\lfloor h \rfloor + 1)} - X_{(\lfloor h \rfloor)}), \quad h = (N-1)p + 1$$

Canonical Institutional / Literature Reference: *Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. The American Statistician.*

Active in Platform Sections: `2.1`, `3.1`

16.11.13 Independent Seasonal Standardization (Seasonal z-score)

A standardization protocol where OND and MAM seasonal rainfall are transformed into z-scores independently against their own distinct historical seasonal means and standard deviations, ensuring fair multi-season sequence classification.

Mathematical Formulation / Metric:

$$z_{s,y} = \frac{R_{s,y} - \mu_{s(1991\text{–}2020)}}{\sigma_{s(1991\text{–}2020)}}, \quad s \in \{\text{OND}, \text{MAM}\}$$

Canonical Institutional / Literature Reference: *World Meteorological Organization (2017). WMO Guidelines on the Calculation of Climate Normals (WMO-No. 1203); Kenya ENSO Explorer methods monograph, Figure 3.1B.*

Active in Platform Sections: `3.1B`

16.11.14 Leave-One-Out Cross-Validation (LOOCV)

A deterministic cross-validation procedure where each year in the historical hindcast record is iteratively omitted from the training distribution and used solely as the out-of-sample evaluation target, preventing artificial hindcast overconfidence.

Mathematical Formulation / Metric:

$$\text{Skill}_{\text{LOOCV}} = \frac{1}{N} \sum_{y=1}^{N} \mathcal{S}(\hat{P}_{(-y)}, O_y)$$

Canonical Institutional / Literature Reference: *Wilks, D. S. (2011). Statistical Methods in the Atmospheric Sciences. Academic Press.*

Active in Platform Sections: `0.0`, `2.2`, `6.1`

16.11.15 Modal Hit Rate (Modal Hit Rate)

The percentage of historical verification seasons where the forecast's highest-probability (modal) category exactly matched the tercile category that actually occurred.

Mathematical Formulation / Metric:

$$\text{HR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I} \left(\arg\max_k p_{i,k} == \arg\max_k o_{i,k} \right) \times 100\%$$

Canonical Institutional / Literature Reference: *Jolliffe, I. T., & Stephenson, D. B. (2012). Forecast Verification: A Practitioner's Guide.*

Active in Platform Sections: `0.0`, `2.2`

16.11.16 Non-Parametric Empirical Quantiles (Empirical Quantiles)

Fractile estimators calculated directly from the sorted empirical cumulative distribution function without assuming a theoretical parametric distribution (such as Gaussian normal). Mandatory for East African rainfall due to positive skewness.

Mathematical Formulation / Metric:

$$\hat{Q}(p) = (1-\gamma)X_{(j)} + \gamma X_{(j+1)}, \quad j = \lfloor pN + (1-p)/3 \rfloor$$

Canonical Institutional / Literature Reference: *Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. The American Statistician.*

Active in Platform Sections: `2.1`, `3.1`, `3.2`

16.11.17 Olympic Mean Baseline (Olympic Mean)

A robust central tendency statistic calculated by dropping the highest and lowest extreme values from a multi-year sample (e.g. 5 or 10 years) and averaging the remaining values, minimizing distortion from black-swan shocks.

Mathematical Formulation / Metric:

$$\bar{X}_{\text{Olympic}} = \frac{1}{N-2} \sum_{i=2}^{N-1} X_{(i)}$$

Canonical Institutional / Literature Reference: *WMO Climatological Practices Guide.*

Active in Platform Sections: `4.1`, `4.2B`

16.11.18 Ordinary Least Squares Linear Regression (OLS Regression)

A parametric linear model estimating the slope and intercept relating continuous oceanic driver z-scores to county rainfall accumulation in mm.

Mathematical Formulation / Metric:

$$\hat{Y} = \beta_0 + \beta_1 X + \epsilon, \quad \min_{\beta} \sum (Y_i - \hat{Y}_i)^2$$

Canonical Institutional / Literature Reference: *Weisberg, S. (2005). Applied Linear Regression.*

Active in Platform Sections: `3.3`

16.11.19 Rainfall Positive Skewness (Positive Skewness)

The asymmetric right-tailed distribution characteristic of East African seasonal precipitation, where the majority of seasons experience near- or below-average totals and a minority experience extreme episodic deluge.

Mathematical Formulation / Metric:

$$\gamma_1 = \frac{\frac{1}{N}\sum_{i=1}^{N}(R_i - \bar{R})^3}{\left(\frac{1}{N}\sum_{i=1}^{N}(R_i - \bar{R})^2\right)^{3/2}} > 0$$

Canonical Institutional / Literature Reference: *Hastings, D., et al. (2020). Skewness in African rainfall extremes. J. Hydrometeorology.*

Active in Platform Sections: `3.1`, `6.1`

16.11.20 Rainfall Terciles (Terciles)

Three equiprobable rainfall classes (Below normal or Drier, Near normal, Above normal or Wetter) separated by the empirical 33.3rd and 66.7th percentiles of the 1991–2020 seasonal totals for a county and season. By construction each class holds one third of the baseline years.

Mathematical Formulation / Metric:

$$\text{Drier} : R \le \hat{Q}(1/3), \quad \text{Near normal} : \hat{Q}(1/3) < R < \hat{Q}(2/3), \quad \text{Wetter} : R \ge \hat{Q}(2/3)$$

Canonical Institutional / Literature Reference: *Hyndman, R. J., & Fan, Y. (1996). Sample quantiles in statistical packages. The American Statistician, 50(4), 361–365.*

Active in Platform Sections: `2.1`, `2.2`, `3.1`, `3.2`

16.11.21 Ranked Probability Score (RPS)

A strictly proper scoring rule for probabilistic forecasts of ordered categories, equal to the sum of squared differences between the cumulative forecast probabilities and the cumulative observation indicator over the categories. For three rainfall terciles only the first two cumulative terms are informative. Forecasts that place probability far from the observed category are penalised more than near misses.

Mathematical Formulation / Metric:

$$\text{RPS} = \sum_{m=1}^{K-1}\left(\sum_{k\le m} p_k - \sum_{k\le m} o_k\right)^2, \quad K = 3$$

Canonical Institutional / Literature Reference: *Epstein, E. S. (1969). A scoring system for probability forecasts of ranked categories. Journal of Applied Meteorology, 8(6), 985–987.*

Active in Platform Sections: `2.2`

16.11.22 Ranked Probability Skill Score (RPSS)

A categorical probabilistic verification metric that measures the relative accuracy of a multicategory probabilistic forecast (such as terciles) compared to a climatological reference forecast.

Mathematical Formulation / Metric:

$$\mathrm{RPS} = \sum_{m=1}^{K} \left(\sum_{k=1}^{m} p_k - \sum_{k=1}^{m} o_k \right)^2, \quad \mathrm{RPSS} = 1 - \frac{\mathrm{RPS}_{\text{forecast}}}{\mathrm{RPS}_{\text{climatology}}}$$

Canonical Institutional / Literature Reference: *Epstein, E. S. (1969). A scoring system for probability forecasts of ranked categories. J. Appl. Meteor.*

Active in Platform Sections: `0.0`, `2.2`, `6.1`

16.11.23 Standard Normal Homogeneity Test (SNHT)

A statistical test applied to climatological time series to detect abrupt shifts or changepoints caused by station relocations, instrument recalibrations, or landcover changes.

Mathematical Formulation / Metric:

$$T_k = k\bar{z}_1^2 + (n-k)\bar{z}_2^2$$

Canonical Institutional / Literature Reference: *Alexandersson, H. (1986). A homogeneity test applied to precipitation data. J. Climatology.*

Active in Platform Sections: `3.1`

16.11.24 Subcounty Apportionment Shares (Subcounty Weights)

Commodity-specific, mass-preserving weights that distribute each statutory county valuation total across the 290 IEBC parliamentary constituencies: MapSPAM 2020 physical crop area for crops and 2019 census household livestock counts for livestock and products, with land-area shares as a flagged fallback where the prior is zero. Shares sum to one within each county and commodity, so constituency values re-aggregate exactly to the county total.

Mathematical Formulation / Metric:

$$w_{sc} = \frac{A_{sc}}{\sum_{k \in c} A_k}, \quad \sum_{sc \in c} w_{sc} = 1, \quad V_{sc} = w_{sc} V_c$$

Canonical Institutional / Literature Reference: *Kenya ENSO Explorer methods monograph, Figure 1.2.*

Active in Platform Sections: `1.2`

16.11.25 Subcounty Spatial Apportionment (Spatial Apportionment)

The mathematical disaggregation of county-level aggregate production and livestock values across subcounties using high-resolution spatial auxiliary weights.

Mathematical Formulation / Metric:

$$V_{sc} = V_c \times w_{sc}, \quad \sum_{sc \in c} w_{sc} = 1.0$$

Canonical Institutional / Literature Reference: *Kenya ENSO Explorer methods monograph, Figure 1.2: mass-preserving dasymetric apportionment over IEBC constituencies (MapSPAM 2020 v1r2 physical area; KPHC 2019 livestock counts).*

Active in Platform Sections: `1.2`

16.11.26 Teleconnection Gradient (Pearson r)

The bivariate linear correlation coefficient quantifying the strength and sign of teleconnection between ocean sea surface temperature anomalies and local county precipitation.

Mathematical Formulation / Metric:

$$r = \frac{\sum (z_x - \bar{z}_x)(z_y - \bar{z}_y)}{\sqrt{\sum (z_x - \bar{z}_x)^2 \sum (z_y - \bar{z}_y)^2}}$$

Canonical Institutional / Literature Reference: *Pearson, K. (1895). Mathematical contributions to the theory of evolution.*

Active in Platform Sections: `3.3`
